



supercurriculum
year11

History

Basic Supercurriculum

Why Supercurriculum Matters in History

Studying beyond the syllabus helps students connect past events with modern contexts, think critically about evidence, and understand how interpretations are shaped. Independent exploration strengthens essay writing, source evaluation, and historical empathy.

1. Reading – Breadth and Comprehension

- Read “**A Short History of the World**” by **H.G. Wells** for a broad timeline perspective.
- Study key chapters from **Norman Lowe’s *Mastering Modern World History***.
- Follow **BBC Bitesize – IGCSE History** topic summaries and quizzes.
- Use **school or local library archives** to find old newspapers about 20th-century events.

2. Writing – Argument and Precision

- Write a **500-word historical commentary** each month (e.g. “Why did the League of Nations fail?”).
- Practise **timed essays** comparing causes and consequences.
- Keep a “History Argument Journal” – one page per topic, stating your argument, counterargument, and evidence.

3. Listening – Interpretation and Empathy

- Listen to **BBC Radio 4 “In Our Time – History”** episodes (e.g. *The Treaty of Versailles*, *Cold War*, *Reformation*).
- Find interviews with historians on **YouTube or podcasts**, noting differing interpretations.

4. Watching – Visual and Applied Understanding

- Watch documentaries such as:
 - *World at War* (ITV)
 - *Apocalypse: World War II* (France 2)
 - *The Century: America’s Time* (ABC News)
- Reflect on how camera angles, narration, and tone influence interpretation.

5. Researching – Independent Inquiry

- Investigate one **local or family story** linked to a major event (e.g. migration, war, reform).
- Use **national archives** (UK or your country’s equivalent) to find primary sources.

- Create a short presentation comparing **two historians' views** on the same event.

6. Student-led Tasks – Collaboration and Leadership

- Lead a group debate: “*Was appeasement a reasonable policy?*”
- Curate a **mini-exhibition or timeline** of one topic (e.g. “Rise of Dictatorships 1919–1939”).
- Peer-mark classmates’ essays using simplified mark schemes.

7. Creative Tasks – Innovation and Expression

- Design a **propaganda poster** from the perspective of a 1930s political party.
- Write a **historical diary entry** or short story set during a chosen event.
- Reimagine a historical speech with a modern message.

8. Reflection – Metacognition and Growth

- After each topic, answer:
 - *What surprised me most?*
 - *Which sources did I trust least and why?*
 - *How does this connect to the modern world?*
- Keep a **learning log** to track progress and note skills gained.

9. Interdisciplinary Connections – Synthesis

- Link History with **Geography** (map borders, resources, migration).
- With **English**, analyse speeches and rhetoric.
- With **Economics**, examine the impact of depression, trade, and recovery.

10. Digital & Career Literacy – Modern Relevance

- Learn how digital archives, data visualisation, and AI tools preserve and analyse historical records.
- Explore careers in **law, diplomacy, journalism, teaching, and museum curation** that depend on historical thinking.

History

Extended Supercurriculum

1) Reading — Depth, Historiography, Contrasting Views

- **Core texts for depth**
 - Norman Lowe, *Mastering Modern World History* (targeted chapters; make margin notes with Q/C/E: *Question, Challenge, Extend*).
 - A.J.P. Taylor vs. Ian Kershaw (short extracts) on causes of WWII — *log where they agree, where they diverge, why*.
- **Monographs by topic** (choose one per term)
 - *The Origins of the Cold War* (John Lewis Gaddis)
 - *The Holocaust* (Christopher Browning or Saul Friedländer)
- **Historiography journal**
 - For each topic, complete a one-page “**Interpretations Sheet**”:
 - Thesis (≤15 words) • Method (economic / political / cultural) • Evidence type • Bias/assumptions • Counter-thesis.

2) Writing — High-scoring Essays & DBQ Technique

- **Essay architecture**
 - Use **PEAC** paragraphs (Point–Evidence–Analysis–Counter). Each paragraph must include **at least two** precise facts (date/name/statistic) and one inference.
- **Timed drills** (alternate weeks)
 - 25-minute **Explain** (8–10 marks style)
 - 45-minute **To what extent** (16–20 marks style)
- **Model intros & conclusions**
 - Intro: context (1–2 lines) + line of argument + factor map.
 - Conclusion: judgement **ranked** (most → least significant factor) + “however” clause.
- **DBQ (source-based) mini-papers**
 - 4–6 sources; write a **comparative synthesis paragraph** that explicitly links at least **three** sources.

Marking checklist (self/peer)

- Relevance sustained; no narrative drift.
- Precise evidence (dates, treaties, acts, names) embedded, not appended.
- Counterargument present in ≥50% of body paragraphs.
- Judgement explicit and justified, not implied.

3) Listening — Source Lectures & Note Mastery

- Curate a playlist (e.g., *In Our Time – History* episodes) mapped to your syllabus options.
- **Cornell notes + 24-hour revision**: within a day, add a **2–3 sentence synthesis** and **two questions to the speaker** (imagined).
- Produce one **audio abstract** (≤90 seconds) per topic explaining “why historians disagree”.

4) Watching — Visual Literacy & Propaganda Analysis

- Documentary set (e.g., *World at War*, *Apocalypse: WWII*). For each:
 - Identify **three editorial choices** (music, sequencing, interviewee selection) and how they shape interpretation.
 - Capture one **statistic/visual** to chase back to its original source (do a micro-provenance check).

5) Researching — Primary Sources & Micro-History

- **Archive task** (school/local/national digital archive): locate a **non-text** primary source (poster, photo, map, speech waveform).
- Apply **OPCVL** to two contrasting sources on the same event:
 - **Origin, Purpose, Content, Value, Limitations** (bullet each with one concrete detail).
- **Micro-history**: 1,000–1,200 word study of a **local person/place** intersecting with a world event (e.g., wartime economy, migration). Include **at least three** primary references.

6) Student-led Tasks — Debates, Exhibitions, Teaching

- Lead a **formal debate** (e.g., “Appeasement was a rational strategy in 1938”). Roles: openers, evidence team, rebuttal, adjudication.
- Curate a **pop-up museum** (6–8 panels): thesis statement, artefact images, captions with provenance.
- **Teach-back**: 10-minute mini-lesson to Year 10 on “How to analyse a source without paraphrasing”.

7) Creative Tasks — Transformations & Counterfactuals

- **Propaganda remix**: reproduce a 1930s poster’s message with modern design tools; append a 250-word critique explaining technique (bandwagon, fear, glittering generalities).
- **Counterfactual** (≤800 words): “How might the Cold War have unfolded if the Marshall Plan had failed?” Include **two** plausible ripple effects and **one** reason it’s speculative.

8) Reflection — Metacognition Routines

- **After-Action Review** following each timed essay:
 - What decision (planning/evidence choice) most improved my mark?
 - What misconception persisted?
 - One micro-goal for the next attempt (e.g., “embed dates in topic sentences”).
- **Bias log**: note when you trusted a source too quickly; rewrite a claim with hedging (likely / suggests / indicates).

9) Interdisciplinary Connections — Maps, Data, Ethics

- **Geography:** redraw borders after a treaty; annotate with resources/ethnicities to explain tensions.
- **Economics:** one page on **inflation, reparations, unemployment** with a small table of figures and a 3-line historical interpretation.
- **Civics/Ethics:** short brief on **rights vs. security** in emergency laws.

10) Digital & Career Literacy — Evidence, Tools, Pathways

- Build a **citation habit** (author, title, year, page/URL date-accessed). Keep an **annotated bibliography** (1–2 lines per entry: usefulness & limits).
- Use timelines/mind-mapping tools to visualise causation vs. correlation.
- Explore careers: law, civil service, diplomacy, journalism, museums, policy analysis. Draft a **100-word role profile** linking History skills to each.

Extension Projects (choose 1 per term)

1. **Comparative Interpretations Dossier** (6–8 pages): two historians, one event. Include thesis map, evidence table, and your adjudication.
2. **Local Archive Exhibition:** digitise 5 artefacts with captions and a 300-word curator's essay.
3. **Oral History:** record and transcribe an interview; append source-critique on memory reliability.

Exam Craft: Weekly Micro-Plan (8 weeks)

- **Week 1–2:** Causation vs. consequence drills; 25-min “Explain” essays ×3.
- **Week 3–4:** Source evaluation ladder; write two synthesis paragraphs linking 3+ sources.
- **Week 5:** Two full timed essays (different question stems).
- **Week 6:** Mixed DBQ mini-paper (30–40 min).
- **Week 7:** Whole paper under time; review with mark scheme and error codes.
- **Week 8:** Personal weak-area bootcamp; one final mock.

Quick Rubrics (student-friendly)

Source Analysis (8–10 style)

- 0–2: paraphrase; limited provenance.
- 3–5: some inference; basic OPCVL; limited linkage.
- 6–8: confident inference + provenance; links sources; addresses purpose/bias.
- 9–10: integrated comparison; weighs reliability; conclusion aligns with evidence.

Essay Judgement (16–20 style)

- Thesis explicit; factor ranking justified; counterargument integrated; precise evidence throughout; no narrative padding.

High-yield Prompts (practice rotation)

- “How far do you agree that **economic** factors were the main cause of X?”
- “Which was more significant: **leader decisions** or **structural constraints** in Y?”
- “Assess the **reliability** of Sources A–D on Z; how and why do they differ?”

Toolkit (print and keep)

- **PEAC** sentence starters; **OPCVL** checklist; **Dates & Names** mini-deck (10 cards per topic); **Error code sheet** (EV = vague evidence; CJ = unclear judgement; CT = missing counter).

Want me to adapt this to your **specific option choices** (e.g., 19th-century: *Development of Modern Nation States*; 20th-century: *International Relations, Germany, Russia, USA*, etc.) and produce **topic-specific drills** and **fact banks**?

Είπες:

guide

To ChatGPT είπες:

Here’s a **Student Guide** for **History** (fits any options; UK spellings). It’s practical, printable, and pairs with your Basic/Extended supercurriculum.

History: Student Guide

1) Course Map & Exam Overview

- **What you study:** Key topics from 19th–20th century History (International Relations plus one or more Depth Studies, depending on your centre).
- **Skills you’re tested on (AOs):**
 - **AO1 Knowledge:** Recall, select, deploy accurate facts.
 - **AO2 Causation/Consequence/Change:** Explain and analyse.
 - **AO3 Source Skills:** Inference, reliability, purpose, comparison, contextual evaluation.
- **Typical papers (check your syllabus code):**
 - **Paper 1 (Core Content & Depth):** Structured knowledge/analysis essays.
 - **Paper 2 (Source Paper):** Analysis of a source set on a focused topic.
 - **Paper 4 or Coursework (centre dependent):** Depth Study essay under timed conditions.

Always confirm your exact paper combination with your teacher.

2) Command Words—What They Really Mean

- **Describe / Outline:** Give key facts in order (no explanation).
- **Explain / Why:** Give cause–effect links; use **because/therefore/so**.
- **How far / To what extent:** Argue both sides, then **rank/judge**.
- **Assess / Evaluate reliability:** Use **provenance + content + context**.
- **Compare:** Directly link similarities/differences; don’t write two mini-essays.

3) High-Scoring Essay Structure (Paper 1 / Depth)

PEAC paragraphing

Point (answer the question) → Evidence (precise facts: dates, names, stats) →

Analysis (how/why it matters) → Counter (weigh against an alternative).

Intro (3–4 lines):

- 1 line context; 1-line thesis; brief factor map (most → least important).

Conclusion (4–5 lines):

- Final **judgement** with **ranked factors** and a “however” clause that shows nuance.

Checklist before you finish:

- Each paragraph has 2+ **precise facts**.
- At least **one counter** per paragraph or a dedicated counter paragraph.
- The conclusion **answers the exact question stem**, not a generic summary.

4) Source Paper Method (Paper 2) — OPCVL+

Use for each source, briefly:

- **Origin** (who/when/where)
- **Purpose** (why produced; audience)
- **Content** (what it says/shows; key inferences)
- **Value** (what it’s good for answering this question)
- **Limitations** (gaps, bias, scope, timing)

Comparison/Synthesis Tips:

- Link sources **explicitly**: “Source B challenges A by...”
- **Contextualise** with 1–2 outside facts to test reliability/purpose.
- Finish with a **mini-judgement**: which source(s) best address the question and why.

5) Evidence Mastery: Building Your Fact Bank

Create a one-page fact sheet per topic with:

- **Timeline (10–12 dates)**
- **Key actors** (3–5 leaders/organisations with one-liners)
- **Data points** (e.g., percentages, treaty terms, casualties)
- **Keywords** (appeasement, deterrence, hyperinflation, totalitarianism, etc.)

Keep cards small; **precision beats volume**.

6) Weekly Study Rhythm (sample)

Mon: 20 min timeline drill + 25 min “Explain” question

Tue: Read 6–8 pages + annotate (Q/C/E: Question–Challenge–Extend)

Wed: Source pair (inference + reliability) 30 min

Thu: Plan a “How far do you agree...?” (8-minute plan → 35-minute write)

Fri: Fact-bank update (5 key stats) + 10-min flash review

Weekend: 1 documentary segment + 200-word reflection (what changed my view?)

7) Note-taking That Works (Cornell + Recall)

- **Left column:** Questions/key terms.
- **Right column:** Notes in short lines (no paragraphs).
- **Bottom summary (3–4 lines):** “What caused change? What limited it?”
- **Next day:** Close notes; **recall** from memory in 5 bullets, then check.

8) Common Pitfalls & Fixes

- **Narrative drift:** Fix by underlining your **link-back** sentence each paragraph.
- **Vague evidence:** Replace “many people” with **named groups/dates/figures**.
- **One-sided essays:** Insert a **counter** in every body paragraph.
- **Paraphrasing sources:** Force yourself to write **inference stems**: “This suggests...”, “The author aims to...”, “Given the date, the audience likely...”.

9) Mini-Rubrics (use for self/peer marking)

Essay (16–20 style):

- Clear thesis answering the stem []
- Accurate, precise evidence throughout []
- Analysis explains **how/why** evidence proves the point []
- Counterargument integrated (not bolted on) []
- Judgement ranked and justified []

Source Response:

- Inferences beyond paraphrase []
- Provenance + purpose used analytically []
- Context facts applied to test reliability []
- Comparison/synthesis across sources []
- Focus on the exact question []

10) Revision Sprint Plan (4 weeks pre-exam)

Week 1: Core Content timelines + 6 “Explain why” drills

Week 2: Source skills pack: inference, reliability, comparison (3 sets)

Week 3: Depth Study essays: 2 × “How far...” + 1 plan bank

Week 4: Full mock under timed conditions + error analysis + target fixes

11) Quick Templates (copy/paste)

Essay intro (adapt):

“While several factors contributed to [issue], the **most significant** was [factor] because [reason]. [Factor 2] also mattered due to [reason], whereas [Factor 3] had a **more limited** impact given [reason].”

Source judgement sentence:

“Given its [origin] and [purpose] at [date], Source X is **valuable for**

understanding [aspect], but its [limitation] means it is **less reliable** for [other aspect].”

12) Materials & Tools

- Folder with dividers: **Core, Depth, Sources, Fact bank, Plans.**
- Flashcards (digital or paper) for **dates/names/terms.**
- Past paper pack + mark schemes for **deliberate practice.**
- Timeline/mind-map tool (any simple app is fine).

13) Wellbeing & Exam Day

- Sleep 8 hours before mocks/exams; last 24 hours = light drills only.
- In the exam: **Plan first** (5–8 minutes). Use the plan to keep paragraphs tight.
- If stuck: write a **mini-paragraph** with one strong fact + one inference; move on.

Want this tailored to your exact Option(s) (e.g., Germany 1918–45, USA 1919–41, Russia 1905–41, Cold War)?

Tell me your option set and I’ll add **topic-specific fact banks, source packs, and timed essay prompts.**

Είπες:

11 Geography basic

To ChatGPT είπες:

Excellent — here’s the **Basic Supercurriculum Guide for Geography** (Cambridge International, 0460 or 0976 syllabus).

It’s written in the same structure and tone as your other subject guides, ready for student use or integration into your learning app / study manual.

□ Geography

Basic Supercurriculum

Why Supercurriculum Matters in Geography

Supercurricular work helps students **go beyond memorising case studies** and instead explore **real-world processes, patterns, and perspectives**. Geography connects science, society, and citizenship — preparing students to analyse data, interpret maps, and make informed global decisions about sustainability, development, and risk.

1. Reading – Breadth and Comprehension

- Read **BBC Bitesize – IGCSE Geography** topic summaries and test sections.
- Use **Cambridge Geography Student Book** for key concepts and diagrams.
- Read news features in **National Geographic, The Guardian Environment**, or **BBC Future Planet** about urbanisation, migration, and climate.

- Explore short nonfiction works like *Prisoners of Geography* by Tim Marshall or *Factfulness* by Hans Rosling.

□ Tip: Keep a “Geo Journal” — for each article, note the place, issue, causes, and impacts.

2. Writing – Argument and Precision

- Practise **explaining causes and effects** in 3-sentence paragraphs (Cause → Effect → Example).
- Keep a **case study diary**: 1 page per example (Location, Key facts, Causes, Effects, Responses).
- Write short essays comparing two contrasting regions (e.g. rich vs. poor countries; high vs. low population growth).
- Use **connectives** like *therefore*, *as a result*, *in contrast*, *however* to link ideas clearly.

3. Listening – Interpretation and Empathy

- Listen to **BBC Earth Podcast** or **National Geographic Audio** stories about people and the environment.
- Watch school geography lectures or YouTube explainers (e.g. *Time for Geography*).
- Summarise what you hear in a 5-point note set (what, where, why, result, future).

4. Watching – Visual and Applied Understanding

- Watch geography documentaries such as:
 - *Planet Earth II* (BBC)
 - *The Impossible Planet* (National Geographic)
 - *Our Planet* (Netflix)
- Observe **patterns and processes** — e.g. river erosion, deforestation, urban growth — and sketch a labelled diagram after each episode.

5. Researching – Independent Inquiry

- Conduct a **mini fieldwork project** near your school or home (traffic flow, litter, land use, or microclimate).
- Use **Google Earth / Google Maps** to trace rivers, coasts, or city growth over time.
- Research how your local area has changed in population, transport, or housing in the last 20 years.
- Present your findings as a short report (Question → Method → Data → Conclusion).

6. Student-led Tasks – Collaboration and Leadership

- In small groups, prepare a **poster or slideshow** explaining a natural hazard and how people manage it.

- Debate: “*Should all countries limit tourism to protect their environment?*”
- Create a **Geography Quiz** for classmates (10 multiple choice questions on global facts).

7. Creative Tasks – Innovation and Self-expression

- Design an infographic on **global inequality**, **water scarcity**, or **renewable energy**.
- Build a 3D model of a river valley or volcano using recyclable materials.
- Write a short story from the perspective of a **migrant**, **farmer**, or **climate refugee** showing human impact.

8. Reflection – Metacognition and Growth

- After each topic, write:
 - “What was the key process?”
 - “How is it linked to human or natural causes?”
 - “What could change in 50 years?”
- Track your confidence on key skills (map reading, graph interpretation, data analysis).

9. Interdisciplinary Connections – Synthesis

- **Science:** climate, ecosystems, energy, plate tectonics.
- **Economics:** trade, development, resource distribution.
- **History:** colonialism, industrialisation, migration.
- **ICT:** mapping, data visualisation, and satellite imagery.

10. Digital & Career Literacy – Global Futures

- Learn to use free online tools: **Google Earth**, **ArcGIS StoryMaps**, or **Our World in Data**.
- Explore careers that rely on geography: urban planning, environmental science, meteorology, cartography, tourism, policy analysis.
- Write a short profile (100 words) linking geography skills to one job you might enjoy.

Geography

Extended Supercurriculum

1) Deep Reading & Case-Study Mastery

Core texts & portals

- Geography (Student Book/Revision Guide) + examiner reports.
- Our World in Data topic pages (population, energy, climate).
- National Geographic/Guardian Environment long-reads (annotate with “C–E–E”: *Cause–Evidence–Effect*).

Case-study menu (pick 1 per theme; build a 1-page fact bank)

- **Tectonics:** Japan 2011 (EQ/tsunami) • Montserrat 1995–97 (volcano)
- **Weather hazards:** Cyclone Idai 2019 (Mozambique) • Typhoon Haiyan 2013 (Philippines)
- **Coasts/Rivers:** Nile Delta erosion • Holderness coast (hard/soft engineering) • The Ganges–Brahmaputra floodplain
- **Population/Migration:** China/India population policies • Syrian migration to EU • Rural-urban migration in Nigeria
- **Urban/Industry:** Lagos or Dhaka (opportunities/challenges) • Bangladesh garment industry (globalisation)
- **Energy/Water:** Kenya geothermal • Three Gorges Dam (benefits/trade-offs) • Cape Town Day Zero (water stress)

Fact bank template (printable, 1 page)

- **Location:** map sketch + coordinates/region
- **Key stats:** 5–7 concrete numbers with dates
- **Causes:** physical & human (2/2)
- **Impacts:** social/economic/environmental (≥2 each)
- **Responses:** short/long term + evaluation (what worked/what didn't)

2) Writing for Top Marks (P1/P2/P4)

Paragraph engines

- **PEEL** (Point–Evidence–Explain–Link) for 4–6 mark explainers.
- **CAUSE–EVIDENCE–SO WHAT?** for “why” questions.
- **Balance & Judgement** for “To what extent”: present both sides, then rank significance.

Intros/Conclusions

- **Intro (2–3 lines):** define scope, name case study, state line of argument.
- **Conclusion (3–4 lines):** ranked factors + conditional (“...depends on...”).

Self-check (tick before handing in)

- Named place + 3+ **precise stats/dates**.
- Both **process** and **place** explained.
- At least one **counter-point** & mini-judgement.

3) Data, Graphs & GIS (skill boosters)

- **Transform** one dataset three ways (e.g., bar → line → proportional circles) and write a sentence on what each best reveals.
- **Map skills:** draw/interpret cross-sections, 4/6-figure grid refs, bearings, long profile of a river.
- **Stats you should use confidently:** mean, median, mode, range, **IQR**, simple % **change**, **spearman-lite** (rank → comment on correlation, no calc if not taught).
- **GIS mini-tasks (free tools):**
 1. Delineate a coastal cell in Google Earth; measure a 1 km transect; add 4 geotagged photos.
 2. Create a StoryMap slide with *map + photo + 2-line annotation* for an urban land-use pattern.

4) Fieldwork Excellence (Paper 3 Coursework or Paper 4 Alt)

Investigation cycle (memorise these steps)

1. **Question/Hypotheses** (e.g., “River velocity increases downstream.”)
2. **Methods** (sampling, equipment, timing, risk & ethics)
3. **Data presentation** (located bar charts, field sketches, cross-sections, scatter, choropleths)
4. **Analysis** (patterns, anomalies, link to theory)
5. **Conclusion** (answer hypotheses, with evidence)
6. **Evaluation** (limitations, improvements, next steps)

Go-to techniques

- **Rivers:** width, depth, velocity (float/flow-meter), bedload size/roundness (Power’s scale).
- **Coasts:** beach profiles, sediment size, groyne measurements, wave counts.
- **Urban:** land-use transects, environmental quality surveys (EQI), traffic/pedestrian counts, bi-polar building quality.

Presentation mix (choose 3 per study)

- Located proportional symbols • Cross-section/profile • Scatter with best-fit line • Bi-polar bar • Field-sketch + annotations.

Evaluation stems

- “Our sample size of __ limits reliability because __.”
- “Systematic sampling every __ m introduced bias as __; stratified would reduce this by __.”
- “Instrument error likely $\pm$ __; repeat readings would __.”

5) Watching/Listening with Purpose

- **Time for Geography** (rivers, coasts, urban)—after each video, sketch a labelled diagram + 2 bullet links to your case study.
- **Podcasts:** BBC Earth/Science in Action—write a 60-second audio abstract & one policy implication.

6) Student-Led & Creative Extensions

- **Host a geo-debate:** “Managed retreat is the only sustainable coastal strategy.” Roles: proposer/opposer/rebuttal/judges; use a rubric.
- **Design an infographic** on one SDG (e.g., water, cities, climate) combining 3 charts + a mini-policy brief (120 words).
- **Photo transect:** 10 photos along a high-street; annotate changes in land use, sphere of influence, footfall.

7) Reflection & Metacognition

- **After-Action Review** (post homework or timed essay):
 - What did I do that raised the mark?
 - Which stat/process was weak?
 - One concrete change for next time.
- **Error codes** (write in margin): EV=vague evidence; MAP=missing map; BAL=one-sided; JP=judgement poor.

8) Exam Craft (P1/P2/P4)

Command words—how to hit them

- **Describe:** patterns/trends with figures & place names.
- **Explain:** causes/processes → *therefore/so/this leads to*.
- **Discuss/To what extent:** both sides + judgement.
- **Evaluate a strategy:** criteria (cost, effectiveness, time, environment, equity) + ranked verdict.

Paper 2 (skills) quick wins

- Always **quote figures** from graphs/maps.
- Locate with **direction + distance** (“NE, ~5 km from...”).
- For photos: **foreground/mid/background**, processes, evidence.

Paper 4 (Alt to Coursework) scaffold

- HYPOTHESES → METHODS (with sampling) → DATA PRESENTATION → ANALYSIS → CONCLUSION → EVALUATION.
- Use at least **3 presentation types** and make **explicit links** to hypotheses.

9) 8-Week High-Yield Plan

- **W1–2:** Case-study fact banks (6 sheets) + daily 10-min stats/graph drill.
- **W3:** Rivers/Coasts processes + one fieldwork write-up (mini).
- **W4:** Urban/Industry + Paper 2 skills set (map/photo/data).
- **W5:** Hazards/Climate & development links + 2 x 6-mark explainers (timed).
- **W6:** Full Paper 1 section under time; mark with checklist.
- **W7:** Paper 4 mock (Alt to Coursework) with full evaluation.
- **W8:** Weak-spot bootcamp; mixed-topic retrieval every day.

10) Quick Rubrics (student-friendly)

6-mark Explain/Discuss

- Clear process/causal chain []
- Two precise place/number examples []
- Balanced & linked sentences (connectives) []
- Mini-judgement or prioritisation []

Fieldwork Response

- Hypotheses stated & tested clearly []
- Methods fit question; sampling justified []
- Appropriate, accurate presentation []
- Analysis uses data (figures, trends, anomalies) []
- Conclusion answers hypotheses; evaluation realistic []

11) Printable One-Pagers (ready to lift)

- **Case-Study Fact Bank** (template above)
- **Graph/Map Sentence Starters** (“The data show... rises from... to...; the highest value is... located...”)
- **Fieldwork Planning Sheet** (question, sites, risks, methods, presentation, analysis prompts)
- **Command Word Cheatsheet** (Describe/Explain/Discuss/Evaluate exemplars)

Geography: Student Guide

1) Course Map & Exam Overview

- **Core themes:** Population & settlement • The natural environment (rivers, coasts, weather, climate, ecosystems) • Economic development (industry, energy, tourism) • Global interdependence • Resource management.
- **Assessment objectives (AOs):**
 - **AO1 Knowledge** (places, processes, case studies)
 - **AO2 Understanding & application** (explain, link, evaluate)
 - **AO3 Skills** (maps, photographs, data, fieldwork)
- **Typical papers (check your syllabus code):**
 - **Paper 1** (Geographical Themes): structured questions on Core + case studies.
 - **Paper 2** (Geographical Skills): map/graph/photo interpretation and short explanations.
 - **Paper 4** (Alternative to Coursework) or **Coursework (Paper 3):** fieldwork methods, data presentation/analysis, conclusions, evaluation.

Confirm with your teacher which options and papers your centre uses.

2) Command Words — What They Really Mean

- **Describe:** say what you see; use data, direction, pattern, and place names.
- **Explain / Why:** causes → processes → outcomes; use connectives (*therefore, so, leads to*).
- **Compare:** similarities/differences in paired sentences (not two mini-essays).
- **Discuss / To what extent / Evaluate:** both sides + criteria (cost, effectiveness, time, environment, equity) + judgement.
- **Identify / State:** brief, factual points only.

3) Case-Study Mastery (what markers look for)

- **One page per case study** with: Location map • 5–7 precise stats/dates • Causes (physical/human) • Impacts (S/E/E) • Responses (short/long term) + **evaluation** (what worked/what didn't).
- Name **a real place** and quote **numbers**. Accuracy beats volume.

Quick menu examples

- **Tectonics:** Japan 2011; Montserrat 1995–97
- **Weather:** Typhoon Haiyan; Cyclone Idai
- **Coasts/Rivers:** Holderness coast; Ganges–Brahmaputra floods
- **Population/Urban:** China OCP; Lagos megacity
- **Industry/Globalisation:** Bangladesh garment industry
- **Energy/Water:** Kenya geothermal; Three Gorges Dam; Cape Town “Day Zero”

4) High-Scoring Writing Frames

- **PEEL** for 4–6 mark explainers: **Point** → **Evidence** (place/number) → **Explain** (process) → **Link** back.
- **CAUSE–EVIDENCE–SO WHAT?** for “Why...?” questions.
- **Evaluate/To what extent:** present both sides with criteria, then **rank** the most significant factor or best strategy.

Intro (2–3 lines): define scope, name place, state line of argument.

Conclusion (3–4 lines): ranked judgement + “depends on...” condition.

5) Paper 2 (Skills) — Fast Wins

- **Topographic maps:** 4/6-figure grid references, contours, relief, bearings, distance (scale), direction.
- **Graphs & data:** always **quote figures**; describe **trend** + **anomaly**; use % change where relevant.
- **Photographs:** describe **foreground/mid/background**, processes (erosion, deposition, urban land use), and evidence.
- **Diagrams:** sketch labelled cross-sections (beach profile/river long profile) with one explanatory sentence.

6) Fieldwork Method (Coursework or Paper 4)

Investigation cycle you must write in this order:

1. **Question/Hypotheses** (testable, directional)
2. **Methods** (sites, sampling, equipment, timing, risk/ethics)
3. **Data presentation** (located bars, scatter, cross-sections, proportional symbols, choropleths)
4. **Analysis** (patterns, relationships, anomalies, link to theory)
5. **Conclusion** (answer hypotheses with evidence)
6. **Evaluation** (limitations + realistic improvements)

Go-to techniques

- **Rivers:** width, depth, velocity, bedload size/roundness.
- **Coasts:** beach profiles, sediment size, wave counts, groyne measurements.
- **Urban:** land-use transects, EQI, traffic/pedestrian counts, bipolar surveys.

Evaluation stems

- “Sample size (n=__) limits reliability because __.”
- “Systematic sampling every __ m introduced bias; stratified by zone would __.”
- “Measurement error $\pm$ __; repeats/averages would __.”

7) Weekly Study Rhythm (sample)

- **Mon:** 20 min facts drill + 10 min map skill.
- **Tue:** Read 6–8 pages; add 3 stats to a case-study sheet.
- **Wed:** One Paper 2 task (map/graph/photo) — 30 min.
- **Thu:** 6-mark explainer using **PEEL** — 25 min.
- **Fri:** Update glossary (5 key terms) + 10-min recap quiz.

- **Weekend:** Fieldwork mini-task or documentary + 150-word reflection.

8) Common Pitfalls & Fixes

- **Vague place:** always **name the place** and give **numbers**.
- **Process missing:** add a causal link (*because ... therefore ...*).
- **One-sided evaluation:** present **both sides** then rank.
- **Weak skills answers:** forget to **quote the figure** or give **direction/distance** — fix by underlining your numbers.

9) Revision Sprint (4 weeks pre-exam)

- **Week 1:** Build/refresh 6 case-study one-pagers.
- **Week 2:** Paper 2 drills (maps/graphs/photos) — 30 min daily.
- **Week 3:** Theme essays (2 × “explain/discuss” per theme) under timed conditions.
- **Week 4:** Full mock of Paper 1 + Paper 2; mark with checklist; repair weak spots.

10) Quick Templates (copy/paste)

Describe a trend:

“The data **increase** from ___ to ___ between ___ and ___, peaking at ___. An **anomaly** occurs at ___.”

Explain a process:

“[Factor] causes [process] because ___; **therefore** ___, which **leads to** ___.”

Evaluate a strategy:

“[Strategy] is **effective** because ___; however, **cost/time/environment/equity** concerns reduce impact in ___. **Overall**, [ranked judgement] because ___.”

Fieldwork conclusion:

“The results **support/refute** hypothesis ___ because ___ (cite figures); exceptions at site ___ due to ___.”

11) Toolkit & Materials

- Folder sections: **Themes, Case Studies, Skills (P2), Fieldwork, Plans**.
- Flashcards (terms, processes, stats).
- Free tools: **Google Earth, Our World in Data**, any simple mind-map app.
- Past papers + mark schemes for deliberate practice.

12) Wellbeing & Exam Day

- Sleep, hydrate, light review only on the last day.
- In exams: **Plan first** (3–5 minutes); underline **place names and numbers**; leave space to return to tricky parts; finish with a 3-line judgement where needed.

Global Perspectives

Basic Supercurriculum

Why Supercurriculum Matters in Global Perspectives

Global Perspectives trains students to **see issues from multiple viewpoints**, understand **local and global connections**, and develop **critical, evidence-based thinking**.

Supercurricular study helps you move beyond classroom examples — to explore real current events, evaluate sources, and become an informed global citizen.

1. Reading – Breadth and Comprehension

- Read international news from **BBC World**, **Reuters**, or **Al Jazeera English**.
- Explore youth-friendly sources like **UNICEF Voices of Youth** and **The Conversation**.
- Read one short global issues book each term, e.g.
 - *Factfulness* by Hans Rosling (data literacy)
 - *Half the Sky* by Nicholas Kristof (gender & development)
 - *No One Is Too Small to Make a Difference* by Greta Thunberg (climate action).
- Keep a **Global Issues Journal** — summarise each article: *issue, stakeholders, viewpoints, evidence, solution ideas*.

2. Writing – Argument and Precision

- Practise **writing balanced paragraphs**:
 - Start with *one clear claim*.
 - Give *two pieces of evidence* from reliable sources.
 - Add *a counter-view*.
 - End with a *mini-judgement*.
- Write short reflections (150–200 words) on questions such as:
 - “Is technology widening or reducing inequality?”
 - “Should education be a human right for all ages?”
- Review your writing for **bias, clarity, and evidence quality** (fact vs. opinion).

3. Listening – Interpretation and Empathy

- Listen to **TED Talks** or **BBC World Service – The Inquiry**.
- After each, note:
 - The main question.
 - Two perspectives.
 - Evidence given for each.
 - Your own evaluation: *Who was most convincing and why?*

4. Watching – Visual and Applied Understanding

- Watch documentaries such as:

- *Before the Flood* (environment)
 - *He Named Me Malala* (education & gender)
 - *The True Cost* (fast fashion & ethics)
- Record key facts and human stories — link them to your themes (e.g., “Education for all,” “Sustainability and the environment,” “Poverty and inequality”).

5. Researching – Independent Inquiry

- Choose one global issue (e.g., *water scarcity*, *online privacy*, *child labour*).
- Collect **two contrasting perspectives** from different countries or organisations.
- Make a short fact sheet with:
 - What the issue is.
 - Who is affected.
 - Different viewpoints.
 - Possible solutions and their pros/cons.
- Use **reliable**, **current** sources — cite the title, author, date, and website.

6. Student-Led Tasks – Collaboration and Leadership

- In pairs or groups:
 - Plan a **mini-project** on a local version of a global issue (e.g., recycling, inclusion, wellbeing).
 - Divide tasks (research, interviews, visuals, report).
 - Present findings with slides, posters, or short video.
- Reflect as a team: *What went well? What was challenging? What changed your perspective?*

7. Creative Tasks – Innovation and Expression

- Design a **poster campaign** or social-media infographic about a global challenge.
- Create a **1-minute awareness video** (with subtitles) explaining why the issue matters.
- Write a **speech for youth parliament** presenting your solution to a current problem.

8. Reflection – Metacognition and Growth

After each unit or project, answer:

1. What did I learn about this issue?
2. How did my opinion change?
3. Which sources were reliable or biased, and why?
4. What action could I take next?

Keep these reflections in a **digital portfolio** or notebook — they help prepare for your **Individual Report** and **Reflective Paper**.

9. Interdisciplinary Connections – Synthesis

- **Science:** climate change, pandemics, technology impacts.
- **Economics:** inequality, trade, globalisation.
- **Geography:** migration, resources, sustainability.
- **Ethics/Philosophy:** human rights, justice, fairness.
- **ICT/Media:** digital citizenship and misinformation.

10. Digital & Career Literacy – Global Citizenship Skills

- Practise evaluating websites (author, date, domain, bias, evidence).
- Learn basic **referencing** (author, year, title, link).
- Build collaboration skills through cloud tools (Google Docs, Teams, Canva).
- Explore careers where global thinking matters: journalism, policy, education, NGOs, diplomacy, international business.

Global Perspectives

Extended Supercurriculum

1) Deep Reading & Perspective-Building

- **Issue menu (rotate each term):** Climate justice • Digital privacy & AI • Migration • Global health • Fast fashion & ethics • Water security.
- **Balanced reading rule:** for any issue, read **3 sources**: one intergovernmental (UN/WHO/OECD), one NGO or think tank, and one quality newsroom (BBC/Reuters).
- **Historiography touch:** add one **opinion/editorial** and one **data-driven** piece; note how conclusions differ by method.

1-page “Issue Dossier” template

- Problem in ≤40 words • Stakeholders (5) with interests • 5 key facts (with dates) • Two competing solutions + trade-offs • Open questions.

2) Advanced Writing: Arguments that Score

- **Paragraph engine (BACE):**
Balanced claim → Authoritative evidence (cite) → Counter-view (steelman it) → Evaluative judgement (which is stronger and why).
- **Short forms to practise:** 150-word policy brief; 90-second speaking script; 6-slide explainer deck (Issue • Why it matters • Two views • Evidence • Options • Your conclusion).
- **Common fixes:** avoid vague verbs (“affects”, “impacts”); replace with precise mechanisms (“raises prices by...”, “limits access to...”).

3) Source Quality & Bias — Triangulate

- **FACT test (fast):** Factual? Authorial intent? Credibility (domain/org)? Timeliness (date)?
- **Deeper check (OPAL):** Origin (who/where), Purpose (why), Audience (for whom), Limitations (what’s missing).
- **Triangulation rule:** never rely on a **single** statistic; confirm with **two independent** sources.
- **Misinformation drill:** rewrite a sensational claim neutrally; find the original study or dataset.

4) Research Design (Individual & Team)

Question types

- **Causal:** “To what extent does X lead to Y in Z context?”
- **Comparative:** “How do perspectives differ between A and B?”
- **Evaluative:** “Which solution is more effective, and by what criteria?”

Hypothesis & criteria

- State a tentative answer and **evaluation criteria** (e.g., cost, feasibility, equity, environment, timescale).

Methods menu

- **Desk research:** reports, datasets, policy papers.

- **Field methods (ethical):** online surveys (parental consent if underage), semi-structured interviews, focus groups, observation logs.
- **Data visuals:** bar/line charts, stacked bars for pros/cons, decision matrix.

Ethics quick code

- Informed consent • anonymity • right to withdraw • store data securely • avoid harm or pressure.

5) Data Literacy & Visuals

- Always record: **source, date, unit, sample size**.
- Show % **change** and **confidence limits** if available; avoid implying causation from correlation.
- **Best-fit match:**
 - Comparisons → bar chart
 - Over time → line chart
 - Trade-offs → 2×2 matrix
 - Option choice → weighted decision table (score 1–5 per criterion; show totals).
- **Caption rule:** every figure has a one-sentence takeaway in plain English.

6) The Three Assessed Components (exam-aligned stretch)

A) Individual Report (1,500–2,000 words)

- **Plan:** question • perspectives (at least two countries/regions) • methods • criteria for judgement.
- **Body:**
 1. Background & why it matters (150–200)
 2. Perspective 1 with evidence + mini-judgement
 3. Perspective 2 with evidence + mini-judgement
 4. Optional third perspective (policy/industry/citizen)
 5. Comparative evaluation using your **criteria**
 6. Conclusion: justified overall judgement + limits + next steps
- **Appendix:** data tables, consent note, instrument (survey/interview guide).
- **Mark gainers:** clear comparison between perspectives; explicit evaluation criteria; credible citations.

B) Team Project (Presentation + GP Skills)

- **Roles:** Project lead • Research lead • Evidence checker • Product designer • Presenter(s).
- **Product choices:** campaign toolkit, school policy proposal, awareness video, infographic series, mini-website.
- **Collaboration log:** who did what, when; decisions and why; problems + solutions.
- **Impact check:** define audience, success metric (views, signatures, policy meeting secured).

C) Reflective Paper

- **Prompt bank:** “How did evidence challenge my initial bias?” “Which perspective was hardest to understand and why?”
- **Structure (600–800 words):** Situation → What I thought → What I found → How I changed → Next actions.

7) Teamwork & Project Management

- **One-page charter:** goal • roles • milestones (dates) • communication rules • conflict pathway.
- **Weekly stand-up (10 min):** progress, blockers, next task.
- **Risk register:** top 5 risks with mitigations (e.g., low survey returns → extend window, use stratified outreach).
- **Version control:** date file names; keep a references spreadsheet.

8) Speaking & Presenting (persuade, don’t overwhelm)

- **Story arc:** Problem → Stakes → Perspectives → Evidence → Options → Your recommendation → Call to action.
- **Slide craft:** ≤6 lines per slide • large numbers • one visual per point • source in small text.
- **Delivery drill:** 90-second elevator pitch; record once; improve pause, pace, emphasis.

9) Rubrics (student-friendly, tick before submitting)

Argument Quality

- Clear, balanced claim []
- Two credible sources cited []
- Counter-view steel-manned []
- Justified judgement using criteria []

Source Use

- Triangulated key stats []
- Dates & authors noted []
- Bias/limitations discussed []

Team Evidence

- Roles & contributions explicit []
- Decisions justified with evidence []
- Product tailored to audience []

Reflection

- Specific change in understanding []
- Links to evidence, not feelings only []
- Next steps realistic []

10) Eight-Week High-Yield Plan

- **W1:** Pick issue; build 1-page dossier; agree criteria.
- **W2:** Source hunt (6 items); FACT/OPAL evaluation; trim to best 3.

- **W3:** Draft argument map; run BACE paragraphs; gather local data (survey/interview) ethically.
- **W4:** Data visuals; decision matrix; mini-judgement.
- **W5:** Individual Report draft; peer review with rubric.
- **W6:** Team product build; audience test; revise.
- **W7:** Presentation rehearsal; reflective notes snapshot.
- **W8:** Final polish; references check; reflective paper write-up.

11) Ready-to-Use Templates (copy/paste)

Decision Matrix (example)

Option	Cost	Feasibility	Equity	Environment	Time	Total
Plastic tax	3	4	4	5	3	19
Bottle deposit	4	5	4	4	4	21
Education only	5	5	3	3	5	21

(Score 1–5; justify each score in 1 line)

Interview Guide (semi-structured)

- “From your experience, what is the most significant effect of ___ on ___?”
- “Which solution seems workable locally? Why?”
- “What trade-offs would you accept?”
- Close: “Anything we haven’t asked that matters?”

Reference Note (minimal Harvard)

- Author/Org (Year). Title. *Site/Publisher*. URL (accessed DD Mon YYYY).

12) Wellbeing & Professionalism

- Timebox socials; set **hard stops** for drafting.
- Be courteous to participants; keep data private; thank contributors.
- Celebrate small wins; reflect on one growth point per week.

Want this adapted to your **chosen issue set** (e.g., *AI & privacy*, *fast fashion*, *migration*) with **issue dossiers**, **decision matrices**, **slide skeletons**, and a **model Reflective Paper**?

Είπε:

guide

To ChatGPT είπε:

Here’s a practical, printable **Student Guide for Global Perspectives (o457)**. UK spellings used.

Global Perspectives: Student Guide

1) Course Map & Assessment Overview

- **What you learn:** how to analyse global issues from **personal**, **local**, **national**, and **global** perspectives; evaluate evidence; build balanced arguments; collaborate; reflect.
- **Assessed components (check your centre’s route/timings):**
 - **Individual Report (IR)** — 1,500–2,000 words on a chosen global issue with contrasting perspectives and a justified judgement.

- **Team Project (TP)** — collaborative research + a practical product/presentation addressing a local need linked to a global issue; individual and team marks.
- **Written Examination** — source-based analysis of perspectives, evidence quality, reasoning, and conclusions.

2) Command Words — What the Examiner Wants

- **Identify/State:** brief, factual points.
- **Describe/Summarise:** what the source says (no opinion).
- **Analyse/Explain:** how/why; give reasons, mechanisms, consequences.
- **Evaluate/Assess reliability:** weigh evidence (author, date, method, bias, limitations).
- **Compare/Contrast:** direct links between viewpoints (not two mini-essays).
- **Justify/Judge:** reasoned conclusion using stated **criteria**.

3) Perspective Thinking (the 4 lenses)

For any issue, collect notes under:

- **Personal:** values, experiences, risks to individuals.
- **Local/National:** community or country policies, culture, capacity.
- **Global:** international organisations, treaties, cross-border effects.
- **Future/Intergenerational:** long-term trade-offs, sustainability.

Tip: Build a 4-box matrix; place **evidence** (not opinions) in each.

4) Source Quality & Referencing (fast but rigorous)

- **FACT test:** Factual? Author/organisation credible? Citations/data shown? Timely and relevant?
- **OPAL deeper dive:** Origin • Purpose • Audience • Limitations.
- **Triangulate:** confirm key stats with **2 independent** sources.
- **Reference mini-Harvard:** Author/Org (Year). Title. *Site/Publisher*. URL (accessed DD Mon YYYY).

5) Writing Frames that Score

BACE paragraph (use everywhere):

Balanced claim → Authoritative evidence (cite) → Counter-view (steelman it) → Evaluative judgement.

Short forms to practise

- 120–150 word **policy note** (Problem → Two options → Your recommendation + criterion).
- 6-slide **explainer** (Issue • Why it matters • Perspectives • Evidence • Options • Conclusion).

6) Individual Report (IR) — Structure & Checklist

Structure

1. **Question & scope** (≤120 words)
2. **Background & why it matters** (context + key terms)
3. **Perspective A** (country/group) — evidence + mini-judgement
4. **Perspective B** — evidence + mini-judgement
5. **Comparative evaluation** using explicit **criteria** (e.g., effectiveness, cost, equity, environment, timescale)
6. **Conclusion** — justified overall judgement; limits; next steps
7. **References** (+ appendices if needed)

Marks come from: clarity of question; quality/variety of sources; comparison of perspectives; explicit criteria; justified judgement; correct referencing.

7) Team Project (TP) — Plan, Product, Proof

Team charter (one page): goal • roles • timeline • communication rules • risk plan.

Product ideas: campaign toolkit, policy proposal, awareness video, infographic pack, micro-site, school action plan.

Evidence you must keep: minutes, drafts, role logs, feedback from users/audience.

Impact check: define target audience + success metric (e.g., 40 survey responses, 100 pledges, meeting with council).

8) Exam Technique (Written Paper)

- **Read the claim carefully;** underline the **question stem** (cause? reliability? comparison?).
- **Quote the source** briefly, then analyse (don't paraphrase whole chunks).
- Use **OPAL** to evaluate reliability (who/why/for whom/limits).
- **Compare** perspectives explicitly: "Source B challenges A because..."
- Finish with a **reasoned judgement** linked to criteria or evidence strength.

9) Weekly Study Rhythm (sample)

- **Mon:** Read one high-quality article; complete a 4-lens matrix (10–15 mins).
- **Tue:** Write one **BACE** paragraph (120–150 words).
- **Wed:** Source evaluation drill (FACT + one OPAL paragraph).
- **Thu:** Team/IR task (notes, interview questions, or data table).
- **Fri:** Update references list; 5-minute reflection (How did my view change?).
- **Weekend:** One documentary/TED talk → 150-word reflection + one data visual.

10) Common Pitfalls & Fixes

- **Opinion without evidence** → add two cited facts or drop the claim.
- **One-sided argument** → add a counter-view and explain why it's less convincing.
- **Weak sources** → switch to intergovernmental/academic/credible news; show date & method.
- **No criteria in judgements** → pick 2–3 criteria and rank options against them.
- **Messy referencing** → update your reference list **as you go**.

11) Quick Templates (copy/paste)

Decision Matrix (score 1–5; justify each score):

Option	Cost	Feasibility	Equity	Environment	Time	Total
--------	------	-------------	--------	-------------	------	-------

A: ____	–	–	–	–	–	–
---------	---	---	---	---	---	---

B: ____	–	–	–	–	–	–
---------	---	---	---	---	---	---

C: ____	–	–	–	–	–	–
---------	---	---	---	---	---	---

Interview guide (semi-structured):

- “From your experience, what’s the most significant effect of ____ on ____?”
- “Which solution seems workable locally? Why?”
- “What trade-offs would you accept?”
- “Anything important we haven’t asked?”

Reflective sentence starters:

- “Initially I thought..., but evidence from [source] showed... Therefore I now...”
- “The perspective I found hardest to understand was..., because... I addressed this by...”

12) Mini-Rubrics (tick before submitting)

Argument Quality

- Balanced claim [] • Two credible sources [] • Real counter-view [] • Justified judgement with criteria []

Source Use

- Dates/authors shown [] • Triangulated key stats [] • Bias/limits explained []

Teamwork Evidence (TP)

- Roles clear [] • Decisions documented [] • Product tailored to audience & shows impact []

Reflection

- Specific change in understanding [] • Links to evidence [] • Sensible next action []

13) Wellbeing & Professionalism

- Timebox research; take screen breaks; keep data secure and anonymised.

- Be courteous to participants; log consent; store files in shared folders with dates.
- Aim for **progress, not perfection**—iterate based on feedback.

Want this adapted to your **chosen issue set** (e.g., *AI & privacy, fast fashion, migration, water security*) with **ready-made dossiers, decision matrices**, and a **model IR outline**?

Είπες:

YEAR 11 Business basic

To ChatGPT είπες:

Awesome — here's the **Basic Supercurriculum Guide for Business Studies** (Cambridge International, 0450). It matches the style of your other basics and is ready to drop into your packs.

Business Studies

Basic Supercurriculum

Why Supercurriculum Matters in Business

Going beyond the syllabus builds **commercial awareness, numeracy with context, and decision-making**. Students learn to connect **people, operations, finance, marketing, and external influences** to real firms and economies.

1) Reading — Breadth & Comprehension

- Regularly read concise business news (executive summaries) on topics like inflation, exchange rates, supply chains, and tech disruption.
- Skim company press releases/annual report highlights (mission, markets, risks).
- Keep a **glossary** of core terms (stakeholder, break-even, USP, cash flow, economies of scale, PED/YED).

2) Writing — Argument & Precision

- Weekly 200–300 word response to a prompt (e.g., “Should Firm X outsource?”).
- Use a **decision paragraph**: *Option → Benefits → Drawbacks → Judgement with condition*.
- Practise short calculations with one-sentence business interpretation (not just numbers).

3) Listening — Interpretation & Empathy

- Short founder/manager interviews or podcasts: note the **problem, metric, decision**.
- Create a 5-bullet “board note” after each: what changed, risk, next step.

4) Watching — Visual & Applied Understanding

- Case videos on marketing mixes, lean operations, or HR.
- After each, sketch a **mini value chain** (inputs → process → outputs → feedback) and mark where profit/leakage occurs.

5) Researching — Independent Inquiry

- Pick a **local SME** (café, gym, online store). Build a one-page profile:
 - Customer segment, value proposition, key resources, main costs, main risks.
- Track one KPI for a month (footfall estimate, price changes, social engagement) and link to an external factor (seasonality, competition).

6) Student-led Tasks — Collaboration & Leadership

- Team debate: *“Price promotion beats product improvement for short-term growth.”*
- Run a **mini market-research survey** (≤8 questions). Present findings with 2 charts + recommendation.

7) Creative Tasks — Innovation & Expression

- Design a **lean product canvas** for a school micro-enterprise (problem, solution, channels, costs, revenue).
- Create a simple **advert concept** (headline + 3 features + CTA) and explain how it targets a segment.

8) Reflection — Metacognition & Growth

After each topic, answer:

- *Which assumption did I challenge?*
- *Which number mattered most and why?*
- *What would change my decision (a trigger/threshold)?*

9) Interdisciplinary Connections — Synthesis

- **Maths:** percentages, indexes, break-even, elasticity basics.
- **Geography/Econ:** exchange rates, trade barriers, development stages.
- **ICT:** dashboards, simple spreadsheets, presentation design.

10) Digital & Career Literacy — Modern Relevance

- Build basic **spreadsheet fluency** (formulas: +, −, ×, ÷, %, SUM).
- Understand entry-level roles: sales, ops assistant, accounts clerk, social media coordinator.
- Write a 100-word role profile linking Business skills (analysis, communication, ethics) to one job.

Quick Practice Menu (rotate weekly)

- **Finance:** contribution, break-even, margin %, cash-flow effect of a scenario.
- **Marketing:** 4P mini-mix for a chosen segment; one change and expected effect.
- **Operations:** identify a bottleneck and a lean improvement.
- **HR:** job ad + 3 interview questions aligned to a job spec.
- **External factors:** exchange-rate shift or tax change → who wins/loses and why.

Want the **Extended Supercurriculum** (with full case templates, mini-P&L/cash-flow drills, elasticity intuition, and exam-style decision writing), or a **Student Guide** aligned to Paper 1/2 techniques and calculation checklists?

Είπες:

yes

To ChatGPT είπες:

Perfect — here's the **Extended Supercurriculum** for **Business Studies (0450)**. It builds on your *Basic* version with deeper case analysis, structured exam-writing practice, and applied commercial insight.

Business Studies

Extended Supercurriculum

1) Deep Reading & Business Awareness

- **Weekly business digest** (choose one):
 - *BBC Business Daily, The Economist Espresso, Financial Times for Schools, Reuters Business.*
- Build a “**5-Firm Tracker**”: pick five real companies (local + international). For each, update a 1-line headline per week under:
Finance, Marketing, HR, External (economic/political), Ethical/environmental.
- Each term, write a **mini case brief (300 words)**: business name, challenge, decisions, results, your analysis.

2) Writing — Decision & Evaluation Mastery

IGCSE evaluation paragraph model (L3–L4 style):

- State the option and key benefit.
2. Add numerical or contextual evidence (link to the case).
 3. Include one drawback or risk.
 4. Make a conditional judgement: “This is effective **if...**”
 5. Add a final line comparing to an alternative.

Example:

Expanding online could increase sales by reaching new markets and reducing rent costs. However, initial setup and delivery costs are high. It is a good strategy if cash reserves are strong and demand forecasts hold; otherwise, outsourcing may be safer short-term.

Writing drill:

- Pick one syllabus area per week (Marketing, Finance, Ops, HR, External).
- Write a **100-word explain**, a **200-word analyse**, and a **250-word evaluate** version of the same issue.

3) Numerical & Analytical Fluency**You must be fluent with:**

- Revenue, cost, profit, contribution, break-even formula.
- Average costs and productivity ratios.
- Percentage changes, gross/net margin.
- Cash-flow effects of events.
- Interest and exchange-rate impacts.

Applied numeracy routine:

- Recreate 1 real company metric (e.g., Tesla margin, Starbucks expansion rate) in a small spreadsheet.
- Change one variable (price, units, cost) and note how results move — this builds elasticity intuition.

4) Case Study Deep Dive Tasks**Mini case types (rotate monthly):**

- 1. **Startup challenge** – e.g., limited capital, break-even uncertainty.
- 2. **Growth dilemma** – e.g., outsourcing vs. in-house, expansion risk.
- 3. **Financial strain** – e.g., cash-flow deficit, credit terms.
- 4. **HR issue** – e.g., motivation vs. pay, training ROI.
- 5. **External shock** – e.g., exchange-rate rise, new law, recession.

For each:

- Identify stakeholders (owners, staff, suppliers, customers, government).
- Write a **T-chart** of advantages/disadvantages of each option.
- Rank decisions (1 = best).
- End with a 2-line judgement: “Best option = __ because __, assuming __.”

5) Listening & Watching with Analysis

- **BBC “The Boss”** interviews: identify decisions, KPIs, leadership style.
- *Planet Money* (NPR) or *Business Daily* (BBC World Service): 15-min topics on trade, pricing, innovation.
 - Note 3 numbers, 2 causes, 1 consequence per episode.
- **Documentaries:**
 - *Inside the Factory* (operations)
 - *Dirty Money* (ethics)

- *Made on Earth* (globalisation)

6) Research & Enterprise

- **Field mini-project:**
 - Survey 15–20 people on brand choice or pricing.
 - Present data in 2 charts (bar + pie).
 - Write 1 paragraph linking results to *price elasticity, income levels, or branding*.
- **Micro-enterprise simulation:**
 - Create a £50 startup plan (imaginary or real).
 - Record startup costs, unit cost, price, expected break-even.
 - Evaluate feasibility (payback period, cash gap, risk).

7) Creative & Leadership Tasks

- Design a **marketing mix (4Ps)** for a new school product or café.
- Role-play a **board meeting** deciding between two strategies — assign CEO, Finance, Marketing, HR.
- Create a **poster** comparing **motivational theories** (Maslow, Herzberg, Taylor, Mayo) with examples.

8) Reflection & Growth

After each topic, reflect:

- What decision had no perfect solution?
- What new trade-off did I notice?
- How confident am I explaining cause + effect numerically (score 1–5)?
- What would I do differently as a manager?

Keep a short “**Business Reflection Log**” with these answers monthly.

9) Interdisciplinary Connections

- **Economics:** supply, demand, elasticity, market failure, inflation.
- **Mathematics:** ratios, percentages, interpreting graphs.
- **ICT:** spreadsheet modelling, digital marketing tools.
- **Geography:** trade routes, location analysis.
- **Ethics:** CSR, sustainability, human rights in production.

10) Digital & Career Literacy

- Practise simple financial models in **Excel/Google Sheets**.
- Use **Canva or PowerPoint** for concise pitch decks.
- Learn LinkedIn basics: company profiles, job skills, industry trends.
- Explore entry careers: marketing assistant, operations coordinator, finance trainee, HR intern.
 - For each, note 2 skills + 1 business concept you’d use daily.

11) Eight-Week Extension Plan

W1-2: Business environment & ownership types → research two firms with different structures.

W3: Marketing → design an ad + small survey on audience perception.

W4: Finance → full break-even & cash-flow sheet.

W5: Operations → production process sketch + lean idea.

W6: HR → motivation theory poster + pay structure comparison.

W7: External factors → mini PEST analysis on one country.

W8: Integrated case → mock Paper 2 (one long decision essay, 40 min).

12) Exam Craft & Writing Style (P1 & P2)

Command word	What to do	Typical marks	Example
Identify / State	short fact/term	1-2	Define “added value”
Explain	cause-effect + context	3-4	Explain why profit might fall
Analyse	connect causes and effects	5-6	Analyse impact of wage rise on costs
Evaluate / Recommend	weigh both sides + conclude	7-12	Recommend option after comparing pros & cons

Evaluation sentence stems:

- “The most effective option is... because... however, this depends on...”
- “In the short term..., but in the long term...”
- “If the economy/interest rates change..., this decision would need review.”

13) Quick Rubrics (Self/Peer)

Short Questions

- Uses context []
- Includes key term []
- Clear cause-effect chain []

Case Study (Evaluate)

- Both sides discussed []
- Contextual evidence used []
- Conditional conclusion []
- Structure (intro → analysis → judgement) []

14) Ready-to-Print Templates

PEEL paragraph

P: Firm X should ____.

E: Evidence (data, fact).

E: Explain cause → effect → stakeholder.

L: Link to question.

Business Summary Table

Function	Key Concept	Example	Real Company
Marketing	Market segmentation	Demographics	Coca-Cola
Finance	Cash-flow forecast	Timing inflow/outflow	Tesla
HR	Motivation	Fringe benefits	Google
Ops	Lean production	Just-in-time	Toyota

Business Studies: Student Guide

1) Course Map & Assessment Overview

The Core Themes

1. **Understanding Business Activity** – enterprise, objectives, stakeholders, ownership, growth.
2. **People in Business** – motivation, training, recruitment, leadership, communication.
3. **Marketing** – customer needs, market research, segmentation, marketing mix (4Ps).
4. **Operations Management** – production methods, productivity, lean, quality, location, logistics.
5. **Financial Information and Decisions** – sources of finance, cash flow, break-even, profit, ratios.
6. **External Influences** – economic, legal, social, technological, environmental, and ethical factors.

Exam Structure

Paper	Focus	Marks	Time	Weight
Paper 1	Short-answer & structured questions	80	1 hr 30 min	50%
Paper 2	Case study with data and evaluation	80	1 hr 30 min	50%

2) Command Words — What the Examiner Wants

Command Word	Meaning	Typical Marks	Tip
Identify/State	Name a fact, term or example	1–2	Keep it short.
Describe	Say what something is or does	2–4	Use one point per mark.
Explain	Show cause and effect	4–6	Use connectives (<i>because, therefore</i>).
Analyse	Show how and why something happens	6–8	Link points in sequence (no lists).
Evaluate / Recommend	Weigh up pros & cons and make a decision	8–12	Include a final justified judgement.

Sentence frame for evaluation:

“This option is effective because... however, it depends on... Overall, it is better/worse than... because...”

3) Case-Study & Exam Writing Skills

Paper 1

- Each question is separate; plan quick, concise answers.
- Use **PEEL** structure:
 - **P**oint – state the idea
 - **E**xample – add context or figure
 - **E**xplain – cause/effect
 - **L**ink – back to the question

Example:

Higher wages can motivate staff (**P**), for example, bonuses at Toyota (**E**). Motivation increases productivity, which lowers average costs (**E**). Therefore, profits may rise (**L**).

Paper 2

- Context is key. Always refer to the **business name, data, and situation**.
- Use **balanced arguments** before concluding.
- For 12-mark evaluative questions, plan with a **T-chart** (advantages vs disadvantages).

4) Calculations You Must Know

Concept	Formula	Tip
Revenue	Price $\times$ Quantity Sold	Label units (\$/£).
Total Cost	Fixed + Variable Costs	Always show both.
Profit	Revenue – Total Cost	State if it's gross or net.
Break-even Output	Fixed Cost $\div$ (Price – Variable Cost)	Round to whole units.
Margin of Safety	Actual Output – Break-even Output	Higher = safer.
Gross Profit Margin (%)	(Gross Profit $\div$ Revenue) $\times$ 100	Use % symbol.
Net Profit Margin (%)	(Net Profit $\div$ Revenue) $\times$ 100	Compare year-on-year.
Average Cost	Total Cost $\div$ Output	Use for economies of scale.

5) Case Study Analysis (Paper 2)

- **Step 1:** Identify the issue (e.g. location, source of finance, marketing change).
- **Step 2:** Extract data – sales, costs, trends, % change.
- **Step 3:** Discuss advantages & disadvantages (use context).
- **Step 4: Evaluate** – make a reasoned judgement:
 - Short-term vs long-term impact
 - Which stakeholder benefits most
 - What it depends on (e.g. economy, demand, finance)

Example template:

Expanding production may reduce costs through economies of scale (lower average cost per unit). However, the business will face high upfront capital costs and cash-flow pressure. If demand remains strong, expansion is likely to improve profit long-term; otherwise, it could increase debt risk.

6) Case Study Toolkit

Analytical Tools (you can mention them without drawing)

- **SWOT:** strengths, weaknesses, opportunities, threats.
- **PEST:** political, economic, social, technological.
- **Break-even Chart:** show costs/revenue intersection.
- **Flow Diagram:** illustrate production or decision process.
- **Decision Matrix:** rank options (criteria × score 1–5).

7) Weekly Study Rhythm (sample)

Day	Focus	Task
Monday	Marketing	Create 4P mini-mix for a known brand.
Tuesday	Finance	3 calculation drills + interpretation.
Wednesday	Operations	Diagram a production process.
Thursday	HR	Motivation theory + real example.
Friday	External	Read one BBC business story and summarise 3 impacts.
Weekend	Exam Practice	One Paper 1 short question + one 12-mark evaluation.

8) Common Pitfalls & Fixes

Weakness	Fix
Listing facts without link	Always add “because...”
Ignoring the case context	Name the business + data
Missing numbers	Use figures wherever available
One-sided answer	Add an opposite argument
No conclusion	Always end with “Overall...”
Weak time management	Spend ≈1 mark = 1.2 minutes

9) Revision Checklist by Topic

Topic	Core Terms & Skills
Business Activity	Entrepreneur, Objectives, Stakeholders, Growth methods
People in Business	Motivation, Recruitment steps, Appraisal, Leadership styles
Marketing	Market research, Segmentation, 4Ps, Product life cycle
Operations	Productivity, Quality, Lean/JIT, Economies of scale
Finance	Cash flow, Break-even, Profitability ratios, Sources of finance
External Influences	Exchange rate, Inflation, Taxation, Legislation, Ethics, Environment

Tick off each concept once you can define, explain, and give one real-life example.

10) Exam Week Plan (4 Weeks Before)

- **Week 1:** Relearn formulas + practise 10 calculation questions.
- **Week 2:** Paper 1 timed questions (25–30 mins each).
- **Week 3:** Paper 2 case studies (at least two 12-mark evaluations).
- **Week 4:** Mixed review — rotate topics daily, practise under time, simulate full paper.

11) Quick Templates

PEEL paragraph

Point → Evidence → Explanation → Link.

Raising prices may increase profit if demand is inelastic, because total revenue rises even if units sold fall slightly. Therefore, this depends on price sensitivity and competitor action.

Evaluation (12-mark)

“Although Option A offers short-term cost savings, Option B provides better long-term stability because _____. Overall, Option B is preferred if _____.”

Break-even interpretation

“The business will break even at ____ units. Producing more increases profit; producing fewer causes losses.”

12) Self-Assessment Rubric

Criteria	Yes	Needs Work
I define and apply business terms accurately	☐	☐
I can calculate and interpret results	☐	☐
I use real-world examples or case context	☐	☐
I evaluate both sides and justify my decision	☐	☐
I manage time and structure under pressure	☐	☐

13) Tools & Resources

- **Business Studies** (Hodder or CUP Textbook)
- **BBC Bitesize Business**
- **Tutor2U IGCSE Business**
- **BusinessEd.co.uk worksheets**
- **YouTube channels:** Two Teachers • Bizconsesh • IGCSE Business World.

14) Wellbeing & Exam Day

- Sleep well, plan answers, write clearly.
- **Under time pressure:** If stuck, move on and return later.
- **Show method in calculations** – marks are awarded for working.
- **Underline final answers** in calculations and key words in essays.

Literature

Basic Supercurriculum

Why Supercurriculum Matters in Literature

Reading beyond the set texts builds **interpretation**, **close analysis**, and **critical voice**. It strengthens exam skills for **poetry**, **prose**, **drama**, including **unseen** responses.

1) Reading — Breadth & Comprehension

- Read **one extra text per term** outside the syllabus (rotate: modern novel, classic drama, poetry collection).
- Build a **Context Card** for each set text: author, date, movement, big themes, key symbols, historical backdrop ($\leq \frac{1}{2}$ page).
- Keep a **Quote Bank** (10–15 lines per text) with line refs and 1-line gloss.

2) Writing — Argument & Precision

- Use the **PEEL** paragraph: Point → Evidence (quotation) → Explain (method + effect) → Link (to Q/theme).
- Practise **topic sentences** that answer the question directly.
- End essays with a **judgement sentence** (“Overall, the text suggests... through...”).

3) Listening — Interpretation & Voice

- Listen to **author readings** or **actors’ performances** of scenes/poems.
- After each, jot 3 bullets: tone, pacing, emphasis — how these **shift meaning**.

4) Watching — Visual & Applied Understanding

- Watch credible stage/film adaptations (one scene you’ve studied).
- Note **director choices** (lighting, music, camera/stage space) and how they **reinterpret** the text.

5) Researching — Independent Inquiry

- 1 page per term: **Mini-context brief** on a movement (Romanticism, Modernism) or issue (war poetry, post-colonial voice).
- Read **one critic paragraph** (review, essay snippet); summarise the claim and write a 2–3 sentence **response/stance**.

6) Student-led Tasks — Collaboration & Leadership

- Run a **seminar circle**: each student leads 5 minutes on a passage; peers ask 2 questions.
- Create a **theme map** on a large sheet: quotations + devices clustered by theme.

7) Creative Tasks — Innovation & Expression

- Write a **pastiche**: 14 lines in the style of a studied poet, with a 120-word commentary on techniques imitated.
- **Monologue rewrite** of a scene from another character's viewpoint.

8) Reflection — Metacognition & Growth

After each unit, answer:

- What insight changed my reading most?
- Which **device**→**effect** link did I prove clearly?
- One target for next essay (e.g., “embed shorter quotes,” “analyse verbs/adjectives more precisely”).

9) Interdisciplinary Connections — Synthesis

- **History**: period context shaping themes/voices.
- **Art/Music**: imagery and motifs across media.
- **PSHE/Ethics**: moral choices, power, identity, empathy.

10) Digital & Career Literacy — Modern Relevance

- Keep tidy **citations** (author, text, act/scene or line numbers).
- Use a simple **annotation system** (colour for theme, shape for device, margin codes).
- Careers linked: law, journalism, teaching, editing, media, marketing — all value argument, audience awareness, and tone control.

High-Yield Practice Menu (rotate weekly)

- **Unseen Poetry (25 min)**: Meaning in one sentence → 3 devices → effect each → mini-judgement.
- **Passage-Based (30–40 min)**: Line-by-line comments on diction, imagery, sound; link out to whole-text theme.
- **Whole-Text Essay (40 min)**: Plan 3 claims; one counter-reading; conclusion with refined thesis.

Quick Exam Frames (copy/paste)

Unseen opener (2–3 lines):

The speaker's attitude to ____ is created through [**device**] and [**tone**], moving from ____ to ____.

Method sentence starters:

- “The **verb/adjective/noun** ‘____’ suggests...”
- “Through **enjambment/caesura/alliteration** the poet emphasises...”
- “The **stage direction/silence/aside** reveals...”

Conclusion stem:

Overall, the writer invites us to view ____ as ____, using [**method cluster**] to shape our sympathy or judgement.

Minimal Checklist (tick before you hand in)

- Quotation embedded and short []
- Device named precisely (not “shows”) []
- Effect linked to **question** and **theme** []
- Alternative reading or nuance offered []
- Conclusion returns to the exact wording of the question []

Want the **Extended Supercurriculum** next (unseen mastery drills, critic snippets, comparative essays, and model planning templates) or a **Student Guide** aligned to the exact paper route (open/closed books, set text list) you’re using?

Είπε:

extended

To ChatGPT είπε:

Excellent — here’s your **Extended Supercurriculum** for **Literature in English** (0475).

It’s designed to deepen analytical sophistication, unseen confidence, and comparative essay craft — all fully aligned with the IGCSE structure (Poetry, Prose, Drama; Passage-based and Essay questions; Unseen option).

☐ **Literature in English**

Extended Supercurriculum

1) Deep Reading & Literary Awareness

Core texts + beyond

- Read your **set texts** *twice*: once for plot/theme, once for language/form/structure.
- Read **one companion text per genre**:
 - *Poetry*: Carol Ann Duffy, Seamus Heaney, Wilfred Owen, Langston Hughes.
 - *Prose*: Chimamanda Ngozi Adichie, George Orwell, Kazuo Ishiguro.
 - *Drama*: Shakespeare (compare themes), Miller, Hansberry.
- Keep a “**Compare & Contrast Table**” for tone, conflict, and resolution across genres.

Context capsules (1 paragraph each)

- Author biography – 3 sentences max: era, worldview, voice.
- Historical / social moment – how does it shape character or theme?
- Literary movement – e.g. Modernism, Postcolonialism, Realism, Romanticism.

2) Analytical Writing Mastery

Essay skeleton for top-band answers (Band 1–2):

Introduction – question restated, thesis, mini map of 3 ideas.

2. Main paragraph x3–4 – analytical depth (quote + device + interpretation + context).

3. **Counter-reading paragraph** – acknowledge alternative view (“Some may argue... however...”).

4. **Conclusion** – return to thesis, refine not repeat.

Micro-analytical drill (10 minutes):

- Choose a **5-line extract**, underline **4 key words**, write one sentence of effect per word.
- Add one **mini-comparison** (“Similarly/In contrast to...”).

Language precision bank:

Imagery • Symbolism • Diction • Syntax • Caesura • Enjambment • Motif • Foreshadowing • Antithesis • Sibilance • Dramatic irony • Register • Free indirect discourse • Pathos.

Avoid: “shows,” “makes the reader feel.”

Use: “conveys,” “implies,” “foregrounds,” “juxtaposes,” “evokes sympathy,” “invites reflection.”

3) Unseen Poetry & Prose Mastery

Step-by-step strategy (15–20 min planning)

1. **First read:** overall meaning/emotion.
2. **Second read:** mark *what* changes (tone/voice/tense).
3. **Third read:** underline 6–8 techniques (sound, structure, diction, imagery).
4. Write **3 topic sentences**:
 - Meaning / Voice
 - Imagery & sound
 - Structure & tone shift.
5. Conclude with 2 sentences: meaning + how achieved.

Practice bank (alternate weeks):

- 1 unseen **poem** (15 lines)
- 1 unseen **prose extract** (200–250 words)
- Reflect: “What made this writer’s voice distinctive?”

Sentence frames:

“The repetition of ____ heightens the sense of ____.”

“The shift from ____ to ____ mirrors the speaker’s changing ____.”

“Through the metaphor of ____, the author invites us to consider ____.”

4) Comparative and Contextual Thinking

Comparative paragraph model:

Both writers present [**theme**], but while **Writer A** uses [**device/method**] to emphasise ____, **Writer B** suggests ____ through [**technique**].

Extension task: choose 2 texts (poem + prose) with linked theme; produce a 1-page “double grid”:

Theme	Text A Evidence	Text B Evidence	Shared message	Key difference
-------	-----------------	-----------------	----------------	----------------

5) Critical & Scholarly Engagement

How to engage with critics (simply)

- Read 1 short academic comment or review per term (e.g., SparkNotes “Critics’ View” or Guardian Review).
- Write a **3-sentence summary**:
 - What is their claim?
 - Do I agree/disagree? Why?
 - What evidence from the text supports or undermines it?

Mini exercise:

Find two critics with opposing views on one character (e.g. Lady Macbeth as “ambitious manipulator” vs “tragic partner”).

Write a 100-word synthesis paragraph reconciling both.

6) Performance & Dramatic Interpretation

- Read one act **aloud**; annotate pauses, pace, stage movement.
- Watch 2 versions of a scene (film/stage) and note how performance changes meaning.
- After each, record 3 bullets:
 - Tone shift
 - Actor’s emphasis
 - Impact on audience empathy.
- Write a short evaluation: “This portrayal highlights ____; in the text, the line could also suggest ____.”

7) Creative Extensions

- **Alternative viewpoint diary** (one entry per set text character).
- **Adaptation task**: rewrite a poem as a monologue; add 5-line commentary on what changed.
- **Poetic mirror**: respond to a studied poem in your own voice; mimic 2 techniques, invert its tone.

8) Reflection & Metacognition

- After every essay, complete a **self-assessment log**:
 - What is my strongest paragraph? Why?
 - Where did I over-explain the obvious?
 - Which quotation was most powerful?
 - What next step would move me one band higher?

Personal growth reflection:

“How has my empathy changed through reading this text?”

“What part of this work challenged my own assumptions?”

9) Interdisciplinary Links

- **History**: time period influences (e.g., gender, class, war).
- **Psychology**: motivation, guilt, identity, perception.
- **Philosophy**: morality, freedom, existentialism.
- **Art**: visual metaphors; colour, mood, texture parallels.

- **Music:** rhythm, repetition, silence = poetic equivalents.

10) Digital & Career Literacy

- Practise **annotating digitally** (Google Docs comments / OneNote highlighters).
- Use **AI tools** or online glossaries for vocabulary support (but interpret in your own words).
- Explore careers that rely on interpretive clarity: journalism, media analysis, law, editing, teaching, screenwriting.

11) 8-Week High-Yield Plan

Week	Focus	Key Task
1	Review set text themes	Write 3 thesis statements per text.
2	Unseen Poetry	2 timed analyses (25 min each).
3	Character study	Create “quotes by mood” table.
4	Context	One-page era summary per author.
5	Comparative Writing	One essay linking two texts.
6	Unseen Prose	Analyse narrative voice + imagery.
7	Practice Exam (Paper 1/2)	Full timed paper, self-mark.
8	Reflection & Revision	Rewrite weakest essay paragraph.

12) Quick Rubrics (student-friendly)

Band	Features
Band 1 (Top)	Perceptive argument, detailed textual evidence, confident stylistic analysis, nuanced tone.
Band 2-3	Clear, structured, relevant, detailed references, some insight.
Band 4-5	Some understanding, basic comment on effects, general rather than precise.

13) Self-Checklist (tick before submission)

- Embedded short quotes (under 8 words) []
- Literary term precise []
- Contextual note relevant, not bolted on []
- Alternative reading attempted []
- Conclusion refines argument []

14) Resources

- **Literature Textbook (Hodder or CUP)**
- **BBC Bitesize Literature**
- **Poetry Foundation / Poetry Archive**
- **SparkNotes / York Notes** for overview, not memorisation.
- **YouTube:** Mr Bruff, First Rate Tutors, BBC Teach.

Literature in English: Student Guide

(Syllabus 0475)

1) Course Overview & Assessment Map

Paper	Focus	Time	Marks	Weight
Paper 1	Poetry and Prose – essay and passage-based questions on set texts	1 hr 30 min	50	50%
Paper 2	Drama & Unseen – essay and passage-based questions, or one unseen poem/prose	1 hr 30 min	50	50%

Some centres sit **Paper 3 (Unseen)** separately – same skills apply.
Confirm with your teacher which *set texts* and *papers* your school uses.

2) Aims of the Course

By the end of Year 11, you should be able to:

- **Analyse** how writers use language, structure, and form.
- **Interpret** meanings, themes, and character motives.
- **Compare** texts and ideas.
- **Evaluate** how context shapes meaning.
- **Express** personal, critical responses with evidence.

3) Command Words & What They Require

Command Word	What It Means	Tip
Analyse	Break down methods and effects	Use literary terms precisely
Discuss / Explore	Explain with argument and examples	Cover both meanings and methods
How does the writer...	Focus on <i>technique</i> → <i>effect</i> → <i>meaning</i>	Always quote concisely
Compare / Contrast	Link similarities & differences in method and idea	Alternate between texts, don't write two mini-essays
In what ways...	Identify several approaches	Cover 3–4 key points
To what extent...	Evaluate with balanced argument	Finish with a justified judgement

4) Exam Paper Breakdown

□ Paper 1: Poetry & Prose

- Two questions (one on poetry, one on prose) — choose **one**.
- Each question has **two options**:
 1. **Passage-based**: analysis of an extract (language & structure).
 2. **Essay-based**: whole-text question (themes, characters, ideas).

□ Paper 2: Drama & Unseen

- One **drama question** (same structure: passage-based or essay).

- One **unseen text** (poem or prose): respond analytically with no preparation.

5) Essay & Passage-Based Response Structure

Essay Plan (whole text)

Intro (3–4 lines): Restate question + clear thesis + 3 focus areas.

2. Body Paragraphs (x3–4):

- Topic sentence → Quotation → Technique → Effect → Link.
 - Use precise literary terms (e.g. *enjambment*, *motif*, *irony*, *caesura*).
3. **Alternative View (optional paragraph):** “However, some might argue...”
4. **Conclusion (4–5 lines):** Refine thesis, summarise insight, avoid repetition.

Passage-Based Plan

Brief **context sentence** (where in text).

2. Analyse **language & imagery** (3–4 examples).
3. Comment on **structure / shifts** (tone, pace, punctuation).
4. Link extract to **whole-text themes**.
5. End with **judgement** on how the extract shapes the reader’s understanding.

6) Unseen Poetry/Prose Strategy (Paper 2 or 3)

Step-by-step (35–40 min):

1. **First read:** what is it about? (theme & tone).
2. **Second read:** mark shifts (tense, tone, imagery, stanza change).
3. **Third read:** underline 6–8 devices.
4. **Plan 3 paragraphs:**
 - Meaning & voice
 - Imagery & language
 - Structure & tone change
5. **Conclude:** message + how it’s created.

Starter sentence:

“The writer explores ____ through the use of ____, creating a tone that moves from ____ to ____.”

Common methods list:

Imagery • Symbolism • Contrast • Alliteration • Rhythm • Rhyme • Enjambment • Irony • Juxtaposition • Repetition • Caesura • Stage direction (for drama).

7) Paragraph Model — P.E.E.L.

Point – Make a clear claim linked to the question.

Evidence – Embed a short quotation (≤8 words).

Explain – Analyse language, structure, or form.

Link – Return to theme or question focus.

Example:

Shakespeare portrays Macbeth's ambition as corrupting. The verb "o'erleaps" suggests a reckless desire that ignores limits, creating tragic irony as ambition leads to downfall. Thus, ambition becomes both his strength and fatal flaw.

8) Timing Strategy

Task	Time	Tip
Read question + plan	10 mins	Identify command words; write 3 bullet points.
Write intro	5 mins	Thesis clear and concise.
Body (3-4 paras)	55 mins	15 min each; 1 quote every 3-4 lines.
Conclusion	5 mins	Refine argument.
Review & proofread	5 mins	Check spelling of character names, literary terms.

9) Key Literary Terms to Master

Language	Structure	Form
Diction, imagery, connotation, tone, symbol, motif, simile, metaphor, personification	Juxtaposition, contrast, repetition, climax, shifts, pacing, syntax	Sonnet, soliloquy, dialogue, narrative voice, stanza form, rhyme, metre

You don't need to label every term — just choose the right one for the effect you describe.

10) Common Pitfalls & Fixes

Pitfall	How to Fix
Retelling the story	Focus on <i>how</i> the writer shapes meaning.
Quoting too much	Use 2-4 word quotes, integrated smoothly.
Generic verbs ("shows", "tells")	Replace with "conveys", "implies", "reveals".
Ignoring structure	Comment on shifts in tone, stanza, punctuation.
Forgetting conclusion	Summarise the writer's purpose & reader effect.

11) Weekly Study Routine (Example)

Day	Focus	Activity
Monday	Set text re-reading	2 pages annotated, update quote bank.
Tuesday	Poetry	Analyse one poem (3 devices, 3 effects).
Wednesday	Prose	Write one PEEL paragraph from past question.
Thursday	Unseen practice	One short unseen poem (25 min).
Friday	Drama	Watch one scene, note tone & character change.
Weekend	Review	Create one page of theme connections across texts.

12) Revision Sprint Plan (Final 4 Weeks)

Week 1: Key themes & context summaries.

Week 2: 2 unseen poems, 1 prose extract (timed).

Week 3: 3 past-paper questions (one per genre).

Week 4: Mock Paper 1 + Paper 2 under timed conditions.

13) Self-Assessment Rubric

Skill	Excellent (✓)	Developing (-)
Focuses on question throughout		
Selects concise, relevant quotations		
Explains effects of language precisely		
Analyses structure & tone		
Evaluates alternative views		
Writes fluently, uses literary terms accurately		

14) Quick Templates

Essay Intro (3 lines):

In *[Text]*, *[Author]* explores *[theme]* through *[character/device]*, suggesting that *[interpretation]*. Across the text, *[summary of how]*.

Mini Conclusion (2 lines):

Overall, *[writer]* presents *[theme]* as *[idea]*, leaving the reader to question *[implication]*.

Unseen Opening (2 lines):

This poem explores *[theme]* through *[device]*, creating a *[tone]* that develops from *[A]* to *[B]*.

15) Resources

- Literature Textbook (Hodder or CUP)
- BBC Bitesize: English Literature
- Poetry Archive / Poetry Foundation
- York Notes & SparkNotes (for revision, not memorisation)
- YouTube: Mr Bruff • Stacey Reay • BBC Teach • Massolit (short lectures)

16) Exam-Day Reminders

- Write in paragraphs — not bullet points.
- Underline text titles; spell author names correctly.
- Use quotation marks for every quote.
- Leave **5 minutes to check grammar & coherence**.
- Stay calm: show *interpretation*, not memorisation.

Drama basic YEAR 11

DRAMA

Basic Supercurriculum

Why Supercurriculum Matters in Drama

Drama is both **art and communication**. Extending learning beyond lessons builds confidence, empathy, and critical evaluation — essential for devising, performing, and understanding theatre from multiple perspectives. Supercurricular exploration deepens creativity, movement control, and interpretation.

Reading – Breadth and Comprehension

- Read **three plays** outside the syllabus (contrast genres: tragedy, comedy, modern realism).
 - *Examples: A Doll's House* (Ibsen), *Blood Brothers* (Russell), *An Inspector Calls* (Priestley), *Antigone* (Sophocles).
- Research **practitioners**: Stanislavski, Brecht, Artaud — note their rehearsal techniques and performance aims.
- Keep a **Play Logbook**:
 - Title & playwright
 - Themes
 - Key scenes
 - Performance ideas / audience effect.

2. Writing – Reflection and Analysis

- After each rehearsal, write a **short log (100–150 words)**:
 - What we aimed to achieve.
 - What worked / what needs improvement.
 - How my performance choices communicated meaning.
- When analysing plays, use **PEEL**:
 - **Point** – what technique or choice?
 - **Evidence** – line, gesture, direction.
 - **Explain** – what emotion or meaning created?
 - **Link** – how it supports the theme or practitioner's style.

3. Listening – Interpretation and Empathy

- Listen to **radio plays** or **BBC Sounds Drama podcasts**.
- Note how voice, silence, and sound effects replace movement or staging.
- Reflect: *How does the actor's tone shape your understanding of character?*

4. Watching – Visual and Applied Understanding

- Watch stage and screen productions of your set play.
- Compare **two interpretations** of the same scene (e.g. *Macbeth* 1970s vs 2015 versions).

- Record in your log:
 - Acting style
 - Use of space / lighting
 - Audience impact.

5. Researching – Independent Inquiry

- Investigate one **theatre style** (Realism, Physical Theatre, Expressionism, Greek Chorus).
- Create a **one-page info sheet**:
 - Origins & context
 - Main features (movement, sound, stage, purpose)
 - One practitioner who influenced it.
- Visit a **local performance** or view a recorded play and write a 200-word review.

6. Student-led Tasks – Collaboration & Leadership

- Take turns directing a short scene (2–3 mins).
- Lead a warm-up or improvisation for your group.
- Peer-evaluate using prompt cards: voice, movement, focus, audience engagement.

7. Creative Tasks – Innovation and Expression

- Devise a **1-minute solo piece** expressing an emotion using only movement and sound.
- Create a **tableau** (frozen image) showing tension or conflict from your set play.
- Rewrite a **scene in a new style** (e.g. Brechtian version of *Romeo and Juliet*).

8. Reflection – Metacognition and Growth

After each performance or workshop, ask:

- What did I discover about character or space?
- How did the audience respond?
- What will I change next time?

Keep this in your **Drama Journal**; it prepares you for the **practical coursework log**.

9. Interdisciplinary Connections – Synthesis

- **English Literature**: script analysis, theme, subtext.
- **History**: political or social context shaping the play.
- **Music/Art**: sound, colour, rhythm, visual atmosphere.
- **Psychology**: motivation, status, body language.

□ Digital & Career Literacy – Theatre in the Modern World

- Watch National Theatre or Globe online productions.

- Explore careers in **acting, directing, stage design, lighting, sound, arts management**.
- Create a short **digital portfolio** (photos, reflections, mini clips) showing rehearsal development.

☐ **Practice Menu (rotate weekly)**

- **Voice:** pitch, pace, pause, projection drills.
- **Movement:** gesture, stance, spatial awareness.
- **Improvisation:** 2-min story with random object.
- **Text work:** choose 10 lines, perform with 3 different emotions.
- **Peer direction:** give one note to each performer (specific, kind, actionable).

☐ **Quick Evaluation Checklist**

Area	Key Question	✓
Voice	Did I vary tone, pace, and volume to suit emotion?	
Movement	Was my physicality clear and intentional?	
Focus	Did I maintain character throughout?	
Interaction	Did I listen and react to others?	
Space & Levels	Did I use the stage effectively?	
Audience Awareness	Was my performance clear, engaging, controlled?	

Drama

Extended Supercurriculum

1) Practitioner Toolkit (do-learn-apply)

Stanislavski (naturalism)

- *Do*: Given circumstances; “Magic If”; objectives/tactics; units & beats; circles of attention.
- *Apply*: Mark objectives in margins; swap at beat changes; add a private moment before entrances.

Brecht (epic theatre)

- *Do*: Direct address; placards; gestus; visible tech; songs/commentary; scene titles.
- *Apply*: Insert a freeze + caption that reframes the action; narrate a stage direction out loud to distance the audience.

Artaud (Theatre of Cruelty)

- *Do*: Sensory assault (soundscape, light surges), ritualised movement, vocal extremes.
- *Apply*: 60-second non-verbal sequence exploring a fear using breath, percussion, and proxemics.

Boal (Forum/Legislative)

- *Do*: Image theatre; spect-actor interventions; replay with different choices.
- *Apply*: Freeze a conflict image → invite audience interventions → replay with altered power dynamics.

Physical Theatre (Frantic Assembly)

- *Do*: Push-pull, counterbalance, chair duets, lifts with breath cues.
- *Apply*: Create a movement motif (8 counts) for a relationship; vary via canon, retrograde, accumulation.

2) Devising System (from blank page to product)

A. Stimulus → Brainstorm (20 mins)

- Word-cloud; 10 images; 3 sounds; 1 contrasting perspective.

B. Concept → Logline (1 sentence)

- “When X faces Y, they must Z, but **complication.**”

C. Structure (choose one)

- 3-act arc • Episodic montage (Brecht) • Non-linear fracture (memory loop) • Chorus-led sequence.

D. Creation Loops (repeat x3)

1. *Make* (5–8 mins on feet)
2. *Show* (no apologies)
3. *Respond* (WWW/EBI: what worked well / even better if)
4. *Refine* (keep, cut, heighten)

E. Score the piece

- Map *energy* (low→high), *status shifts*, *space patterns* (zones/levels), and *audience address* moments.

F. Lock the draft

- Script or scene map; timecode; cue list (LX/SFX); costume/prop plot; risk assessment.

3) Rehearsal Craft (professional habits)

Warm-up stack (12–15 mins)

- Body: joint roll-down; spine waves; shake-out.
- Voice: breath on “sss/fff”; humming; 1–3–5–8 sirens; tongue twisters (articulation).
- Ensemble: mirroring; status walk (1–10); name/gesture.

Notes that land (SBI)

- *Situation* (“In Scene 4, apology line”), *Behaviour* (“pace flattens”), *Impact* (“tension drops”). Then 1 actionable change.

Line-to-action conversion

- Underline **verbs**; pick a **tactic** per line (tempt, threaten, plead, mock, seduce, deflect). Switch tactics on a beat change.

Fixing pace & focus

- *Pace*: cue-tight runs; “no gaps” drill; cut indulgent pauses.
- *Focus*: sightlines check; eyeline rules; share stage picture (triangles/levels).

4) Voice & Movement Intensives

Voice ladder

- Breath: 4-in / 8-out → 6/12; add text on the out-breath.
- Resonance: chest → mask → head; switch per intention.
- Articulation grid: p/t/k; b/d/g; f/v; s/z; th/ð words in rhythm.

Movement clarity

- Laban efforts (punch/press/wring/dab/flick/float/glide/slash) — assign one per emotion; change effort at turning points.
- Proxemics drill: play the scene with **fixed 2 metres apart**, then **0.5 m**; choose the most truthful distance.

5) Design & Tech Literacy (actor-friendly)

Lighting (LX)

- Angles: front (flat), 45° (natural), sidelights (sculpt), back (silhouette).
- Colours: warm (amber), cool (steel blue).
- *Task*: Build a 4-state cue list: *Preset* → *Natural interior* → *Tension build (cool side)* → *Finale silhouette*.

Sound (SFX + underscoring)

- Cue sheet: cue no. • name • in • out • level • operator.
- *Task*: Create a 60-second soundscape with 3 layers (environment, rhythm, motif).

Set/Props

- Triangulate furniture; ensure clear sightlines; prop table labelled L/R.
- *Risk*: tape sharp edges; “show and stow” routine after each run.

Costume/Make-up

- Quick-change plan; neutral base with add-ons for multi-role; footwear = silent, safe, in-character.

6) Written Analysis & Logbook Excellence

Performance log (each rehearsal/performance, 120–180 words)

- *Intention*: what meaning I aimed to communicate.
- *Method*: choices (voice, movement, space, tactic).
- *Evidence*: audience/peer response or video moment.
- *Next step*: one precise change.

Text analysis paragraph (PEEL-Drama)

Point (choice) → Evidence (line/stage action) → Effect (audience/meaning) → Link (theme/practitioner).

Design justification (mini)

- “We chose **cool sidelights** to isolate characters, echoing Brecht’s distancing; the **projected headline** reframed the scene politically.”

7) Assessment Alignment (create • perform • evaluate)

- **Creating**: originality, structure, effective ensemble, practitioner influence evident.
- **Performing**: vocal clarity, physical control, character consistency, audience connection, pace/energy, proxemics.
- **Evaluating**: clarity of intention, analysis of choices, use of subject language, insight into impact.

Top-band habits

- Bold, repeatable choices; clear intention; refined detail; thoughtful reflection with evidence.

8) Rehearsal Room Etiquette

- On time, warmed, lines learned; phones away.
- “Yes, and...” in devising; critique the *choice*, not the person.
- Log changes on a whiteboard or shared doc; version-control scripts.

9) Eight-Week High-Yield Plan

Week	Focus	Output
1	Stimulus & logline	1-sentence concept + 2 practitioner ideas
2	Scenes 1–2 devised	4–6 mins rough cut; WWW/EBI notes
3	Stanislavski detail	Objectives/tactics labelled; beats marked
4	Brecht/Physical layer	Caption/gestus or movement motif added
5	Tech sketch	LX/SFX cue list v1; prop/costume plot
6	Full draft run	Video + note sweep; cut 10%
7	Audience test	Feedback form; fix pace/sightlines
8	Polish & paperwork	Final cue stack; risk check; logbook summary

10) Quick Rubrics (student-friendly)

Devised Piece (10)

- Concept & structure clear (0-2)
- Practitioner influence purposeful (0-2)
- Ensemble timing & focus (0-2)
- Vocal/physical precision (0-2)
- Audience impact & coherence (0-2)

Monologue/Scene (10)

- Objective/tactics evident (0-2)
- Voice (audibility, modulation) (0-2)
- Movement/proxemics (0-2)
- Text detail (beats, pace) (0-2)
- Commitment & truth (0-2)

Evaluation/Log (10)

- Intentions & methods clear (0-3)
- Specific evidence (moments, cues) (0-3)
- Insight & next steps (0-2)
- Subject vocabulary (0-2)

11) Ready-to-Use Templates

Cue Sheet (sample)

Cue	In/When	Action	Notes
LX1	Preset	Warm wash @60%	Audience in
SFX1	Scene 1 start	Street ambience -20 dB	Fade 8"
LX2	Line "Stop!"	Snap to cool side @70%	Isolate attackers
SFX2	End Scene 1	Cut ambience	Follow LX2

Director's Note (SBI)

- *S*: Scene 3 confession
- *B*: Pace droops after "I didn't know"
- *I*: Tension dips; audience coughs
- *Change*: Cut 2-sec pause; cross DS earlier; raise pitch on "didn't"

Risk Check (mini)

- Lifts: spotter assigned; practice with mats.
- Cables: taped; gaffer check pre-show.
- Props: non-breakable where possible; glass → plastic.

12) Stretch Tasks (for distinction-level polish)

- **Dual-style scene**: play once Stanislavski, repeat Brechtian with captions—reflect on audience distance vs empathy.
- **Status swap**: improvise scene twice with inverted status to uncover power dynamics.
- **Silent score**: perform a whole scene with no text; carry meaning via gesture and rhythm; reintroduce key lines only.

13) Reflection Prompts (to level-up evaluation)

- Which *single* change most raised clarity today?
- Where did I see the practitioner influence most clearly?
- What do I want the audience to think/feel at minute 3—did we achieve it?
How do I know?

If you tell me your **set text** (if any), **group size**, and whether you're submitting **coursework or the alternative route**, I can tailor **scene starters**, **cue stacks**, **logbook pages**, and a **15-minute warm-up** specific to your piece.

Είπες:

guide

To ChatGPT είπε:

Here's your **Drama (0411) Student Guide**, matching the style of your other guides. It's practical, printable, and exam-aligned.

□ Drama: Student Guide

(Syllabus 0411)

1) Course Overview & Assessment Map

You will **create, perform, and evaluate** drama, individually and in groups, exploring style, form, and practitioner influence.

Component	Focus	Mode	Weight
C1: Written Exam	Analysis & evaluation of performance, staging, and devised choices	1 hr 30 min (short/extended responses)	40%
C2: Practical Coursework	Devised and scripted performance + logbook/evidence	Internally assessed, externally moderated	60%

Check your centre's exact brief (group sizes, monologue/duologue options, and submission dates).

2) What Examiners Want (Core Skills)

- **Creating:** clear concept, structure, and purposeful practitioner influence.
- **Performing:** vocal clarity, physical control, focus, presence, proxemics, pace.
- **Evaluating:** specific, evidence-based analysis using drama vocabulary.

3) Command Words — Do Exactly This

Word	Do	Tip
Describe	Say what is done/seen/heard	Be precise (voice, movement, space, tech)
Explain	Give reason/cause/effect	"Because... therefore... so that..."
Analyse	Break down choices & effects	Method → effect → audience impact
Evaluate/Assess	Judge effectiveness using	Use "depends on...", compare

	criteria	alternatives
Justify	Defend your choice with reasons	Link to style/theme/intention

4) Performance Technique — Quick Wins

- **Voice:** breath support, articulate consonants, vary pitch/pace/volume; land operative words.
- **Movement:** clear posture/gesture; motivated blocking; use **levels** and **distance** to show status.
- **Focus:** fix sightlines; share stage picture; stay alive in silence.
- **Pace/Energy:** cue-tight; cut dead air; build to climaxes deliberately.
- **Proxemics:** choose distances that tell the relationship story.

5) Practitioner Anchors (use in devising & analysis)

- **Stanislavski:** objectives/tactics, given circumstances, beats, circles of attention.
- **Brecht:** direct address, captions/placards, gestus, visible tech to provoke thought.
- **Artaud:** sensory impact, ritualised movement, non-verbal sequences.
- **Physical Theatre (Frantic Assembly):** contact work, counterbalance, motifs/variation.

In your writing, name the choice **and** why that practitioner's method suits the intention.

6) Devising Workflow (exam-ready)

1. **Stimulus → Concept:** 1-line logline ("When X faces Y, they must Z, but...").
2. **Structure:** three-act / episodic / chorus-led / non-linear.
3. **Make–Show–Respond–Refine** cycles (keep WWW/EBI notes).
4. **Score the piece:** map energy, status shifts, and audience address.
5. **Lock draft:** script/scene map; LX/SFX cue list; prop/costume plot; risk check.

7) Logbook (what to write, every time, 120–180 words)

- **Intention:** what meaning I aimed to communicate.
- **Method:** voice/movement/space/tactics; practitioner reference.
- **Evidence:** specific moment (line, cue, audience reaction).
- **Next Step:** one precise change (e.g., "raise stakes before beat change on 'Go!'").

Sentence frame:

"We used [technique] to [effect] because [reason]. This aligns with [practitioner], who emphasises [principle]."

8) Written Exam Technique (C1)

Short responses (4–6 marks): one clear point → specific example → effect on audience.

Longer responses (8–15 marks):

- Paragraph model: **Point** → **Evidence (moment/cue)** → **Method** → **Effect** → **Why it works (or not)** → **Link to intention/style**.
- Compare alternatives and finish with a **reasoned judgement** (“most effective if...”).

Timing (90 min):

- Read & plan: 10 min
- Q1: 20–25 min
- Q2: 20–25 min
- Q3 (longest): 30–35 min
- Proofread: 5 min

9) Tech Literacy (actor-friendly)

- **Lighting:** front 45° (natural), side (sculpt), back (silhouette); warm vs cool palettes.
- **Sound:** environment bed + motif + spot SFX; document in a cue sheet.
- **Set/Props:** sightlines, safe levels, practicals tested; prop table L/R.
- **Costume:** character-appropriate, movement-safe; quick-change plan.

Cue Sheet (mini template):

Cue	In/When	Action	Notes
LX1	Preset	Warm wash @60%	Audience in
SFX1	Scene 1 start	Street ambience -20 dB	Fade 8"
LX2	“Stop!”	Snap to cool side @70%	Isolate attacker

10) Weekly Rehearsal Rhythm (sample)

- **Mon:** Warm-up 12 min → scene work (beats/objectives) → notes.
- **Wed:** Physical/voice drill → integrate tech beats → run scene.
- **Fri:** Cue-to-cue (tighten pacing) → record 10-min run → reflective log.

11) Common Pitfalls & Fixes

Pitfall	Fix
Rushing lines / dropped ends	Support breath; mark operative words; “full stop finish” drill
Flat staging	Add levels/diagonals; re-block for sightlines
Generalised emotion	Switch tactics at beat changes; choose strong verbs
Dead space between cues	“No-gaps” drill; pre-set crosses; sharpen pick-ups
Vague evaluation writing	Name moment + technique + audience effect + judgement

12) Quick Rubrics (self/peer)

Performance (10): Voice (2) • Movement/Proxemics (2) • Focus/Pace (2) • Character truth (2) • Audience impact (2)

Devising (10): Concept/structure (2) • Practitioner use (2) • Cohesion (2) • Originality (2) • Risk & safety (2)

Evaluation (10): Specific evidence (3) • Subject language (2) • Analysis depth (3) • Reasoned judgement (2)

13) Ready-to-Use Prompts

- **Director note (SBI):** *Scene/line → what happened → why it matters → change.*
- **Actor tactic list:** tempt, threaten, plead, mock, seduce, deflect, confront, console.
- **Evaluation starter:** “At [time/moment], we used [technique] which [effect]; this was **effective/limited** because [reason].”

14) Exam-Day Reminders

- Warm up (breath/voice/body) and walk the space early.
- Pre-set props; check cues; confirm spike marks.
- In writing: **avoid plot retell** — analyse **choices and effects**.
- Finish answers with a **clear judgement** (“most effective when... because...”).

**English First Language
BASIC SUPERCURRICULUM**

Why Supercurriculum Matters

Extending learning beyond class helps you think like a writer, not just a student. You'll read more widely, refine precision and style, and understand audience, tone, and purpose — the essence of IGCSE success.

Reading – Breadth and Comprehension

- Read across genres: journalism, essays, short stories, speeches.
- Weekly habit: one *editorial*, one *feature article*, one *fiction piece*.
- Identify **writer's viewpoint, bias, and method** (how tone and language shape attitude).
- Keep a **Language Log**:
 - 1 new word per day (definition, root, example).
 - 1 sentence that impressed you (note structure and punctuation).

2. Writing – Argument and Precision

- Practise 3 styles each fortnight:
 - **Transactional** – letters, reports, articles, speeches.
 - **Descriptive** – vivid sensory language, imagery, mood.
 - **Narrative** – character, perspective, twist.
- Keep a *100-Word Journal*: short entries using different tones (formal, persuasive, reflective).
- Read your work aloud for rhythm and flow.

3. Listening – Interpretation and Empathy

- Listen to TED-Ed or BBC Radio 4 clips; summarise in 3 bullet points: *tone, purpose, audience*.
- Recreate a speech segment in your own words using synonyms and rhetorical balance.

4. Watching – Visual and Applied Understanding

- Watch short news or film clips without subtitles.
- Identify mood shifts, body language, and persuasive devices.

5. Researching – Independent Inquiry

- Choose a **topic of personal interest** (e.g. environment, technology, equality).
- Read 3 sources, identify bias, and summarise arguments in one paragraph.
- Present findings as a short class talk (1 minute).

6. Student-Led Tasks – Collaboration & Leadership

- Peer edit a classmate's article using the **3C model**: *Content – Clarity – Connection*.
- Lead a mini “language lab” on one grammar or punctuation feature.

7. Creative Tasks – Innovation & Expression

- Write a diary from a new perspective (e.g. “The day the internet disappeared”).
- Adapt a news report into a monologue.
- Recast a short story as a letter to the editor.

8. Reflection – Metacognition

After each writing piece:

- What was my purpose and audience?
- Which sentence achieved my aim best?
- Which technique could I develop further?

9. Interdisciplinary Connections

- **History:** bias and propaganda.
- **Psychology:** tone and persuasion.
- **Art/Media:** imagery, symbolism.
- **ICT:** digital literacy and source reliability.

□ Digital & Career Literacy

- Learn email etiquette, résumé style, and concise online writing.
- Explore communication careers: journalism, marketing, editing, law, diplomacy.

📖 English First Language

EXTENDED SUPERCURRICULUM

Deep Reading & Writer Awareness

Weekly discipline: one opinion column, one short story, one non-fiction essay.

- Highlight **verbs, connectors, and modifiers** that reveal viewpoint.
- Identify **rhetorical patterns:** contrast, repetition, parallelism, tricolon, irony.
- Write a **150-word commentary:** “How the writer’s choices create tone.”

Challenge texts: Orwell, Atwood, Adichie, Sedaris, Guardian opinion pieces.

2. Advanced Writing Mastery

Goal: write with control, argument, and flair.

Use this precision checklist:

- Vary sentence length (simple → compound → complex).
- Replace “good/bad” with precise adjectives.
- Embed quotations naturally.
- Use **semi-colons** and **dash contrasts** for rhythm.

Writing cycle: Plan → Draft → Edit → Reflect → Redraft.

Each stage must have a purpose, not just correction.

3. Argumentative & Discursive Writing

Structure:

1. Introduce issue + stance clearly.
2. Develop 3 arguments (each with example/evidence).
3. Counterpoint paragraph: opposing view briefly considered.
4. Conclusion with forward-looking insight.

Sentence starters:

“It is often claimed that...”

“While some argue that..., it is important to consider...”

“Ultimately, the strength of this argument lies in...”

4. Descriptive & Narrative Enrichment

- Practise **five-sense openings** (visual, auditory, tactile, olfactory, gustatory).
- Apply **zoom-lens technique**: wide shot → mid shot → close detail.
- Use **motifs** (recurring image or sound) to unify description.
- End narratives with a **shift** (mood, revelation, circular echo).

Mini exercise: write 100 words describing the same place *in joy* vs *in fear*.

5. Editing & Technical Accuracy

Proofing checklist:

- No comma splices.
- Dialogue punctuation correct.
- Apostrophes accurate (possession vs contraction).
- One ambitious punctuation mark per paragraph (semicolon, dash, colon).
- Homophones checked (their/there/they're).

6. Listening, Watching & Speaking Practice

- Watch a TED Talk → outline speaker's structure (hook, point, evidence, call-to-action).
- Record a 90-second persuasive pitch; replay and assess diction, pacing, and variation.

7. Research & Real-World Literacy

- Compare two contrasting news sources on the same issue.
- Identify **loaded language**, **statistical framing**, and **tone shift**.
- Write a 200-word editorial summarising your stance, emulating the newspaper's voice.

8. Reflection & Self-Assessment

After each major piece:

- What techniques worked best for purpose/audience?
- How effective is my structure?
- Is my tone authentic or forced?

- What one sentence would I publish proudly?
- Keep a **Writer's Growth Log** (3 entries per term).

9. Interdisciplinary Links

- **Economics:** data interpretation and logical argument.
- **Ethics/Philosophy:** debate, rhetoric, persuasion.
- **Literature:** figurative language, motif, imagery.
- **Science:** clarity, concision, precision.

□ Digital & Career Literacy

- Create a 1-page portfolio with your best argument, description, and narrative.
- Practise concise writing for online audiences (blog or article, ≤600 words).
- Career applications: journalism, advertising, PR, copywriting, diplomacy, policy.

□ English First Language

STUDENT GUIDE

Exam Overview

Paper	Focus	Time	Marks	Weight
Paper 1	Reading (Extended or Core) – comprehension, summary, analysis	2 hours	80	50%
Paper 2	Directed Writing & Composition – transactional + creative	2 hours	80	50%

Some centres use 0990 (identical structure with minor mark scaling).

2. Key Objectives

- **R1:** Demonstrate understanding of explicit and implicit meanings.
- **R2:** Analyse how writers use language and structure.
- **W1:** Communicate clearly, effectively, appropriately.
- **W2:** Use a wide range of vocabulary and sentence structures.
- **W3:** Organise ideas coherently with correct grammar, punctuation, and spelling.

3. Paper 1 — Reading

Question Types

1. **Comprehension:** recall and inference.
2. **Writer's Effects:** analyse language and imagery.
3. **Summary:** concise synthesis of key points in own words.

Quick Strategy

- Read text once for meaning; again for details.
- Highlight verbs, adjectives, imagery.

- Use **Point** → **Evidence** → **Effect** format.

Writer's Effects Example

"The metaphor 'iron curtain' suggests separation and fear; the hard consonants reinforce the sense of finality."

Summary Tips

- Use bullet-style notes.
- Count points (usually 10–12).
- Paraphrase — never copy phrases.

4. Paper 2 — Directed Writing & Composition

Section A: Directed Writing (transactional)

- Task: respond to a passage (e.g. article, letter, speech).
- Assess understanding of ideas + own writing style.

Structure:

1. Intro – engage audience.
2. 3 body paragraphs (points + development).
3. Conclusion – balanced or persuasive closure.

Tone Register Bank

Type	Features
Formal report	passive verbs, factual tone, subheadings
Speech	direct address, rhetorical questions, emotive lexis
Article	catchy intro, varied sentence length, balanced argument

Section B: Composition (creative)

Choose **Descriptive** or **Narrative**.

Descriptive: focus on mood, imagery, sensory detail, structure by time or perspective.

Narrative: clear conflict, build-up, climax, resolution or open ending.

Planning Formula (10 min):

- 5-bullet structure plan
- 3–4 key images or moments
- Decide tense, POV, tone

5. Command Words Glossary

Command Word	Expectation
Identify	Give factual information directly from text
Explain	Give cause/effect, reasons for meaning
Analyse	How language/structure creates meaning
Summarise	Condense key ideas in own words
Compare	Show similarities/differences clearly
Evaluate	Judge effectiveness, justify opinion

6. Style Markers (Band 1 writing)

- Controlled tone and register.
- Sophisticated vocabulary but natural phrasing.
- Logical paragraphing with connective flow.
- Ambitious punctuation (dash, colon, semicolon) accurately used.
- Varied sentence openings.
- Precise spelling and grammar.

7. Time Management

Paper	Section	Suggested Time
Paper 1	Q1–2 comprehension	40 min
Paper 1	Q3 summary	20 min
Paper 2	Directed Writing	1 hr
Paper 2	Composition	1 hr

Leave 5 minutes to reread and correct punctuation.

8. Common Pitfalls & Fixes

Mistake	Correction
Quoting entire sentences	Select short, sharp phrases
Drifting off topic	Re-read question every paragraph
Repetition of adjectives	Build a synonym bank
Weak openings	Start with image, question, or contrast
Ignoring audience	Picture who you're addressing; adjust tone

9. Revision Plan (4 Weeks)

Week	Focus	Task
1	Reading Paper skills	Past paper comprehension & summary
2	Directed Writing	2 tasks: article & speech
3	Creative Writing	2 narratives, 1 description
4	Mixed review	Timed mock under exam conditions

☐ Self-Assessment Rubric

Criteria	Band 1	Band 3	Band 5
Content & Style	Clear voice, precise, original	Relevant but uneven tone	Limited awareness of audience
Organisation	Logical, fluent transitions	Some coherence	Lapses in structure
Language	Varied & ambitious	Mostly accurate	Simple & repetitive
Technical Accuracy	Rare errors	Occasional slips	Frequent basic errors

1 Practical Revision Tools

- **Past Papers:** Cambridge & SaveMyExams.

- **BBC Bitesize, Tutor2U, Mr Salles, First Language IGCSE YouTube channels.**
- Use colour-coded annotation (Language = blue, Structure = green, Tone = orange).
- Record yourself reading aloud – improves flow and punctuation awareness.

12. Exam-Day Checklist

- Bring black pen, highlighter, ruler, watch.
- Read all questions *twice*.
- Annotate but don't rewrite entire passages.
- Leave 3 lines between answers.
- Use a confident, mature voice — avoid slang.
- Proofread for sense, not just spelling.

Year 11 – Cambridge IGCSE English First Language Basic Supercurriculum

Why Supercurriculum Matters

English First Language is about **clarity, control, and creativity**. Reading beyond the syllabus and writing regularly builds the ability to communicate ideas powerfully — essential for academic success, interviews, and real-world confidence.

Reading – Breadth & Comprehension

- Read **one article + one short story each week**.
 - *Non-fiction*: BBC, Guardian, National Geographic.
 - *Fiction*: Dahl, Atwood, Orwell, Adichie, Bradbury.
- After each text, note:
 - What is the **writer's viewpoint or message**?
 - What techniques create **tone or emotion** (verbs, imagery, punctuation)?
- Keep a **Reading Log**:
 - Date | Title | Tone | Purpose | 1 favourite sentence.

2. Writing – Argument & Precision

- Practise **3 main forms**:
 1. **Transactional**: letters, reports, speeches, articles.
 2. **Descriptive**: vivid, sensory, atmospheric.
 3. **Narrative**: clear structure, conflict, and resolution.
- Use the **PEE rule** in every paragraph:
Point → Evidence → Explain.

- Start a **Writer's Journal**: 2 paragraphs a week on any topic (e.g., a memory, a local issue, a story idea).

3. Listening – Interpretation & Empathy

- Listen to podcasts or radio news (*BBC Learning English, TED-Ed*).
- Note how tone and pace change for effect.
- Summarise the speaker's **purpose, emotion, and message** in 3 bullet points.

4. Watching – Visual & Applied Understanding

- Watch short news clips or film scenes without subtitles.
- Write 5 sentences describing what you noticed about **tone, mood, and body language**.
- Then write the same scene as a short narrative (100–150 words).

5. Researching – Independent Inquiry

- Pick a current issue (climate change, social media, fast fashion).
- Read 2 articles with different opinions.
- Create a 1-paragraph summary showing the **difference in tone, bias, or perspective**.

6. Student-led Tasks – Collaboration & Leadership

- Swap short pieces with classmates and mark using 3 steps:
 - **Star**: what works well.
 - **Wish**: what could improve.
 - **Question**: something you'd like to know more about.
- Organise a mini "Writing Club" – one topic, 15 minutes, one page each.

7. Creative Tasks – Innovation & Expression

- Write a diary entry from a different point of view (e.g., "a tree watching the seasons").
- Turn a poem or song lyric into a short descriptive paragraph.
- Rewrite a news story as a personal letter or short dialogue.

8. Reflection – Metacognition

After each piece:

- What was my **purpose** (to inform, argue, describe, entertain)?
- Who was my **audience**?
- Which **sentence or image** am I most proud of?
- What could I improve next time?

9. Interdisciplinary Connections – Synthesis

- **History**: analysing speeches, propaganda, and perspective.
- **Psychology**: tone, emotion, persuasion.

- **Media Studies:** bias, structure, and audience targeting.
- **ICT:** writing for websites, posts, and digital communication.

□ Digital & Career Literacy

- Practise writing **emails, proposals, and personal statements.**
- Explore online articles for structure and tone.
- Discover careers using strong communication:
 - Journalism • Marketing • Law • Public Relations • Education • Editing.

□ Weekly Challenge Menu

Day	Task	Output
Monday	Read a news article	Write a 50-word summary
Tuesday	Describe a photo	1 paragraph, sensory detail
Wednesday	Listen to a speech	Note 3 persuasive devices
Thursday	Write a mini-article	100 words, 1 viewpoint
Friday	Reflection	2 sentences: what I learned this week

⇒ Self-Check Before Submitting

Skill	Ask Yourself
Content	Does my writing have a clear point or argument?
Structure	Do ideas flow logically, paragraph by paragraph?
Language	Have I used varied vocabulary and sentence length?
Accuracy	Are punctuation and spelling correct?
Style	Is my tone appropriate for the audience?

Year 11 – Cambridge IGCSE English First Language Extended Supercurriculum

1. Purpose

To move from writing **clearly** to writing **powerfully** — through analytical precision, advanced structure, and original expression.

This programme strengthens control of **voice, rhythm, and argument** across all IGCSE paper tasks.

Deep Reading & Writer Awareness

Goal: read like a writer — study craft, not plot.

- Read one **editorial**, one **short story**, and one **literary non-fiction text** each week.
- Annotate:
 - 3 key verbs → tone/attitude
 - 3 images/metaphors → effect
 - 3 structural choices → shift or contrast
- Keep a “**Style Bank**” – sentences worth imitating. Example:
“The world narrowed to the sound of my breath.”
→ *Lesson*: sentence rhythm mirrors focus.

Challenge Texts: Orwell (*Essays*), Didion, Adichie, George Saunders, The Economist columns.

2. Advanced Writing Craft

A. Structure (Macro-level)

- Begin with a *hook line*: contrast, question, image, or fact.
- Develop one idea per paragraph with clear transitions.
- Use a cyclical ending — link final line back to opening tone/image.

B. Sentence Control (Micro-level)

- Vary rhythm:
 - *Short = impact.*
 - *Long = reflection.*
- Combine with punctuation control:
 - *Colon* → reveal or explain.
 - *Dash* → contrast or emphasis.
 - *Semicolon* → balance two related ideas.
- Practise rewriting one paragraph in three tones: *formal, reflective, ironic.*

3. Argumentative & Discursive Mastery

Structure Template:

1. **Introduction** – define issue + stance clearly.
2. **Argument 1** – logic & evidence.
3. **Argument 2** – empathy & example.
4. **Counterpoint** – consider opposing side.
5. **Conclusion** – insight, not repetition.

Language Moves:

“It is often assumed that...”

“However, one must recognise...”

“Ultimately, the issue rests on...”

Mini task:

- Take a controversial headline (e.g. “Should social media be banned for under-16s?”).
- Write both sides, then one **balanced paragraph** using connectives: *Although, whereas, therefore, nevertheless.*

4. Descriptive & Narrative Elevation

Descriptive Writing:

- Replace lists of adjectives with **verbs + imagery**:
“The sea breathed against the rocks” (dynamic).
- **Zoom-lens** technique: wide → mid → close.
- Build **motifs** (e.g., light/dark, silence/sound).
- Layer **contrast** (warm/cold, movement/stillness).

Narrative Writing:

- Focus on **moment, not plot**.
- Create character through **voice and action**, not description.
- End with a **shift** (emotion, perspective, tone).
- Write a story with only 300 words — one setting, one decision, one consequence.

5. Precision Editing

After every draft:

1. Circle weak verbs (is/was/had) → replace with active choices.
2. Check paragraph openers: none should repeat structure.
3. Read aloud — listen for rhythm breaks.
4. Run a **3-pass edit**:
 - Pass 1: Structure flow
 - Pass 2: Style/syntax
 - Pass 3: Punctuation & spelling

Self-Feedback Formula:

“This paragraph works because...”

“To improve, I need to...”

6. Analysis & Commentary Practice

- Choose a 150–200 word passage from an article.
- Underline **five language features** (imagery, contrast, alliteration, repetition, tone).
- Write a short paragraph:
“The writer uses [technique] to create [effect], implying [idea]. This engages the reader by [reason].”

Sentence Starters for Language Analysis:

“The choice of the adjective “**conveys**...”

“**The repetition of** “ reinforces...”

“The structure moves from ____ to ____, mirroring...”

7. Listening, Speaking, & Persuasion

- Watch a **speech** (e.g. Greta Thunberg, MLK, or Emma Watson at the UN).
- Note:
 - Hook / emotional trigger
 - 3 persuasive devices (rhetorical question, anaphora, contrast)
 - Ending call-to-action
- Record your own **90-second persuasive talk** and listen back for:
 - Intonation
 - Pacing
 - Clarity of structure

8. Research & Source Evaluation

- Compare **two sources** on the same topic (e.g., AI, climate, youth culture).
- Identify tone: formal / emotive / sarcastic / neutral.
- Write 1 paragraph explaining bias and credibility:
“Source A uses emotional appeal through [evidence]; Source B relies on factual tone using [data].”

Task: Summarise both in ≤100 words using *your own language*.

9. Reflection & Growth

At the end of each half-term:

- Review your **3 best paragraphs**.
- Identify what makes them effective (word choice, rhythm, structure).
- Choose one **weak paragraph** and rewrite it using:
 - Stronger verbs
 - Varied sentence openings
 - Punctuation emphasis
 - A clearer tonal purpose

Keep a **Reflection Portfolio** (3–5 pages per term).

□ Interdisciplinary & Digital Literacy

- **History / Politics:** analysing speeches & propaganda.
- **Media:** adapting tone for audience.
- **Science:** clear explanation writing.
- **Digital:** short-form writing — blog posts, summaries, captions.
- Career links: journalism, law, marketing, diplomacy, creative writing, editing.

□ Independent Practice Menu

Week	Focus	Task
1	Argument Writing	Write a balanced 400-word article on a social issue.
2	Descriptive Writing	Describe one moment in detail using all five senses.
3	Narrative Writing	300-word micro-story: conflict and change.
4	Reading Paper	Analyse language in a short article; 2 paragraphs.

5	Editing	Rewrite one piece to Band 1 accuracy.
6	Oral Expression	Record a 1-minute persuasive speech.

□ **Self-Assessment Rubric (Extended Level)**

Band	Characteristics
Band 1 (Excellent)	Original, controlled, varied vocabulary, flawless mechanics, confident tone, logical structure.
Band 2 (Very Good)	Clear, thoughtful, consistent tone, mostly accurate grammar, effective punctuation.
Band 3-4	Relevant but uneven tone or structure; limited vocabulary; occasional lapses.

English First Language — Student Guide (Yr 11, IGCSE 0500/0990)

1) Exam Map

Paper	Focus	Time	Marks	Weight
Paper 1	Reading: comprehension, writer's effects, summary	2h	80	50%
Paper 2	Directed Writing (from a text) + Composition (descriptive or narrative)	2h	80	50%

2) Command Words (do exactly this)

- **Identify/Give/State** → short, factual point.
- **Explain** → cause/effect using *because/therefore/so*.
- **Analyse / Writer's effects** → quote **shortly**, name **technique**, explain effect.
- **Summarise** → condense key ideas **in your own words**.
- **Compare** → link similarities/differences directly.
- **Evaluate** → judgement with **criteria** (accuracy, bias, relevance, tone).

3) Paper 1: Reading Skills

Q1-Q2 Comprehension & Effects

PEE mini-frame:

Point → Evidence (2-5 words) → Effect (*implies/contrasts/creates...*).

Language menu: imagery, contrast, alliteration, metaphor, personification, listing, sentence length, punctuation for impact, tone shift.

Q3 Summary (own words)

1. Bullet key points while reading (aim 10-12).
2. Group by topic (e.g., *dangers / solutions*).
3. Write concisely; **no quotations**, no repetition.
Sentence starters: "The text shows that...", "Additionally...", "In contrast..."

Common slips to avoid: copying phrases; adding opinions; narrative retell.

4) Paper 2: Writing Skills

Section A: Directed Writing (from a passage)

- **Purpose/Audience/Form:** decide **tone** first (report / speech / letter / article).
- **Plan 3 points** taken from the passage → **develop with your own voice**.
- **Integrate** evidence by paraphrasing (do **not** lift whole lines).

Quick structures

- **Speech:** hook → 3 developed points (with signposting) → call to action.
- **Report:** subheadings, objective tone, logical sequence.
- **Article:** engaging intro, varied paragraph lengths, balanced view, crafted ending.

Section B: Composition

Choose **Descriptive** or **Narrative** (not both).

Descriptive (show, don't tell):

- 3-4 **focus moments**; five senses; precise verbs; zoom (wide→close).
- Cohesion by **motif** (light/shadow, sound/silence).

Narrative (controlled, not sprawling):

- One clear **conflict**; 3-part shape (set-up → shift → resolution/echo).
- Character through **action & dialogue**, not long backstory.
- Finish with a **change in perspective** or image echo.

5) Style & Accuracy (Band-1 markers)

- Varied sentence length; confident punctuation (colon, semicolon, dash) used accurately.
- Precise vocabulary; avoid clichés and “very/really”.
- Paragraphing that guides the reader (signpost connectives).
- Spelling/grammar near perfect; pronouns and tenses consistent.

Edit pass: Structure → Style → Spelling/Punctuation (**read aloud**).

6) Time Plan (per paper)

- **Paper 1:**
 - Read & annotate: **15 min**
 - Q1-2: **45 min**
 - Q3 Summary: **25 min**
 - Check: **5 min**
- **Paper 2:**
 - Directed Writing plan: **8 min** | write: **40-45 min**
 - Composition plan: **7 min** | write: **40-45 min**
 - Check: **5 min**

7) Ready-to-use Frames

Writer’s Effects (2–3 lines):

The verb/adjective “___” [**conveys/suggests**] ___; combined with [**device**], it [**creates effect**], shaping a [**tone**] of ___.

Directed Writing paragraph:

Point from text → **Paraphrased support** → **Your extension/reasoning** → **Audience-appropriate sentence**.

Descriptive sentence upgrades:

- “It was cold” → “The wind needled the gaps between my fingers.”
- “The street was busy” → “Wheels hissed; conversation braided and broke.”

Narrative opening options: image | in-media-res action | single-line contrast | brief dialogue.

8) Frequent Errors & Fixes

Error	Fix
Lifting phrases in summary	Paraphrase; swap nouns/verbs; compress ideas
Over-quoting in analysis	Use 2–5 word quotes only
Flat tone in directed writing	Decide audience + purpose first; mirror appropriate register
Plotty narratives	Focus on one moment; show internal shift
Comma splices	Use full stop/semicolon/conjunction appropriately

9) Four-Week Sprint (final run-up)

- **W1:** Paper 1 skills — 2 × writer's effects + 1 summary (timed).
- **W2:** Directed writing — one **speech** + one **report**.
- **W3:** Composition — 1 descriptive + 1 narrative (timed, 45 min each).
- **W4:** Full mock of Paper 1 & 2; mark with rubric; rewrite weakest paragraph.

10) Self-Assessment (tick before submitting)

- Answer stays **on the exact question** throughout []
- Quotations are **short & embedded** (Paper 1) []
- Summary is **own words** and concise []
- Writing shows **purpose, audience, tone** control (Paper 2) []
- **Accuracy:** punctuation/grammar checked []

11) Quick Resource List

- Past papers & mark schemes (Cambridge).
- BBC Bitesize (Reading & Writing skills).
- Mr Salles / First Language IGCSE (YouTube).
- A good dictionary + thesaurus (use sparingly, for precision).

Want a **2-page printable revision sheet** (writer's-effects menu, summary checklist, directed-writing structures, descriptive/narrative sentence starters)? I can generate it to match your other subject handouts.

Είπες:

English Second Language YEAR 11 basic

To ChatGPT είπες:

Perfect — here's a short, practical **Basic Supercurriculum** for **English as a Second Language (E2L, 0510/0511/0991)**. It matches the style of your other basics and is ready to print or drop into a study pack.

ENGLISH AS A SECOND LANGUAGE

Year 11 — Basic Supercurriculum

Why Supercurriculum?

To move from **accurate** English to **confident, natural** English. You'll build real-world communication: reading for gist/detail, clear writing, fluent speaking, and purposeful listening.

1) Reading — Breadth & Comprehension

- Read **one short article + one longer feature** weekly (science/tech, environment, people).
- While reading, mark: **main idea, supporting details, signpost words** (however, therefore, whereas), **tone**.
- Keep a **Vocabulary Pad**: new word • part of speech • collocation • example sentence.
- Try graded readers (B1–B2) or news-in-levels for confidence, then step up to authentic articles.

2) Writing — Clarity & Organisation

Practise the **core task types**:

- **Email/Letter** (informal/formal): greeting → purpose → details → polite close.
- **Article/Blog post**: catchy intro → 2–3 clear paragraphs → short conclusion.
- **Report/Review**: headings, objective tone, recommendations.

Always plan: audience + purpose + 3 bullet points.

PEEL mini-paragraph: Point → Example → Explain → Link.

Accuracy targets

- Tenses: present simple/continuous; past simple/perfect; future (will/going to).
- Punctuation: commas for clauses/lists; full stops; paragraphing.
- Common errors to watch: subject–verb agreement, articles (a/an/the), prepositions, run-on sentences.

3) Listening — Gist, Detail, Note-taking

- Listen **daily** (5–10 min): BBC Learning English, TED-Ed, short podcasts.
- First listen = **gist** (topic + purpose). Second = **detail** (numbers, names, reasons).
- Note with **abbreviations & arrows** (cause → effect). Check predictions before replaying.

4) Speaking — Fluency & Pronunciation

- 3× per week, speak for **90 seconds** on a prompt (e.g., travel, health, technology). Record yourself.

- Use **PREP**: **P**oint • **R**eason • **E**xample • **P**oint again.
- Pronunciation drills: sentence stress (content words louder), final consonants, -ed endings (/t/, /d/, /ɪd/).
- Build **discourse markers**: *Firstly, in my view, for example, however, to conclude.*

5) Language Growth — Practical Grammar & Lexis

- Learn **chunks/collocations**: *take part in, make a decision, in charge of.*
- Upgrade adjectives/adverbs: *important → crucial, very good → highly effective.*
- Modal verbs for tone: *should, might, must, could.*
- Connectors for coherence: *in addition, as a result, on the other hand, for instance.*

6) Student-Led Tasks

- **Peer edit swap**: check for task fulfilment, organisation, accuracy, range.
- **Mini debate**: “Homework should be limited to 30 minutes.” Use PREP and polite turn-taking.
- **Vocabulary quiz**: each student teaches 5 collocations from their reading.

7) Creative Tasks

- Rewrite a news paragraph as a **60-second speech**.
- Turn a travel photo into a **descriptive postcard** (80–100 words).
- Record a **voice note** summarising a documentary in 6 sentences.

8) Reflection — Metacognition

After each task, jot 3 lines:

1. What I did well (e.g., paragraphing, connectors).
2. One error to fix (e.g., articles).
3. One new word/chunk I can use next time.

9) Interdisciplinary Links

- **Geography/Science**: summaries of topics with data.
- **Business/ICT**: formal emails, reports, presentations.
- **Global Perspectives**: balanced arguments, evaluating sources.

□ Weekly Practice Menu (30–40 mins/day)

Day	Focus	Quick Task
Mon	Reading	1 article → 5 bullet summary (own words)
Tue	Writing	120–150 word email/report paragraph
Wed	Listening	8–10 min clip → gist + 6 key facts
Thu	Speaking	90-sec PREP talk (record)
Fri	Language	Grammar micro-drill + 6 collocations

Weekend	Review	Edit one piece; update vocab pad (10 items)
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11) Self-Check Before Submitting (E2L rubric in mind)

- **Task achievement:** Did I answer the exact prompt and cover all bullet points?
- **Organisation:** clear intro, logical paragraphs, linking words.
- **Range:** some complex sentences and varied vocabulary.
- **Accuracy:** few errors; correct tense, articles, prepositions, punctuation.
- **Tone:** appropriate for audience (formal/informal) and purpose.

12) Quick Resource List

- **BBC Learning English, Breaking News English, News in Levels**
- **TED-Ed** (with subtitles → then without)
- **Cambridge E2L coursebook/workbook** (exam practice)
- Free dictionaries with examples: **Cambridge, Longman** (check collocations).

Mini Templates (copy/paste)

Email opening (formal):

Dear Sir/Madam,

I am writing to **[state purpose]**. Firstly, ... In addition, ... Finally, ... I would be grateful if...

Article intro:

Many people believe **[topic]** affects us every day. In this article, I will explain **[two/three points]** and suggest **[recommendation]**.

Conclusion sentence:

Overall, **[restate idea]**, and **[short recommendation or opinion]**.

ENGLISH AS A SECOND LANGUAGE

Year 11 – Extended Supercurriculum

1. Goal

To develop accuracy into **fluency and sophistication** — expressing nuanced opinions, analysing tone and bias, and writing with confident structure and style.

Advanced Reading & Critical Awareness

Read like an examiner — not just “what it says” but *how* it persuades or informs.

Weekly Text Plan

- **Mon:** Short editorial or opinion piece (The Guardian, BBC, National Geographic).
- **Wed:** Report or speech transcript.
- **Fri:** Short story, blog, or cultural article.

Tasks

- Identify:
 1. Purpose (to inform / persuade / describe / entertain)
 2. Tone (formal / sarcastic / optimistic / critical)
 3. Bias (word choice, inclusion, omission)
- Highlight **connectives** (however, therefore, consequently, whereas).
- Paraphrase 3–4 sentences into your own words.

Challenge task:

Compare two articles on the same issue — summarise in ≤100 words showing difference in tone or viewpoint.

2. Advanced Writing Skills — Argument, Accuracy & Style

A. Writing Process

1. **Plan:** audience, purpose, and structure (use bullet outline).
2. **Draft:** 3 paragraphs minimum (intro, development, conclusion).
3. **Edit:** check tenses, connectives, paragraph unity.
4. **Refine:** add one idiomatic expression or sophisticated transition.

B. Genre Skills

Type	Structure	Language Features
Formal Letter / Email	Opening → Purpose → Details → Polite Close	Modals (<i>could, would, might</i>), indirect tone (<i>I would appreciate...</i>)
Report	Title → Introduction → Findings → Conclusion/Recommendations	Passive voice, sequencing (<i>firstly, in addition, finally</i>)
Article / Blog	Hook → Idea 1 → Idea 2 → Final opinion	Varied sentence openings, rhetorical question, balanced tone
Review	Context → Strengths → Weaknesses → Verdict	Adjectives of opinion (<i>gripping, dull, well-</i>

		<i>designed</i>)
Speech	Hook → 3 points → Call to action	Direct address, emotive lexis, repetition

C. Sentence Craft (for higher bands)

- Use **complex connectors**: *provided that, despite, whereas, although, in contrast to*.
- Balance long and short sentences:
“The streets were silent. Then, the rain began.”
- Add rhetorical balance:
“It is not the strongest who succeed, but the most adaptable.”

3. Descriptive & Narrative Precision

Even though EzL focuses on communication, descriptive flair strengthens writing marks.

- Include **five-sense vocabulary**, vivid verbs, and **specific detail** (not “nice,” but “crisp,” “muddy,” “gleaming”).
- Use **figurative language** sparingly: metaphor, simile, personification.
- Avoid over-adjectiving — one strong word is better than four weak ones.

Task:

Describe a busy market without naming it; use sound, smell, and colour instead of the word *market*.

4. Listening Mastery — Detail & Inference

Goal: understand facts + tone + implied meaning.

Practice Routine

1. **Listen once for gist** – What is it about? Who’s speaking?
2. **Listen again for detail** – names, numbers, reasons.
3. **Final check:** tone/emotion (positive, uncertain, angry).
4. **Summarise** in one short paragraph.

Sources

- *BBC Learning English: “6-Minute English”*
- *VOA Learning English*
- *TED-Ed (with subtitles, then without)*

Extension: Shadow reading – pause and repeat phrases exactly to copy intonation and rhythm.

5. Speaking — Fluency & Extended Response

Part 1: Personal questions → practise 4-sentence answers.

Part 2: Topic discussion → express balanced opinion.

Part 3: Conversation → use linking phrases & examples.

Fluency Frame (PREP method):

Point → Reason → Example → Point again.

“I believe online learning is useful **because** it saves time. **For example**, my school used digital tools during lockdown. **Therefore**, it helps students work independently.”

High-level connectors for speaking:

- For example / for instance
- On the other hand / however
- Personally, I feel / It seems to me
- As far as I’m concerned / From my experience
- To be honest / To tell you the truth

Extension tasks:

- Record 2-minute responses weekly.
- Try peer interviews on abstract topics (technology, the environment, future jobs).
- Focus on **intonation and emphasis**, not just grammar.

6. Grammar & Vocabulary Refinement

Focus	Examples	Goal
Complex sentences	“Although it was raining, we went out.”	Use 1 per paragraph.
Passive voice	“The event was organised by the students.”	Formal reports.
Relative clauses	“People who exercise regularly feel better.”	Accuracy + fluency.
Modal verbs	should, could, might, must, may	Tone control.
Collocations	make progress, take part in, keep calm	Natural phrasing.

Challenge: write one short paragraph using 5 *collocations* and 2 *modals*.

7. Research & Real-World Literacy

- Choose a **global issue** (migration, climate change, AI, education).
- Read two sources, summarise in 100 words, and give your **personal response** (agree/disagree and why).
- Keep a “Mini Research Notebook” — topic, sources, vocabulary, and opinion.

8. Reflection & Metacognition

At the end of every week:

- “Which new structure or phrase did I use well?”
 - “What mistake repeated most?”
 - “What can I do to sound more natural next time?”
- Record one 30-second reflection voice note.

9. Digital & Career Literacy

- Practise **formal email writing** (to a company or teacher).

- Use Grammarly or Cambridge Write & Improve for self-feedback.
- Build confidence for professional communication (reports, meeting summaries).
- Career links: tourism, international business, media, diplomacy, customer service.

☐ Weekly Mastery Plan

Week	Focus	Key Output
1	Reading & Summary	Compare two news sources, 100-word summary
2	Formal Writing	120–150 word report with headings
3	Listening & Note-taking	10-min audio → table of key points
4	Speaking	2-min PREP talk + peer feedback
5	Grammar & Vocabulary	10 new collocations + 1 paragraph use
6	Integrated Task	Read article → write response → record summary

☐ Self-Assessment Checklist (Extended Band 1–2)

Skill	Description	✓
Reading	Identifies main ideas, inferences, and tone	
Writing	Fully relevant, logically organised, precise vocabulary	
Language	Few grammar errors, correct tense and word form	
Listening	Understands gist + details, infers attitude	
Speaking	Fluent, coherent, confident pronunciation	

1 Useful Online Tools

- ☐ **BBC Learning English** – grammar, pronunciation, current issues
- ☐ **Breaking News English** – multi-level articles + exercises
- ☐ **Cambridge Write & Improve** – AI feedback on writing
- ☐ **YouGlish / ELSA Speak** – pronunciation practice
- ☐ **The Day / Newsela** – global topics at student level

English as a Second Language — Student Guide (Yr 11, IGCSE 0510/0991)

1) Exam Map (check your entry: Core vs Extended; with/without Speaking endorsement)

Paper	Core Route	Extended Route	Time	Weight/Notes
Reading & Writing	Paper 1	Paper 2	2 h	50%
Listening	Paper 3	Paper 4	~50 min	25%
Speaking (endorsement)	Paper 5	Paper 5	10–12 min	Separate grade (0510). Some centres do with/without speaking (0991 = no endorsement).

Confirm with your teacher: Core/Extended entry, and whether **Speaking** counts as an endorsement at your centre.

2) What Examiners Reward

- **Task fulfilment:** you answered **exactly** what was asked (all bullet points).
- **Organisation & cohesion:** clear paragraphs, logical order, linking words.
- **Range:** varied vocabulary, collocations, some complex sentences.
- **Accuracy:** tense control, articles/prepositions, punctuation, spelling.
- **Appropriate tone & register:** formal/informal, audience-aware.
- **Reading/Listening:** both **gist and detail**, plus **inference** (attitude/tone).

3) Command Words — Do Exactly This

- **Identify/State/Give:** short, direct facts.
- **Select/Match/Complete:** choose from text/audio accurately.
- **Explain/Summarise:** use **your own words**; be concise.
- **Describe/Compare:** details + linking words (*whereas, however*).
- **Write/Report/Article/Email/Review:** produce the required **form + tone**.

4) Reading & Writing Paper — Fast Strategy

A) Reading Tasks (forms vary)

- **Skim** first (titles, first/last lines) for **gist**.
- **Scan** for names, dates, numbers, reasons.
- **Underline signposts:** *however, therefore, whereas, in addition*.
- **Own words** where required; avoid lifting long phrases.

B) Summary/Note Tasks

- Count the **number of points** needed (often 10–12).
- Group by topic (e.g., *problems / solutions*).
- Use short, accurate **paraphrase**; no examples/opinions unless asked.

C) Directed Writing (email/article/report/review)

Plan (3–4 bullets) → Write (120–180 words typical task) → Check.

Mini-frames

- **Email (formal):** Purpose → Details 1–2 → Request/thanks → Polite close.
“*I am writing to enquire about... In addition,... I would appreciate ...*”
- **Article/Blog:** Hook → Idea 1 → Idea 2 → Brief conclusion/opinion.
- **Report:** Heading → Findings (facts) → Recommendations.
- **Review:** Context → 2 strengths → 1 weakness → Verdict.

Linkers bank: *Firstly, moreover, however, in contrast, as a result, for instance, overall.*

5) Listening Paper — Method (3 passes)

1. **Preview** questions (predict topic/words).
 2. **First listen** = **gist** (who/where/what).
 3. **Second** = **detail** (numbers, names, reasons).
 4. **Final check** where provided.
- Write **brief answers while listening**; don't wait.
 - For attitude/tone questions, note words like *delighted, doubtful, concerned*.

Active note symbols: → (cause), ↑/↓ (increase/decrease), ~ (about), yrs/hrs/£.

6) Speaking Test (endorsement)

- **Structure:** warm-up → topic discussion → follow-ups.
- **PREP frame:** Point → Reason → Example → Point again.
- Use discourse markers: *In my view, For example, However, On the other hand, To conclude.*
- **Pronunciation:** stress content words; sound final consonants; clear -ed endings (/t/, /d/, /ɪd/).
- Avoid 1-word answers; aim for **4–6 sentence** responses.

30-sec warm routine: breath + slow first sentence + confident pace.

7) Accuracy & Range — High-Band Check

- **Tenses:** present simple/continuous; past simple/perfect; *will/going to*.
- **Complex sentences:** *Although/whereas/because...*
- **Articles:** a/an/the (watch *the* with unique nouns).
- **Prepositions:** depend **on**, interested **in**, good **at**.
- **Collocations:** *take part in, make a decision, in charge of, highly effective.*

- **Punctuation:** commas for clauses/lists, full stops, paragraphing.

8) Common Pitfalls & Quick Fixes

Pitfall	Fix
Copying phrases in summaries	Paraphrase: swap verbs/nouns; compress ideas
Missing bullet points in writing	Tick off bullets in your plan
Overlong sentences	One idea per sentence; use full stops
Informal tone in formal tasks	Modals + polite forms (<i>I would appreciate...</i>)
Weak linking	Add <i>however, therefore, in addition, for instance</i>
Vague vocabulary	Replace <i>good/bad/nice</i> with precise words

9) Timing Plan (suggested)

Reading & Writing (2h)

- Read + plan: **10 min**
- Reading tasks: **45–50 min**
- Directed Writing: **35–40 min**
- Summary/notes task: **20–25 min**
- Check: **5 min**

Listening (~50 min)

- Preview Qs before each track; answer during; brief check after each part.

Speaking (10–12 min)

- Practise topic banks; aim for **coherent, developed** answers.

10) Weekly Practice Plan (40–50 min/day)

- **Mon:** Article → 6-bullet summary (own words).
- **Tue:** Write a **120–150 word** task (rotate email/report/article).
- **Wed:** Listening (8–10 min) → 6 key facts + tone.
- **Thu:** Speaking (record **90 sec** PREP response).
- **Fri:** Grammar/lexis drill (10 collocations + 5 complex sentences).
- **Weekend:** Edit one piece to accuracy; update vocab pad (10 items).

11) Ready-to-Use Templates

Formal email opening/close

Dear Sir/Madam,

I am writing to **[state purpose]**. Firstly, ... In addition, ... Finally, ...

I would be grateful if you could...

Yours faithfully, **[Name]**

Report paragraph

Findings: The majority of students **prefer...** **However**, several participants **mentioned...** **Therefore**, it is **recommended** that...

Balanced opinion sentence

Although **[view A]** is common, **[view B]** is convincing **because...** **Overall**, ...

12) Self-Assessment (tick before you submit)

- Covered **all bullet points** / answered exactly the question []
- Clear **paragraphs** + linkers []
- **Own words** used in summary []
- **Range:** some complex sentences + varied vocabulary []
- **Accuracy:** tense/articles/prepositions/punctuation checked []
- **Tone:** appropriate for audience and purpose []

13) Quick Resource List

- **Cambridge E2L coursebook/workbook** (your syllabus year)
- **BBC Learning English, Breaking News English, News in Levels**
- **TED-Ed** (with → then without subtitles)
- **Cambridge Write & Improve** (free feedback)
- **YouGlish / ELSA Speak** (pronunciation & stress)

CHEMISTRY

Basic Supercurriculum

Why Supercurriculum?

To move from **remembering facts** to **thinking like a chemist** — planning investigations, handling data, and explaining patterns with particle ideas and equations.

1) Core Knowledge – What to Master

- **Particles & States:** Brownian motion, diffusion, changes of state, heating curves.
- **Atomic Structure & Bonding:** protons/neutrons/electrons, isotopes, ions, ionic vs covalent vs metallic, giant vs simple structures.
- **Stoichiometry:** formulae, equations, **mole** ($n = m/M_r$), **concentration** ($c = n/vol$).
- **Energetics:** exo/endo; simple energy profile diagrams.
- **Rates & Equilibria:** collision theory; factors (surface area, temp, conc, catalysts); reversible reactions & Le Chatelier (qualitative).
- **Acids, Bases & Salts:** pH, indicators, titration ideas, salt prep methods.
- **Electrochemistry:** electrolysis principles, products at electrodes, metal reactivity.
- **Periodic Table & Metals:** trends in Groups 1, 7, 0; transition basics; extraction & corrosion.
- **Organic Intro:** alkanes/alkenes, isomerism (basic), addition vs substitution, simple polymers.
- **Air & Water / Environment:** water treatment, greenhouse gases, nitrogen cycle (industrial), sulfur/acid rain.

2) Do-Tank – 15-minute Weekly Practicals (home/school friendly)

- **Diffusion in gels:** food dye in agar/jelly → note temperature effect.
- **Neutralisation curve (simulated or demo):** indicator colour sequence; practise reading burette scale photos.
- **Rate reaction:** marble chips + HCl under a balloon → compare chip size.
- **Electrolysis demo/sim:** brine/equations; identify products.
- **Chromatography of inks:** R_f calculation practice.

Always follow lab safety rules (goggles, small quantities, label everything).

3) Calculation Mini-Drills (5-10 min)

- **Relative formula mass (M_r)** from formulae.
- **Moles from mass:** $n = m/M_r$ or $n = M_r/m$.
- **Concentration:** $c = n/V$ or $n = cV$ (V in dm^3).
- **Gas volumes (r.t.p.):** $1 \text{ mol} \approx 24 \text{ dm}^3$.
- **Percentage yield & atom economy** (simple).
- **Empirical formula** from % composition.

Habit: write units **every line**; box final answer with unit.

4) Equation Fluency

- Balance 3 equations per study session.
- Add **state symbols** (s,l,g,aq).
- Practise **ionic equations** for precipitation and displacement (cancel spectator ions).

5) Diagrams to Sketch (from memory)

- **Particle diagrams** for state changes & diffusion.
- **Ionic lattice vs simple molecular** (link to mp/bp & conductivity).
- **Energy profile** (exo/endo; activation energy).
- **Electrolysis cell** (label electrodes, ions, products).
- **Titration setup** (burette, pipette, conical flask, indicator).

6) Vocabulary & Linking Phrases

- **Because** particles collide **more frequently/energetically**, therefore **rate increases**.
- **Giant ionic** → **high mp** (strong electrostatic attractions in lattice).
- **Simple molecular** → **low mp** (weak intermolecular forces).
- **Covalent network** (diamond/silicon dioxide) → **very high mp, hard, insulator** (except graphite conducts via delocalised electrons).

7) Quick Case Studies (one per week)

- **Haber process** ($\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$): pressure/temperature/catalyst trade-offs.
- **Contact process** (H_2SO_4): role of V_2O_5 catalyst.
- **Cracking**: breaking long alkanes → alkenes + shorter alkanes (economic reasons).
- **Electrolysis of brine**: Cl_2 , H_2 , NaOH uses.
- **Water treatment**: filtration, chlorination, fluoridation pros/cons.

8) Common Pitfalls & Fixes

Pitfall	Fix
Mixing atoms/molecules/ions	Draw and label with charge and count
Forgetting units in calcs	Write unit at each line; convert $\text{cm}^3 \rightarrow \text{dm}^3$
Confusing bond strength vs intermolecular forces	State the type of force you are talking about
Vague answers in rates/equilibria	Use collision theory or Le Chatelier explicitly
Wrong electrolysis product	Check electrolyte, electrode type, and reactivity series

9) Weekly Study Rhythm (45–60 min/day)

- **Mon:** Content flashcards (10) + 3 balanced equations.
- **Tue:** Calculations set (moles/concentration) + mark scheme check.
- **Wed:** Past-paper **shorts** (6–8 marks mixed).
- **Thu:** Diagram memory sketch + one 15-min practical/demo or video.
- **Fri:** Organic quick quiz + name → formula → reaction.
- **Weekend:** 30-min retrieval quiz + fix-up sheet (errors & correct method).

10) Mini Lab-Book / Evidence (helps Paper 5/6 style)

For any experiment you meet (in class or video), jot:

- **Aim • Variables** (CIV – control/independent/dependent)
- **Method** (bullets) • **Results** (table/graph)
- **Conclusion** (pattern + particle/structure explanation)
- **Evaluation** (errors + improvements)

11) Resources (free & reliable)

- **Chemistry** (coursebook/revision guide for your series)
- **BBC Bitesize (Chemistry GCSE)** for quick refreshers
- **Royal Society of Chemistry LearnChemistry** (animations/experiments)
- **Khan Academy / FuseSchool** (clear videos)
- **Past papers & mark schemes** (your syllabus year)

12) Self-Check Before a Test

- I can **balance** unseen equations []
- I can do **mole, concentration, gas volume** problems with units []
- I can explain **mp/bp and conductivity** using **structure & bonding** (not just names) []
- I can sketch **five key diagrams** from memory []
- I have practised **planning & evaluating** an investigation []

Want the **Extended Supercurriculum** next (calculation ladder, redox & ionic half-equations, titration interpretation, and exam-style planning & evaluation), plus a **Student Guide** (paper breakdown, command words, and timing for Papers 1/2 + 3/4/5/6)?

Είπες:

yes

To ChatGPT είπες:

Σκέφτηκε για 8s

Brilliant — here's your **Chemistry (0620/0971)** package:

1. **Extended Supercurriculum** (skills to push into top bands)
2. **Student Guide** (exam map, timing, command words, and high-yield checklists)

❑ Chemistry — Extended Supercurriculum (Yr 11)

1) Calculation Ladder (build accuracy → speed)

Level 1: Mr, Ar, % by mass

Level 2: $n = \frac{m}{M_r}$, gas volume at r.t.p. ($\approx 24 \text{ dm}^3 \text{ mol}^{-1}$)

Level 3: Solutions: $c = \frac{n}{V}$ (V in dm^3)

Level 4: Empirical → Molecular formula (use Mr)

Level 5: % yield; atom economy

Level 6: Titrations (see §4)

Habit: write a **plan line** before numbers; keep **units every step**; round at the end.

2) Redox, Ionic & Half-Equations (clean method)

- **Oxidation** = O gain / e^- loss / oxidation state $\uparrow$
- **Reduction** = O loss / e^- gain / oxidation state $\downarrow$

Ionic equation recipe (precipitation/displacement):

1. Write full equation → 2) Split (aq) into ions → 3) Cancel spectators → 4) Balance charge & atoms.

Half-equations (electrolysis): e.g. at cathode: $2\text{H}^+ + 2e^- \rightarrow \text{H}_2$; at anode in brine: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2e^-$.

3) Structure, Bonding & Properties — exam phrasing

- **Giant ionic lattice** → *strong electrostatic attractions* → **high mp**, conducts when molten/aqueous (mobile ions).
- **Simple molecular** → *weak intermolecular forces* (NOT “weak bonds”) → **low mp**, non-conductors.
- **Giant covalent** (diamond/ SiO_2) → **very high mp**, **hard**, **insulators**; **graphite** conducts via **delocalised electrons** along layers.
- **Metals:** lattice of positive ions in a **sea of delocalised electrons** → **conducts**, **malleable** (layers slide).

4) Titration Sense-Making (no apparatus? still score)

- **Core idea:** moles acid = moles base $\times$ ratio.
- **Steps:** (i) Write balanced equation. (ii) $n = c \times V_n = c \times V_n$ (dm^3). (iii) Use ratio. (iv) Back-calculate unknown ccc or VVV.

Indicators: strong acid/strong base → almost any; weak acid/strong base → **phenolphthalein**.

Commentary lines examiners like:

- “Titre concordant within 0.10 cm^3 .”
- “Use a **white tile**; read burette at **eye level**; remove **air bubbles** from jet.”
- “Repeat until two concordant titres; average those.”

5) Rates & Equilibrium — model sentences

- **Rates:** Increasing temp **increases frequency and energy of collisions**, so **more collisions exceed E_a** $\rightarrow$ rate $\uparrow$.
- **Le Chatelier (qualitative):** If pressure $\uparrow$, equilibrium shifts to the side with **fewer gas moles**. Catalyst **speeds** both forward and reverse **equally** $\rightarrow$ position **unchanged**.

6) Electrolysis — decide products fast

- **Aqueous at cathode:** least reactive cation (metal below hydrogen) OR H_2 if metal is very reactive (e.g., Na^+).
 - **Aqueous at anode:** halide $\rightarrow$ **halogen**; otherwise typically O_2 from OH^- .
 - **Molten:** simply the ions (e.g., $Pb^{2+} \rightarrow Pb$, $Br^- \rightarrow Br_2$).
- Phrase:** “Product choice depends on **reactivity** (cathode) and **anion present** (anode).”

7) Planning, Results, Evaluation (Paper 5/6 style)

- **Plan:** aim, variables (CIV: Control/Independent/Dependent), risks (goggles, acids/alkalis, hot equipment).
- **Results table:** headings with **units**, sensible sig. figs., repeats/mean.
- **Graphs:** choose **line of best fit**; correct axis units; **anomalies** circled + comment.
- **Evaluation:** “Main errors: reading burette ($\pm 0.05\text{ cm}^3$), heat loss... Improvements: **insulation, smaller intervals, repeat and average.**”

8) Qualitative Tests (must-know lines)

Gases:

- H_2 : squeaky pop with lighted splint
- O_2 : relights glowing splint
- CO_2 : turns **limewater milky**
- NH_3 : pungent; **damp red litmus $\rightarrow$ blue**; white fumes with HCl
- Cl_2 : **bleaches** damp litmus

Cations (aq) — NaOH then NH_3 (aq):

- Cu^{2+} : **blue ppt**, dissolves in excess $NH_3 \rightarrow$ deep blue solution
- Fe^{2+} : **green ppt**, turns brown on standing
- Fe^{3+} : **brown ppt**
- Ca^{2+} : white ppt in NaOH; **no** ppt with NH_3 (or very slight)
- NH_4^+ : on warming with NaOH $\rightarrow$ **NH_3 gas** (test as above)

Anions:

- Cl^- : acidify + $AgNO_3 \rightarrow$ **white ppt** (dilute NH_3 soluble)
- SO_4^{2-} : acidify + $BaCl_2/Ba(NO_3)_2 \rightarrow$ **white ppt**
- CO_3^{2-} : **acid $\rightarrow$ effervescence**; gas turns limewater milky

9) Organic Quick-Wins

- **Alkane** (sat'd) vs **Alkene** ($C=C$): **$Br_2(aq)$ decolourises with alkene** (addition).

- **Additions:** alkene + H_2 (Ni, 150 °C) $\rightarrow$ alkane; alkene + $\text{Br}_2 \rightarrow$ dibromoalkane.
- **Polymers:** name from monomer; repeating unit shows **two carbons** from $\text{C}=\text{C}$.
- **Combustion:** complete $\rightarrow \text{CO}_2 + \text{H}_2\text{O}$; incomplete $\rightarrow \text{CO}$ (toxic) + C (soot).

10) High-Yield Retrieval (8-week loop)

W1–2: Moles ladder + ionic equations

W3: Energetics & rates practical analysis

W4: Equilibria + Haber/Contact **trade-offs**

W5: Electrolysis decisions + half-equations

W6: Titrations (calc + evaluation)

W7: Structure/bonding $\rightarrow$ properties explanations

W8: Full mixed paper under time; error log & fix-ups

□ Chemistry — Student Guide (Yr 11, 0620/0971)

1) Exam Map

Route	Paper	Focus	Marks	Time	Weight
Core	1	Multiple choice	40	45 min	30%
	3	Structured theory	80	1h15	50%
	5	Practical (or 6 Alt to Practical)	40	~1h	20%
Extended	2	Multiple choice	40	45 min	30%
	4	Structured theory	80	1h15	50%
	5/6	Practical / Alt to Practical	40	~1h	20%

You sit **1+3+5** (Core) or **2+4+5** (Extended). Centres may choose **Paper 6** instead of 5.

2) Command Words — What to Do

- **State/Identify:** short fact or number.
- **Describe:** what you see / trend with data.
- **Explain:** *because... therefore...* link to **particles/structure/bonding**.
- **Predict:** apply a rule or trend; give reason.
- **Calculate:** show **method + units**.
- **Evaluate:** judge with data/errors/improvements.

3) Timing Plan

MCQ (P1/2): ~1 min per Q; star and move on; return later.

Theory (P3/4): ~1 min per mark. Leave **5 min** to check units & rounding.

P5 Practical: read whole paper, sketch a plan, work **tidily**.

P6 Alt: write **clear plans**, include **tables with units**, and **reasoned conclusions**.

4) Must-Know Constants, Units, Conversions

- r.t.p. gas volume $\approx 24 \text{ dm}^3 \text{ mol}^{-1}$
- Volume: $\text{cm}^3 \div 1000 = \text{dm}^3$
- Concentration: mol dm^{-3}
- Energy profiles: E_a labelled; exo = products lower than reactants
- Rate units often g s^{-1} , $\text{cm}^3 \text{ s}^{-1}$, or “faster/slower” with reasoning

5) Quick Templates

Balancing & States

Coefficients first, **charges** balanced if ionic, add (s/l/g/aq).

Ionic equation

Write ions (aq), **cancel spectators**, balance **charge** and atoms.

Le Chatelier sentence

“Increasing [**stress**] shifts equilibrium to **reduce** it, so the yield of [**product**] **increases/decreases**.”

Planning practical

“Keep [**controls**] constant; change [**independent**]; measure [**dependent**] using [**apparatus**]; repeat $\times 3$ and average; wear goggles.”

6) Common Pitfalls & Fixes

Pitfall	Fix
Saying “weak bonds between molecules” (for covalent)	Say weak intermolecular forces
No units / wrong power of 10	Convert $\text{cm}^3 \rightarrow \text{dm}^3$; write units every line
Confusing electrolysis products (aq)	Use reactivity + anion rule (see Extended §6)
“More collisions” with temp	Add “ with enough energy to overcome E_a ”
Vague equilibrium answers	Refer to moles of gas and direction of shift

7) Practical Papers — Mark-Winning Habits

- **Tables:** headings with units, consistent sig. figs., spot anomalies.
- **Graphs:** choose best fit; labelled axes + units; sensible scale; use large area.
- **Conclusions:** pattern + particle/structure reason.
- **Evaluation:** identify **systematic vs random** errors; give **practical** improvements (insulation, same volume, calibrated instruments).

8) High-Yield Lists (pin to folder)

- **Gas tests & ion tests** (see Extended §8).
- **Group trends:**
 - **G1:** down group $\rightarrow$ reactivity $\uparrow$, mp $\downarrow$, density $\uparrow$
 - **G7:** down group $\rightarrow$ reactivity $\downarrow$, mp $\uparrow$, colour darkens

- **Reactivity series (snippet):** K Na Ca Mg Al (C) Zn Fe (H) Cu Ag Au
 - Above H → displace H₂ from acid.
- **Industrial icons:** **Haber** (200 atm, ~450 °C, Fe catalyst), **Contact** (V₂O₅ catalyst), **cracking** (Al₂O₃/SiO₂, high T).

9) Four-Week Exam Sprint

- **W1:** Moles & equations (daily 15-min drills).
- **W2:** Structure-bonding properties + electrolysis decisions.
- **W3:** Rates/equilibria + titration maths + evaluative writing.
- **W4:** Full past papers (your series); build an **error log**; redo weak Qs.

10) Final 60-Second Check Before You Hand In

- Answered **every** part? Tick boxes.
- Calculations: **units** each line, **3 s.f.** unless context says otherwise.
- Explanations: include **particle/structure** keyword.
- Graph/table: **titles** + **units** present.
- Practical: **improvement** is **specific** (how it reduces error).

PHYSICS

Basic Supercurriculum

Why Supercurriculum?

Physics explains **how the world works** — from motion and energy to waves, forces, and electricity. Going beyond the textbook helps you think like a scientist: observing, predicting, and testing ideas with logic and evidence.

Core Knowledge — Must Master

1. Motion & Forces

- Speed, velocity, acceleration: $v = \frac{s}{t}$, $a = \frac{v - u}{t}$
- Graphs: gradient = speed/acceleration; area = distance.
- Newton's laws:
 1. Balanced $\rightarrow$ steady motion
 2. $F = ma$
 3. Equal and opposite forces.
- Friction, air resistance, and terminal velocity.

2. Energy & Work

- Energy types: kinetic, potential, thermal, electrical, etc.
- $E_k = \frac{1}{2}mv^2$, $GPE = mgh$, $W = Fd$
- Conservation: total energy stays constant.
- Power = $\frac{E}{t}$ or $P = \frac{W}{t}$
- Efficiency = $\frac{\text{useful output}}{\text{total input}} \times 100\%$

3. Density & Pressure

- density = $\frac{m}{V}$
- $p = \frac{F}{A}$; pressure in liquids: $p = h\rho g$

4. Thermal Physics

- Kinetic particle model: heating = particles move faster.
- Expansion, evaporation, conduction, convection, radiation.
- Specific heat capacity & latent heat (change temp vs change state).

5. Waves & Sound

- Wave types: transverse (light) & longitudinal (sound).
- $v = f\lambda$
- Reflection: angle in = angle out.
- Refraction: bends **towards** normal in denser medium.
- Sound: measured in Hz; needs medium; echoes = distance/2 $\times$ time.

6. Light

- Ray diagrams for reflection/refraction/lenses.
- Concave = converging; convex = diverging.
- Critical angle, total internal reflection (TIR), fiber optics.

7. Electricity

- Charge = current $\times$ time: $Q=It$
- Energy = voltage $\times$ charge: $E=VQ$
- Series: current same; total $R = R_1+R_2$.
- Parallel: voltage same; total R smaller.
- $V=IR$ (Ohm's Law).
- Safety: fuse, earth wire, circuit breaker.

8. Magnetism & Electromagnetism

- Magnetic materials: iron, cobalt, nickel.
- Right-hand rule for current $\rightarrow$ magnetic field.
- Motor effect: $F=BIL$
- Induction: moving conductor cuts field $\rightarrow$ current induced.
- Transformers: $V_p V_s = N_p N_s$

9. Atomic Physics (Radioactivity)

- Types: α ($2p+2n$), β (e^-), γ (wave).
- Effects: ionisation, penetration differences.
- Half-life = time for activity to halve.
- Safety: shielding, distance, time limits.

2. Weekly 15-Minute Practicals (home/school safe)

- Measure acceleration using a **ramp + stopwatch**.
- Find density of a solid/liquid with **mass + volume**.
- Measure current/voltage $\rightarrow$ plot **I-V graph**.
- Use **light box** for reflection/refraction angles.
- Sound echoes: clap $\rightarrow$ stopwatch $\rightarrow$ distance.
- Pendulum timer: find **T^2 vs L graph** $\rightarrow$ slope = $4\pi^2 g$

Always follow lab safety: goggles, no wet hands near electricity, careful with hot lamps.

3. Calculation Drills

Practise these daily:

Topic	Formula	Unit
Speed	$v = \frac{s}{t}$	m/s
Acceleration	$a = \frac{v-u}{t}$	m/s ²
Force	$F = ma$	N
Work	$W = Fd$	J
Power	$P = \frac{E}{t}$	W
Pressure	$p = \frac{F}{A}$	Pa
Density	$\rho = \frac{m}{V}$	kg/m ³
Voltage	$V = IR$	V
Energy	$E = Pt$ or $E = QV$	J

4. Key Graphs to Memorise

- **Distance–Time** → gradient = speed.
- **Velocity–Time** → gradient = acceleration; area = distance.
- **Hooke’s Law:** Force vs Extension (straight line up to limit).
- **I–V characteristics:** resistor (straight), filament lamp (curve).
- **Cooling curve:** flat during state change.

5. Concept Links (for better answers)

Concept	Always connect to...
Motion	Newton’s laws / unbalanced forces
Energy	Conservation & transfer type
Temperature change	Particle motion / energy gain
Sound	Vibrations / medium
Light	Reflection or refraction law
Electricity	Flow of charge per second
Magnetism	Field direction / current interaction

6. Vocabulary Bank

Word	What it means
Displacement	distance in a given direction
Velocity	speed with direction
Acceleration	rate of change of velocity
Efficiency	useful energy ÷ total energy
Refraction	change in direction as wave changes speed
Potential difference	energy transferred per charge
Resistance	opposition to current flow
Induced	produced by changing magnetic field
Ionisation	removal of electrons from atoms

7. Common Pitfalls & Fixes

Mistake	Fix
Mixing mass and weight	Weight = mg (N), mass = kg
Confusing speed vs velocity	Add <i>direction</i> for velocity
Forgetting units	Always write units in every step
Wrong reflection/refraction direction	Use ruler + label <i>normal</i>
Forgetting control variables in practicals	Identify CIV: Control / Independent / Dependent

8. Study Rhythm (45–60 min per day)

Day	Focus	Task
Mon	Formula practice	10 Qs mixed topics
Tue	Graph sketching	3 key graphs from memory
Wed	Past paper Section A (15 marks)	mark & review
Thu	Practical method	outline aim, variables,

		results
Fri	Theory review (light, waves, or radioactivity)	1 page notes
Weekend	Self-test (20 mins) + watch one physics demo video	

9. Quick Reference Diagrams (to sketch weekly)

1. Force diagrams (balanced vs unbalanced).
2. Light ray through prism (label angle of incidence, refraction).
3. Reflected wavefronts / ripple tank setup.
4. Circuit with ammeter/voltmeter positions.
5. Transformer (primary/secondary coils + core).

☐ Resources (free)

- **Physics** (coursebook / revision guide)
- **BBC Bitesize Physics** (GCSE)
- **Khan Academy – Physics playlists**
- **FuseSchool / Cognito / Physics Online** on YouTube
- **PhET Simulations** (<https://phet.colorado.edu>) – motion, waves, circuits.

✓Self-Check Before Any Test

- I can **rearrange formulas** correctly []
- I include **units and labels** on graphs []
- I can describe **energy transfers** in 3 examples []
- I can explain **force** → **acceleration** → **motion** using Newton's laws []
- I can draw **ray/circuit/field diagrams** from memory []
- I revise with **active recall** (not rereading!) []

(Year 11, 0625 / 0972)

Quantitative Mastery

Formula fluency — derive, rearrange, substitute, round.

Topic	Core Equation	Extended Links
Motion	$v = st$ $a = \frac{v-u}{t}$ $a = \frac{v^2 - u^2}{2s}$	SUVAT: $v^2 = u^2 + 2as$ $v = u + at$ $s = ut + \frac{1}{2}at^2$ $s = \frac{v^2 - u^2}{2a}$
Forces	$F = ma$	Momentum: $p = mv$ $F = \frac{\Delta p}{\Delta t}$ $F = \Delta p / \Delta t$
Energy	$E_k = \frac{1}{2}mv^2$ $E_p = mgh$	Work-energy transfer equivalence
Power	$P = \frac{E}{t}$ $P = IV$ $P = I^2R$	Efficiency % = (useful / total) × 100
Density	$\rho = \frac{m}{V}$ $\rho = \frac{m}{V}$	Floating → weight = upthrust
Pressure	$p = \frac{F}{A}$	Liquids: $p = h\rho g$
Electricity	$V = IR$ $Q = It$ $E = VIt$	Resistivity ρ : $R = \rho \frac{L}{A}$ $R = \rho AL$
Waves	$v = f\lambda$	Reflection/refraction law; Snell: $n = \frac{\sin i}{\sin r}$
Magnetism	$F = BIL \sin \theta$	Induction → $\epsilon = Blv$
Gases	$p_1V_1 = p_2V_2$	$pV/T = \text{constant}$

Practice drill: one 10-min session daily — pick 3 random formulas, rearrange for a new variable, add correct SI units.

2. Data & Graph Work

- **Gradients** → rate or constant; **area under curve** → change in quantity.
- **Axes:** include symbol + unit.
- **Uncertainty:** $\frac{1}{2}$ range of repeats; % error = (uncertainty / reading) × 100.
- **Best fit:** ignore outliers only if clearly justified.
- **Proportional reasoning:**
 - Straight line through origin → directly proportional.
 - Constant area/volume → inverse relationship.

3. Advanced Mechanics

- **Momentum conservation:** isolated system → $\Sigma p(\text{before}) = \Sigma p(\text{after})$.

- Elastic → KE conserved; Inelastic → lost to heat/sound.
- **Moments:** $M = Fd$; equilibrium when $\Sigma_{\text{clockwise}} = \Sigma_{\text{anticlockwise}}$.
- **Centre of mass:** balance point — affects stability.
- **Circular motion:** $F = \frac{mv^2}{r}$; direction of force → towards centre.
- **Free-fall:** all bodies accelerate at $\approx 9.8 \text{ m/s}^2$ (ignore air resistance).

4. Energy & Thermal Physics

- **Work done = energy transferred** ($1 \text{ J} = 1 \text{ N m}$).
- **Specific heat capacity:** $E = mc\Delta T$.
- **Latent heat:** $E = mL$.
- Distinguish **heating** (energy in) vs **temperature change** (particle KE).
- **Conduction:** solids via lattice vibrations/electrons.
- **Convection:** fluids via density difference.
- **Radiation:** infrared — no medium; black/dull surfaces best absorbers.

5. Waves — Master Patterns

- **Superposition & interference:** constructive → path diff = $n\lambda$; destructive → $(n + \frac{1}{2})\lambda$.
- **Diffraction** ↑ when $\lambda \approx$ slit width.
- **Refraction:** speed $\propto 1 / \text{refractive index}$.
- **Total internal reflection:** occurs if $i >$ critical angle & going from denser → rarer.
- **Sound:** longitudinal; oscilloscope traces amplitude ↔ loudness, frequency ↔ pitch.

6. Electricity & Circuits (Extended detail)

Concept	Key Relation	Notes
Ohm's Law	$V \propto I$ (if T constant)	non-ohmic: filament, diode
Resistance	$R = \frac{V}{I}$	series → $R_{\text{eq}} = R_1 + R_2$; parallel → $\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2}$
Power	$P = VI = I^2 R = \frac{V^2}{R}$	label fuse by current rating slightly > operating I
Energy	$E = VIt$	convert kWh ↔ J ($1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$)
Magnetism	right-hand grip (current → field)	Fleming's LH

		rule (motor), RH rule (generator)
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7. Electromagnetism, Induction & Transformers

- **Motor effect:** $F = BIL \sin \theta$; increase B, I, or length in field.
- **Generator effect:** moving conductor in B field $\rightarrow$ induced emf.
- **Transformer:** $\frac{V_p}{V_s} = \frac{N_p}{N_s}$
 - Ideal $\rightarrow$ 100 % efficient; real losses = eddy currents + heating.
- **Transmission:** high-voltage $\rightarrow$ low current $\rightarrow$ less I^2R loss.

8. Atomic & Nuclear Physics

- **Isotopes:** same Z, different A.
- **Ionisation:** $\alpha > \beta > \gamma$ (in effectiveness).
- **Half-life:** time for activity $\rightarrow \frac{1}{2}$; independent of initial count.
- **Safety:** limit exposure (time $\downarrow$, distance $\uparrow$, shielding $\uparrow$).
- **Fission:** chain reaction; control rods absorb n.
- **Fusion:** small nuclei combine; huge energy, needs extreme T.

9. Planning, Evaluation, Improvement (Paper 5/6 skills)

1. **Aim:** what variable relationship?
2. **Variables:** identify I/D/C.
3. **Method:** numbered steps + apparatus + safety.
4. **Data:** include **units, repeats, average**.
5. **Analysis:** graph $\rightarrow$ relationship $\rightarrow$ explain pattern using physics law.
6. **Evaluation:** key sources of error + realistic improvements (better resolution, repeat readings, insulation, alignment).

□ Weekly Challenge Cycle (6 weeks)

Week	Focus	Output
1	Kinematics & forces	Graph analysis task
2	Energy & efficiency	Real-life energy audit
3	Waves & optics	Light experiment write-up
4	Circuits & electricity	Design & simulate on PhET
5	Magnetism & induction	Mini-poster on applications
6	Radioactivity	Half-life data analysis

□ Physics — Student Guide

(0625 / 0972)

Exam Map

Route	Paper	Focus	Time	Marks	Weight
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Core	1	Multiple choice	45 min	40	30 %
	3	Theory (short + structured)	1 h 15 min	80	50 %
	5 / 6	Practical / Alt to Practical	1 h	40	20 %
Extended	2	Multiple choice	45 min	40	30 %
	4	Extended theory	1 h 15 min	80	50 %
	5 / 6	Practical / Alt to Practical	1 h	40	20 %

You take either **1 + 3 + 5 (Core)** or **2 + 4 + 5/6 (Extended)**.

2. Command Words

Word	What to Do
State / Define	short fact or definition
Describe	what happens / trend
Explain	give reason — link to law or particle model
Calculate	show formula → substitute → unit
Sketch	shape, axes + labels, no numbers needed
Determine	measure from graph (gradient/area)
Predict / Deduce	apply pattern or rule
Evaluate / Suggest improvements	discuss reliability & error reduction

3. Timing Plan

Paper	Task	Time
MCQ (1/2)	40 Q × 1 min	45 min
Theory (3/4)	80 marks ≈ 1 mark/min + 5 min check	1 h 15 min
Practical (5/6)	Plan 5 min, results 35 min, analysis 20 min	1 h

4. Equations Sheet (must-know)

(memorise even if formula list given)

- $v = \frac{s}{t}$ $v = \frac{s}{t}$ $a = \frac{v - u}{t}$ $a = \frac{v - u}{t}$ $F = ma$ $F = ma$ $F = ma$
- $p = \frac{F}{A}$ $p = \frac{F}{A}$ $W = Fd$ $W = Fd$ $E = Pt$ $E = Pt$ $E = Pt$
 $P = \frac{W}{t}$ $P = \frac{W}{t}$ $P = \frac{W}{t}$
- $\rho = \frac{m}{V}$ $\rho = \frac{m}{V}$ $pV = \text{const.}$ $pV = \text{const.}$ $pV = \text{const.}$
- $E_k = \frac{1}{2}mv^2$ $E_k = \frac{1}{2}mv^2$ $E_p = mgh$ $E_p = mgh$ $E_p = mgh$
 $v = f\lambda$ $v = f\lambda$ $v = f\lambda$
- $n = \frac{\sin i}{\sin r}$ $n = \frac{\sin i}{\sin r}$ $Q = It$ $Q = It$ $Q = It$
 $E = VIt$ $E = VIt$ $E = VIt$
- $F = BIL$ $F = BIL$ $F = BIL$ $V_p V_s = N_p N_s$ $\frac{V_p}{V_s} = \frac{N_p}{N_s}$ $V_s V_p = N_s N_p$

5. Practical-Paper Essentials

Tables: headings + units; consistent sig. figs.

Graphs: >½ page; axes labels; best-fit line; slope calc.

Conclusion: direct link between variables + law.

Evaluation: identify largest uncertainty; give **numerical improvement** (e.g., “repeat $\times 3 \rightarrow$ reduce random error by $\sqrt{n}$ ”).

6. Frequent Errors & Fixes

Mistake	Quick Fix
Mixing mass/weight	Weight = mg (N)
Missing units	Always add after every number
Wrong scale or axes	Use full grid; equal intervals
“More collisions” (rates analogy)	add “with enough energy to overcome E_a ”
Forgetting direction	Add <i>vector</i> arrows & labels

7. High-Band Writing Frames

Explain motion:

When the resultant force acts, the object accelerates because $F=ma$.

Energy question:

Energy is conserved; gravitational potential converts to kinetic.

Practical improvement:

Use digital stopwatch (± 0.01 s) instead of manual to reduce timing error.

Magnetism:

The current in the wire creates a magnetic field; by the LH rule the force acts...

8. Weekly Routine (40 min $\times$ 5)

Day	Focus	Task
Mon	Formula practice	10 Qs mixed units
Tue	Graphs/data	Determine gradient or area
Wed	Theory	2 extended questions
Thu	Practical skills	Write short plan + eval
Fri	Review	Flashcards + self-quiz

9. Final 1-Week Exam Sprint

- Re-write all formulas from memory.
- Redo **3 past papers** timed.
- Review **error log** $\rightarrow$ fix misconceptions.
- Practise **data-to-graph** and **explain-pattern** questions.
- Sleep, hydrate, steady breathing — clarity > cramming.

Quick Reference

Units: N, J, W, V, A, Ω , kg m^{-3} , m/s, Pa.

Prefixes: m (10^{-3}), k (10^3), M (10^6).

Constants: $g = 9.8 \text{ m/s}^2$ (≈ 10), speed of light = $3 \times 10^8 \text{ m/s}$.

MATHEMATICS

Basic Supercurriculum

Why Supercurriculum?

Mathematics builds **logical thinking, problem-solving, and numerical confidence** — essential in science, technology, and everyday life.

Going beyond the syllabus means practising reasoning, pattern recognition, and mental agility, not just formulas.

Core Knowledge — Key Topics

☐ Number

- Four operations with integers, fractions, decimals, and percentages.
- Powers, roots, reciprocals.
- Rounding, significant figures, estimation.
- Ratio, rate, proportion, and scale.
- Percentage change, profit/loss, simple & compound interest.
- Standard form (scientific notation) and order of operations (BIDMAS).
- Time, money, units, metric conversions.

☐ Algebra

- Simplify, expand, factorise expressions.
- Use algebraic fractions and indices.
- Solve linear and quadratic equations.
- Substitute values into expressions and formulae.
- Form equations from word problems.
- Plot and interpret straight-line and quadratic graphs.
- Change subject of a formula.

☐ Geometry

- Angles (lines, triangles, polygons, circles).
- Similarity, congruence, scale factor.
- Pythagoras' theorem and basic trigonometry (SOH-CAH-TOA).
- Circle properties (tangent, chord, arc, sector).
- Bearings, loci, and constructions.

☐ Mensuration

- Perimeter, area, and volume of 2D and 3D shapes.
- Surface area and volume of cuboids, cylinders, cones, and spheres.
- Density, mass, and volume relationships.
- Use of π in calculations.

☐ Coordinate Geometry

- Gradient $m = \frac{y_2 - y_1}{x_2 - x_1}$
- Equation of a line: $y = mx + c$
- Midpoint, distance between two points.
- Parallel and perpendicular lines.

☐ Trigonometry

- Right-angled: SOH-CAH-TOA.
- Use of sine, cosine, tangent buttons on calculator.
- Find missing sides and angles.
- Angle of elevation/depression.

2 Statistics

- Mean, median, mode, range.
- Frequency tables, bar charts, histograms, pie charts.
- Cumulative frequency and box plots.
- Probability: fractions, decimals, percentages, sample space diagrams.

2 Vectors & Transformations

- Column vectors, addition, subtraction, scalar multiples.
- Reflection, rotation, translation, enlargement.

2. Weekly Skills Practice (30–45 min/day)

Day	Focus	Task
Monday	Number & Arithmetic	10 short calculations; aim for speed & accuracy.
Tuesday	Algebra	5 simplifications + 2 equations.
Wednesday	Geometry/Trigonometry	3 diagram questions.
Thursday	Graphs/Functions	Draw or interpret one graph.
Friday	Statistics/Probability	1 data set + 3 probability questions.
Weekend	Review	10 mixed questions + mark scheme check.

3. Core Formulas (memorise & use correctly)

Topic	Formula	Units / Notes
Area of triangle	$A = \frac{1}{2}bh$	base × height
Circle area	$A = \pi r^2$	r = radius
Circumference	$C = 2\pi r$	
Rectangle area	$A = lw$	
Volume (cuboid)	$V = lwh$	
Cylinder volume	$V = \pi r^2 h$	
Cone volume	$V = \frac{1}{3}\pi r^2 h$	
Sphere volume	$V = \frac{4}{3}\pi r^3$	
Pythagoras	$a^2 + b^2 = c^2$	c = hypotenuse
Trig ratios	$\sin\theta = \frac{\text{opp}}{\text{hyp}}, \cos\theta = \frac{\text{adj}}{\text{hyp}}, \tan\theta = \frac{\text{opp}}{\text{adj}}$	
Gradient	$m = \frac{y_2 - y_1}{x_2 - x_1}$	
Distance	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	
Density	$\rho = \frac{m}{V}$	
Speed	$v = \frac{d}{t}$	

4. Calculation Habits

- Write **working clearly** (each step = 1 mark).
- Always state **units** (especially in geometry & measurement).
- Round **only at the end**; use at least 3 significant figures.
- Label diagrams neatly and underline final answers.
- Estimate before using calculator (check reasonableness).

5. Problem-Solving Strategy

R → **P** → **S** → **C** → **E**

Read → **Plan** (what's asked & what's given) → **Solve** → **Check units** → **Explain** (sentence answer if needed).

Example:

Q: A cone has radius 4 cm and height 9 cm. Find volume.

A: $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \times 4^2 \times 9 = 48\pi \approx 151 \text{ cm}^3$
 $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \times 4^2 \times 9 = 48\pi \approx 151 \text{ cm}^3$

6. Mental Fluency Boosters

- Square numbers 1–15.
- Common fractions ↔ decimals ↔ percentages ($\frac{1}{4} = 0.25 = 25\%$).
- 3×, 4×, 6×, 8×, 9×, 12× tables instant recall.
- Powers of 10 conversions (mm–cm–m–km).
- Quick percent: $10\% = \div 10$; $5\% = \div 20$; $1\% = \div 100$.

7. Common Pitfalls & Fixes

Pitfall	Fix
Missing brackets in calculator work	Always check order (BIDMAS).
Mixing formulas (area vs volume)	Write the formula first , then substitute.
Angle in wrong mode	Calculator in DEG for trig.
Forgetting label/units	Write units every step (cm^2 , cm^3 , km/h , etc.).
Poor scale on graphs	Use full grid; label axes clearly.

8. Study Rhythm Example

Time	Activity	Outcome
0–10 min	Review notes / formulas	Recall accuracy
10–25 min	Solve 5 mixed past-paper questions	Application
25–35 min	Check answers & mark	Error awareness
35–45 min	Redo mistakes without notes	Retention

9. Recommended Resources

- **Mathematics (Coursebook & Revision Guide)**
- **Corbett Maths** (videos & worksheets)
- **BBC Bitesize – GCSE Maths** (short reviews)
- **Maths Genie / IXL / Khan Academy** (practice by topic)

- **Past Papers & Mark Schemes** (official Cambridge site)

Self-Check Before a Test

- ☐ I can rearrange equations correctly.
- ☐ I use correct formula & show full working.
- ☐ My calculator is in **DEG** mode.
- ☐ I include **units & 3 s.f.** answers.
- ☐ I've done **one timed past paper** this week.
- ☐ I rechecked my **graph accuracy** and **scale labels**.

1. Aim

To go beyond memorisation — learning to **think mathematically**: reason, connect, and justify.

You'll build mastery across number, algebra, geometry, trigonometry, statistics, and problem-solving through consistent practice and reflection.

Algebraic Mastery

A. Core Principles

- Factorisation patterns:
 - $a^2 - b^2 = (a+b)(a-b)$
 - $a^2 + 2ab + b^2 = (a+b)^2$
 - $a^2 - 2ab + b^2 = (a-b)^2$
- Expand: double brackets and binomials with powers (e.g., $(x+a)^2$, $(x+a)(x+b)$).
- Factorise: common factors, quadratics, difference of squares, grouping.

B. Quadratics

- Solve by:
 - Factorising** → simple roots.
 - Formula**: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
 - Completing the square**.
- Interpret graphically: roots = x-intercepts.
- Discriminant $b^2 - 4ac$: > 0 → two roots, $= 0$ → one root, < 0 → none.

C. Functions & Graphs

- Gradient $m = \frac{y_2 - y_1}{x_2 - x_1}$; intercept form $y = mx + c$.
- Parallel lines → same gradient.
- Perpendicular → product of gradients = -1 .
- Quadratic graphs: identify **axis of symmetry** and **turning point**.
- Transformations: $y = f(x) + a$, $y = f(x) - a$, $y = f(x + a)$, $y = f(x - a)$, $y = -f(x)$, $y = f(-x)$.

2. Geometry & Trigonometry (Extended)

A. Pythagoras & Basic Trig

- $a^2 + b^2 = c^2$, $\sin \theta = \frac{\text{opp}}{\text{hyp}}$, $\cos \theta = \frac{\text{adj}}{\text{hyp}}$, $\tan \theta = \frac{\text{opp}}{\text{adj}}$.
- Angles of elevation/depression**: horizontal reference line always drawn first.

B. Advanced Trigonometry

- Sine rule**: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

- **Cosine rule:** $c^2 = a^2 + b^2 - 2ab \cos C$
 $Cc^2 = a^2 + b^2 - 2ab \cos C$.
- **Area of triangle:** $\frac{1}{2}ab \sin C$
- Use 3 s.f. accuracy; calculator in DEG mode.

C. Circles & Geometry Proofs

- Tangent $\perp$ radius.
 - Angle subtended by diameter = 90° .
 - Opposite angles in cyclic quadrilateral = 180° .
 - Angle at centre = $2 \times$ angle at circumference (same arc).
 - Equal chords $\rightarrow$ equal arcs.
- Task:** prove one circle theorem each week using labelled diagrams.

3. Algebraic Fractions & Inequalities

- Simplify: cancel common factors after factorising numerator/denominator.
- Add/subtract: find LCM of denominators.
- Solve inequalities graphically and algebraically:
 - Linear: $2x+3 < 11$ $2x+3 < 11$ $2x+3 < 11$.
 - Quadratic: sketch curve, identify x-regions.
 - Represent on number lines.

4. Coordinate & Vector Geometry

- **Midpoint:** $M(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$
- **Distance:** $d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$
- **Vectors:**
 - Column form $\begin{bmatrix} x \\ y \end{bmatrix}$
 - Addition/subtraction = component-wise.
 - Scalar multiplication:
 $k \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} kx \\ ky \end{bmatrix}$
 - Parallel vectors $\rightarrow$ one is a scalar multiple of the other.
 - $AB = OB - OA$ (position vectors).

5. Mensuration & 3D Problems

- Solve **compound shapes** (e.g., cylinder + hemisphere).
- Convert between cm^3 and litres ($1 \text{ dm}^3 = 1 \text{ L} = 1000 \text{ cm}^3$).
- Work with **truncated cones**, **spheres**, and **pyramids** (surface area + volume).
- Apply **similar triangles** to find missing heights/lengths.

6. Statistics & Probability (Extended)

- **Mean** = $\Sigma fx / \Sigma f$; **Mode** from highest frequency; **Median** via cumulative freq graph.

- **Histograms:** frequency density = frequency $\div$ class width.
- **Probability:**
 - $P(A') = 1 - P(A)$ $P(A') = 1 - P(A)$
 - $P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$ $P(A \text{ \text{or} } B) = P(A) + P(B) - P(A \cap B)$
 - $P(A \text{ and } B) = P(A) \times P(B)$ $P(A \text{ \text{and} } B) = P(A) \times P(B)$ if independent.
- **Tree diagrams** for multi-stage events.
- **Estimate mean:** midpoint $\times$ frequency $\div$ total frequency.

7. Problem-Solving & Modelling

- Identify **type of problem** (direct proportion, quadratic, simultaneous, etc.).
- Create **equations from context:**
 - “A number increased by 20%” $\rightarrow \times 1.2$.
 - “Speed directly proportional to $\sqrt{\text{height}}$ ” $\rightarrow v = k\sqrt{h}$ $v = k\sqrt{h}$ $v = k\sqrt{h}$.
- Check if answers are **reasonable** (positive, realistic units).
- Use **units and rounding** consistently.

8. Graphical Interpretation

- **Linear:** gradient = rate of change.
- **Quadratic:** symmetrical about vertex; minimum/maximum point.
- **Exponential:** growth/decay curves.
- **Distance–time:** gradient = speed.
- **Velocity–time:** gradient = acceleration; area = distance.

9. Logical Communication

- Justify each step in reasoning:
 - Algebra $\rightarrow$ “because” statements.
 - Geometry $\rightarrow$ reason beside each step.
 - Probability $\rightarrow$ total = 1, independent = multiply.
- Always finish with a **sentence conclusion**, e.g.
Therefore, the height of the cone is **7.5 cm (3 s.f.)**.

Weekly Challenge Rotation (6 weeks)

Week	Theme	Output
1	Algebra & Graphs	Quadratic sketch & equation link

2	Geometry	Circle theorem proof
3	Trig & Mensuration	Real-world application problem
4	Statistics	Histogram & cumulative freq combo
5	Vectors	Geometry problem using vectors
6	Mixed Paper	Timed 40-min past paper

Mathematics — Student Guide
(Year 11, 0580 / 0980)

Exam Map

Paper	Level	Description	Time	Weight
1	Core	Short-answer & structured	1h	35%
3	Core	Extended structured & problem-solving	2h	65%
2	Extended	Short-answer	1h 30	35%
4	Extended	Extended problem-solving	2h 30	65%

Core → grades up to C; Extended → up to A*.

2. Command Words

Term	Meaning
Calculate	Show all steps, include units
Evaluate	Substitute and simplify
Simplify	Reduce to simplest form
Factorise	Write as product of factors
Solve	Find exact value(s) satisfying equation
Plot / Sketch	Accurate shape, key points labelled
Prove / Show	Justify every step logically
Estimate	Round inputs appropriately
Construct	Use compass/ruler with accuracy

3. Time Plan (Extended)

Section	Marks	Suggested Time
Paper 2	70–80	~1.5 min per mark
Paper 4	130–140	~1 min per mark
Review & Check	—	5–10 min

Attempt all questions — no negative marking.

4. Core Formula Bank (must memorise)

Geometry & Trig

- $a^2 + b^2 = c^2$
- $\sin \theta, \cos \theta, \tan \theta$ \sin, \cos, \tan ratios
- $\text{Area} = \frac{1}{2}ab \sin C$
- $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
- $c^2 = a^2 + b^2 - 2ab \cos C$
- Circle area: πr^2 , Circumference: $2\pi r$.

Algebra & Graphs

- Quadratic formula
- Gradient = $\frac{\Delta y}{\Delta x}$; $y = mx + c$
- Midpoint & distance formula
- Simultaneous equations (elimination/substitution).

Mensuration

- Volume cone: $\frac{1}{3}\pi r^2 h$
- Sphere: $\frac{4}{3}\pi r^3$
- Cylinder: $\pi r^2 h$

- Density: $m \times \frac{1}{V}$, Pressure: $F \times \frac{1}{A}$.

Statistics

- Mean = $\frac{\sum fx}{\sum f}$
- Frequency density = $\frac{f}{\text{class width}}$

5. Accuracy & Presentation

- 3 s.f. (unless stated otherwise).
- Angles → **1 d.p.** (°).
- Money → **2 d.p.** (£ or €).
- Label all graphs & diagrams.
- Use **ruler, protractor, compass** carefully (clean lines, clear markings).

6. Frequent Mistakes & Fixes

Mistake	Fix
Using calculator incorrectly	Write formula → substitute → THEN calculate
Rounding mid-step	Round only final answer
Wrong trig mode	Check DEG
Forgetting to square root	Check units: area → m ² , length → m
Missing units	Add to every numeric result
Skipping step	Every method = 1 mark

7. Practical Reasoning Checklist

Before each question:

1. **Identify** topic (Algebra? Trig? Graphs?).
2. **Recall** formula/law.
3. **Substitute** correctly (with units).
4. **Compute** accurately.
5. **Interpret** answer (e.g. “therefore the angle = 37°”).

8. Revision Cycle (4 Weeks)

Week	Focus	Output
1	Algebra & Graphs	2 timed questions/day
2	Geometry & Trig	10 mixed diagrams
3	Statistics & Probability	Histogram + probability tree
4	Mock Paper	1 full paper + review weak areas

9. Resource List

- **Mathematics Coursebook / Revision Guide**
- **Maths Genie, Corbett Maths, BBC Bitesize**
- **Khan Academy / IXL / SaveMyExams** for topic drills
- **Cambridge Past Papers & Mark Schemes**

Exam-Day Quick Tips

Read all instructions first.

Circle what the question *wants* (area? volume? gradient?).

Use clean layout; one method per line.

Double-check calculator mode + rounding.

Leave 5 minutes to scan units, labels, and working.

ADDITIONAL MATHEMATICS

Basic Supercurriculum

Why Supercurriculum?

Additional Mathematics develops **mathematical maturity**, **proof-based thinking**, and **advanced problem-solving** — bridging IGCSE and A Level. Students learn to reason algebraically, model real situations, and manipulate abstract forms confidently.

Core Knowledge – What You Must Master

Algebra

- **Simplifying expressions** using laws of indices and surds.
- **Factorisation** and **expansion** of polynomials (up to cubic).
- **Quadratic equations**:
 - Factorising, completing the square, using the formula
 - Discriminant $b^2 - 4ac$ → nature of roots.
- **Simultaneous equations** (linear + quadratic).
- **Rationalising denominators**.
- **Inequalities**: linear & quadratic → solution sets on number line.

Functions

- Function notation: $f(x)$, $f^{-1}(x)$, $f(g(x))$.
- Domain and range.
- Sketch simple functions: linear, quadratic, reciprocal, exponential, modulus.
- Transformations: $y = f(x+a)$, $y = af(x)$, $y = f(-x)$, $y = -f(x)$.
- Composite and inverse functions — find and interpret graphically.

Coordinate Geometry

- Gradient $m = \frac{y_2 - y_1}{x_2 - x_1}$, midpoint, distance.
- Equation of a line: $y = mx + c$.
- **Perpendicular bisectors** and **circle equations**:
 $(x-a)^2 + (y-b)^2 = r^2$
- **Tangent** to a circle: gradient $\times$ radius = -1 .

Geometry & Trigonometry

- Sine, cosine, tangent rules.
- Area of triangle $A = \frac{1}{2}ab \sin C$.
- Radian measure: arc length $= r\theta$, sector area $= \frac{1}{2}r^2\theta$.
- Graphs of sin, cos, tan; amplitude, period, phase shift.
- Trig identities:
 $\sin^2 x + \cos^2 x = 1$
 $\tan x = \frac{\sin x}{\cos x}$
- Solving trig equations: within $0-360^\circ$ or $0-2\pi$.

❑ Calculus (Basic)

- **Differentiation:**

- Power rule: $\frac{d}{dx}(x^n) = nx^{n-1}$
- Sum, difference, constant multiples.
- Gradient at a point, tangent/normal equations.
- Simple maxima/minima problems.

- **Integration:**

- Reverse of differentiation: $\int x^n dx = \frac{x^{n+1}}{n+1} + C$
- Area under curve between limits.
- Evaluate simple definite integrals.

❑ Vectors

- Vector notation: $\vec{a} = [x \ y]$
- Addition, subtraction, scalar multiples.
- Magnitude: $|\vec{a}| = \sqrt{x^2 + y^2}$
- Unit vector: $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$
- Position vectors and geometric interpretation.

❑ Statistics & Probability

- Mean, median, mode (ungrouped & grouped).
- Probability basics:
 - $P(A') = 1 - P(A)$
 - $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Simple combinatorics: permutations (nPr) and combinations (nCr).
- Expected value & variance (basic concept).

2. Key Formulas (memorise early)

Concept	Formula
Quadratic	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Discriminant	$\Delta = b^2 - 4ac$
Distance	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Midpoint	$M(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2})$
Circle	$(x - a)^2 + (y - b)^2 = r^2$
Gradient of tangent	$\frac{dy}{dx}$
Equation of tangent	$y - y_1 = m(x - x_1)$

Area under curve	$A = \int_a^b y \, dx$
Arc length	$s = r\theta$
Sector area	$A = \frac{1}{2}r^2\theta$
Magnitude of vector	(

3. Weekly Study Plan (45–60 min/day)

Day	Focus	Practice Task
Monday	Algebra & Functions	Simplify 3 expressions + 2 equations
Tuesday	Trigonometry	Sketch sin/cos/tan + 2 identities
Wednesday	Calculus	5 differentiation + 2 integration Qs
Thursday	Geometry	Circle or coordinate geometry Q
Friday	Problem Solving	Real-life modelling task
Weekend	Revision	1 past-paper question + formula drill

4. Mini Investigations (build conceptual depth)

- Explore how **changing parameters (a, b, c)** affects the shape of $y = ax^2 + bx + c$.
- Investigate **gradient function** of $y = x^n$.
- Compare **area under curve** with trapezium rule approximation.
- Model motion using **sine waves** for periodic phenomena.

5. Common Pitfalls & Fixes

Pitfall	Fix
Forgetting radians	Switch calculator to RAD for calculus/trig in radians
Wrong substitution into quadratic	Use brackets when substituting negatives
Misusing tangent/normal gradients	Tangent = dy/dx ; Normal = $-1/(dy/dx)$
Sign errors in trig identities	Always write $\sin^2 x + \cos^2 x = 1$ first
Forgetting +C in integration	Always include +C for indefinite integrals
Missing limits in area calc	Label clearly: aaa, bbb, definite integral

6. Study Rhythm for Retention

- 10 min** daily formula recall (without notes).
- 30 min** topic drills from textbook or past papers.
- 15 min** conceptual reflection:
 - “What does this derivative mean?”
 - “How does changing parameter a affect shape?”

7. Recommended Resources

- *Additional Mathematics* (Coursebook, Cambridge University Press)
- *KMF Learning Hub*, *Save My Exams*, *Maths Genie*, *ExamSolutions*
- *Dr Frost Maths* (A-level foundations)
- *BBC Bitesize Advanced Maths* (for quick reviews)

8. Self-Check Before Any Test

- ☐ I can differentiate and integrate power functions.
- ☐ I can solve quadratic and simultaneous equations accurately.
- ☐ I can find gradients, tangents, and normals correctly.
- ☐ I can apply sine/cosine rule and radian formulas.
- ☐ I can sketch basic functions and apply transformations.
- ☐ I check my calculator is in **DEG or RAD** as required.

9. Study Habits that Boost Results

- Always **show algebraic working**; never skip reasoning.
- Write formulas before substituting.
- Include **units or variables** in final answers.
- Use graph paper for sketches and transformations.
- Redo every wrong question twice: once with notes, once without.

🔍 Reflection Prompt

“Which topics connect most naturally?”

Link **differentiation** ↔ **gradient**, **integration** ↔ **area**, **trig** ↔ **circle geometry**, and **algebra** ↔ **functions**.

Additional Mathematics

Extended Supercurriculum – Year 11

1. Aim

To move from correct methods → *mathematical reasoning, modeling skill, and conceptual fluency*.

By mastering connections between algebra, calculus, and geometry, you'll be ready for A Level Pure Mathematics.

Algebra & Functions – Beyond the Basics

A. Higher Algebra

- **Cubic and polynomial equations:**
Factor by grouping or synthetic division (if one root is given).
- **Remainder Theorem:** If $f(a) = 0$, then $(x-a)$ is a factor.
- **Rational functions:** simplify and determine **asymptotes** and **domain restrictions**.
- **Partial fractions:**

$$\frac{3x+2}{(x+1)(x-2)} = \frac{A}{x+1} + \frac{B}{x-2}$$

$$3x+2 = A(x-2) + B(x+1)$$

$$3x+2 = Ax-2A+Bx+B$$

$$3x+2 = (A+B)x + (-2A+B)$$
Solve for A, B by substitution or equating coefficients.

B. Logarithms & Exponentials

- **Laws:**
 $\log_a(ab) = \log_a a + \log_a b$, $\log_a \frac{a}{b} = \log_a a - \log_a b$, $\log_a a^n = n \log_a a$.
 $\log_a(ab) = \log a + \log b$, $\log_a \frac{a}{b} = \log a - \log b$, $\log_a a^n = n \log a$.
- **Change of base:** $\log_a b = \frac{\log b}{\log a}$
- **Exponential models:**
 $N = N_0 e^{kt}$, where $k > 0$ for growth, $k < 0$ for decay.
- **Linearise data:**
Taking logs $\rightarrow \ln N = \ln N_0 + kt$

C. Function Transformations (precision)

- $y = f(x+a)$ → shift **left** by a.
- $y = f(x) + a$ → shift **up** by a.
- $y = af(x)$ → **stretch vertically** by a.
- $y = f(ax)$ → **compress horizontally** by a.
- Sketch modulus: $y = |f(x)|$

2. Calculus – Full Mastery

A. Differentiation (Advanced)

Function	Derivative
x^n	nx^{n-1}
e^x	e^x
$\ln x$	$1/x$
a^x	$a^x \ln a$
$\sin x, \cos x, \tan x$	$\cos x, -\sin x, \sec^2 x$

- **Product rule:** $\frac{d}{dx}(uv) = u'v + uv'$
- **Quotient rule:** $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v u' - u v'}{v^2}$
- **Chain rule:** $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

Applications:

- Tangent/normal equations.
- Stationary points: $f'(x) = 0$
- Nature test: $f''(x) > 0 \rightarrow \min, f''(x) < 0 \rightarrow \max$

B. Integration (Advanced)

Function	Integral
x^n	$\frac{x^{n+1}}{n+1} + C$
e^x	$e^x + C$
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$\frac{1}{x}$	$\ln x + C$

Techniques:

- **By substitution:** set $u = f(x)$, find $\frac{du}{dx}$
- **By parts:** $\int u \frac{dv}{dx} = u v - \int v \frac{du}{dx}$
Choose u = function that simplifies when differentiated.

Applications:

- Find area under curve $\int_a^b y \, dx$
- Find area between curves (upper – lower).
- Solve motion problems using $v = \frac{dx}{dt}$, $a = \frac{dv}{dt}$

3. Trigonometric Mastery

- **Double-angle identities:**
 $\sin 2A = 2 \sin A \cos A$
 $\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$
- **Product-to-sum:**
 $\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$
 $\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$
 $\cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$
- **Simplify and solve:** turn everything to $\sin$ and $\cos$ before solving.
- **Radian measure:** convert $^\circ \rightarrow \text{rad}$: multiply by $\frac{\pi}{180}$

4. Coordinate Geometry & Circles

- Equation of a tangent to circle at (x_1, y_1) : $x_1(x-a) + y_1(y-b) = r^2$.
- Intersections of lines and circles via simultaneous equations.
- Equation of **circle from general form**: complete the square.

5. Vectors (in 2D & 3D)

- Position vectors: $\vec{OP} = (x, y, z)$
- Vector line equation: $\vec{r} = \vec{a} + t\vec{b}$
- Dot product: $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$
- Perpendicular if $\vec{a} \cdot \vec{b} = 0$
- Projection of $\vec{a}$ on $\vec{b}$: $|\text{proj}_{\vec{b}} \vec{a}| = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$

6. Exponential & Logarithmic Modelling

- Exponential growth/decay**: $y = Ae^{kx}$
- Find A, k from data $\rightarrow$ use $\ln y = \ln A + kx$
- Half-life**: $t = \frac{\ln 2}{|k|}$
- Compound interest: $A = P(1 + \frac{r}{100})^n$

7. Proofs & Reasoning

- Prove trig identities (start with complex side, use standard laws).
- Prove inequalities by completing square.
- Prove gradients relationships ($\perp$ lines $\rightarrow$ product = -1).
- State assumptions clearly (e.g. $x > 0$ for logs).

8. Extended Problem-Solving Framework (PRIME)

Step	Meaning	Tip
P	Problem	Identify topic & what is asked
R	Recall	Formula or principle
I	Implement	Apply correctly with algebra
M	Manipulate	Simplify & check reasonableness
E	Evaluate	Interpret in context

9. Weekly Challenge Cycle (6 weeks)

Week	Focus	Task
1	Logarithms & exponentials	Linearise data & plot ln graph
2	Differentiation	Product, quotient, chain rule mix
3	Integration	Substitution & area problems
4	Trig equations	Solve $0-360^\circ$ + radians

5	Circle geometry	Tangent and line intersection
6	Combined paper	Timed mini past paper

📌 Recommended Enrichment

- **Khan Academy – Calculus 1 (introductory)**
- **ExamSolutions** (Cambridge AddMaths topics)
- **GeoGebra**: for graph transformations.
- **Desmos**: explore calculus visually.
- **Past Paper Challenge**: attempt Paper 2 under timed conditions twice per term.

Additional Mathematics — Student Guide

(0606 / 0980, Year 11)

Exam Structure

Paper	Time	Marks	Weight	Format
Paper 1	2 hours	80	50%	Short-answer & structured questions
Paper 2	2 hours	80	50%	Extended problem-solving

Each question tests multiple areas: algebra → calculus → graphs → trig → modelling.

2. Command Words

Word	Meaning
Show / Prove	Work logically to demonstrate result
Hence	Use previous result to continue
State	Give a concise answer (no working)
Find	Calculate accurately with method
Solve	Obtain exact or simplified value
Sketch	Show shape & key features (intercepts, turning points)
Differentiate / Integrate	Apply calculus techniques
Simplify / Express in form	Manipulate algebra into required structure

3. Time Management

Paper	Task	Target
1	80 marks	~1.5 min per mark
2	80 marks	~1.5 min per mark
—	Checking	Reserve 5–10 min

Tip:

If stuck after 2 min → move on, mark the question, and return later.

4. Formulae to Memorise (non-supplied)

Algebra

- $(a+b)^2 = a^2 + 2ab + b^2$
- $(a-b)^2 = a^2 - 2ab + b^2$
- $(a+b)(a-b) = a^2 - b^2$

Trig & Radians

- $\sin^2 x + \cos^2 x = 1$
- $\tan x = \frac{\sin x}{\cos x}$
- $\sin 2A = 2 \sin A \cos A$, $\cos 2A = \cos^2 A - \sin^2 A$
- Arc length $s = r\theta$, Area $= \frac{1}{2}r^2\theta$

Calculus

- $\frac{d}{dx}(x^n) = nx^{n-1}$
- $\int x^n dx = \frac{x^{n+1}}{n+1} + C$
- Chain, Product, Quotient rules.
- $\frac{d}{dx}(\sin x) = \cos x$, $\frac{d}{dx}(\cos x) = -\sin x$

5. Exam Strategy

- Read entire paper first; begin with confident questions.
- Show all algebraic steps (marks often hidden in method).

- Label diagrams, coordinates, and axes clearly.
- State final answer **in simplest form** (no decimals unless asked).
- Check calculus: differentiate $\rightarrow$ integrate back to test correctness.
- Write *reasons* for any step using identities or laws.

6. Common Mistakes & Fixes

Mistake	Fix
Calculator in DEG when question uses radians	Always check mode
Ignoring domain restrictions in inverse functions	Write " $x \neq 0$ " or similar
Forgetting +C in integration	Always add unless definite
Using $\ln$ instead of log base 10 (or vice versa)	Follow question notation
Dropping negative sign in differentiation	Recheck each step
Not showing substitution in product/chain rule	Write intermediate lines clearly

7. Revision Cycle (4 Weeks)

Week	Focus	Task
1	Algebra & Logs	Simplify & model data
2	Calculus	Mix of derivative/integral problems
3	Trig & Geometry	Equations + radians
4	Mixed Paper	Timed mock + reflection sheet

8. Resource List

- **Additional Mathematics Coursebook (CUP)**
- **ExamSolutions / Save My Exams / Physics & Maths Tutor**
- **Khan Academy – Derivatives & Integrals**
- **Desmos / GeoGebra** (graph exploration)
- **Past Papers (10+ years)** for exam familiarity.

9. Self-Check Before the Exam

- ☐ I can apply all 3 calculus rules correctly.
- ☐ I know logarithmic and exponential properties.
- ☐ I can derive equations of tangents/normals.
- ☐ I can integrate to find areas.
- ☐ I check mode (DEG/RAD) before trig work.
- ☐ I can linearise exponential data and interpret graphs.
- ☐ I revise all identities and can prove at least one from scratch.

📋 Final 60-Second Checklist

- ✓ Formula list memorised
- ✓ Graphs neatly labelled

- ✓Mode checked (DEG/RAD)
- ✓+C included where needed
- ✓Simplest form final answers
- ✓Logical flow of working

Biology

Basic Supercurriculum

Why Supercurriculum?

To move from **recalling facts** to **thinking like a biologist** — observing patterns, explaining processes, interpreting data, and evaluating investigations.

1) Core Knowledge — Must Master

1. Cell Biology

- Cell structure (animal/plant/bacterial), magnification formula, specialised cells.
- Diffusion, osmosis, active transport (direction, energy use, examples).

2. Biological Molecules

- Water, carbohydrates (starch, glucose), lipids, proteins; enzyme action (lock-and-key, temperature/pH effects).

3. Movement of Substances

- Surface area:volume, concentration gradients, partially permeable membranes.

4. Nutrition & Digestion

- Balanced diet, food tests (Benedict's, iodine, biuret, ethanol).
- Human alimentary canal; absorption in ileum (villi adaptations).

5. Plant Nutrition

- Photosynthesis word/symbol equation, limiting factors, leaf adaptations, mineral ions (nitrate, magnesium).

6. Transport

- Plants: xylem (transpiration, cohesion-tension), phloem (translocation).

- Humans: heart, double circulation, blood components, lymphatic system (basics).

7. Respiration

- Aerobic vs anaerobic; lactic acid in muscles; gas exchange in alveoli (features).

8. Coordination & Response

- Nervous system (reflex arc), synapses; endocrine basics (adrenaline, insulin); tropisms in plants.

9. Reproduction

- Asexual vs sexual; human reproductive system; menstrual cycle outline; STI awareness.
- Plants: pollination, fertilisation, seed/germination conditions.

10. Inheritance & Variation

- DNA → genes → chromosomes; dominant/recessive, monohybrid crosses, phenotypes/genotypes; mutations (sources).

11. Ecology & Environment

- Food chains/webs, pyramids, energy loss; cycles (water, carbon, nitrogen basics); population changes; biodiversity and conservation.

12. Human Impacts

- Pollution (air, water, eutrophication), deforestation, sustainability.

2) Do-Tank — 15-Minute Mini Practicals (home/school-friendly)

- **Osmosis in potatoes:** mass change in sugar solutions → plot % change vs concentration.
- **Enzyme rate demo:** catalase (yeast) + hydrogen peroxide → count bubbles vs temperature.
- **Transpiration model:** plastic bag over leaf → measure water loss.
- **Food tests:** starch, glucose, protein, lipid on known samples (safety first).
- **Pulse & exercise:** resting vs post-exercise → recovery curve.

Always follow lab safety: goggles, small quantities, label, dispose correctly.

3) Data & Calculation Mini-Drills

- **Magnification:** $\text{Magnification} = \frac{\text{image size}}{\text{actual size}}$ Magnification = actual size / image size (use scale bars; convert units).
- **Percentage change:** $\frac{\text{final} - \text{initial}}{\text{initial}} \times 100\%$
- **Rates:** change ÷ time (e.g., bubbles/min, mm/day).
- **Food energy (simple):** mass of water × temp rise × 4.2 J g⁻¹ °C⁻¹ (calorimetry demo).
- **Surface area:volume** comparisons (cubes).

4) Diagrams to Sketch from Memory (weekly)

- Animal/plant cell (labels + functions).

- Alveolus with capillary (diffusion path).
- Heart with flow arrows; valve names.
- Neurone + synapse; reflex arc.
- Leaf cross-section; stomata.
- Villi; root hair cell.
- Food web with trophic levels.

5) “Because Biology” Phrases (exam power lines)

- **Diffusion/osmosis:** “moves **down** a concentration/water potential gradient” (osmosis through partially permeable membrane).
- **Enzymes:** “rate changes because enzyme **shape/active site** affected; beyond optimum → **denaturation.**”
- **Gas exchange:** “large surface area, thin moist surface, rich blood supply → **steep gradient** maintained.”
- **Circulation:** “double circuit allows **high pressure** to body, **lower** to lungs.”
- **Energy transfer:** “only ~10% passed on; losses via **respiration/heat, excretion, uneaten parts.**”

6) Common Pitfalls & Fixes

Pitfall	Fix
Mixing diffusion & osmosis	Osmosis = water only via partially permeable membrane
Saying “particles want to move”	Use random movement + net movement down gradient
Enzyme graphs with vague words	Use optimum, denature, active site shape
Heart/circulation arrows wrong	Practise flow: body → right → lungs → left → body
Confusing xylem/phloem	Xylem up (water/minerals), Phloem both ways (sugars)
Genetics terms mixed	Gene vs allele; genotype letters, phenotype trait

7) Weekly Study Rhythm (45–60 min/day)

Day	Focus	Task
Mon	Content recall	15 flashcards (spaced repetition)
Tue	Diagram practice	Sketch + label 2 key diagrams
Wed	Data & maths	6 calculations (magnification/rate/% change)
Thu	Past-paper shorts	8–10 marks mixed topics
Fri	Practical skills	Write a plan (aim, variables, method, risks)
Weekend	Extension	Watch one bio demo/animation + 5 bullet notes

8) Practical/Investigation Writing (Paper 5/6 style habits)

- **Aim** (what relationship?), **Variables** (CIV: control/independent/dependent).
- **Method** (bullets; volumes, times, apparatus).
- **Data** (tables with units; repeats + mean).
- **Graphs** (axes with units; line of best fit; anomalies circled).
- **Conclusion** (pattern + biological reason).
- **Evaluation** (limitations + improvements: e.g., control temperature, use water bath, use same age/size of organisms, more repeats).

9) Vocabulary Bank (learn precisely)

- **Active transport** (energy-using, against gradient, carrier proteins).
- **Denature** (irreversible change in active site).
- **Homeostasis** (maintaining internal conditions).
- **Transpiration/Translocation** (water vs sugars).
- **Pathogen/Antibody/Antigen** (immune basics).
- **Eutrophication** (nutrient run-off → algae bloom → oxygen drop).
- **Mutation** (change in DNA base sequence).

📖 Quick Case Studies (one per week)

- **Exercise & ventilation** (CO₂ increase → rate & depth).
- **Insulin & blood glucose** (pancreas, liver, negative feedback).
- **Nitrate deficiency in plants** (growth stunted, yellow leaves).
- **Vaccination** (primary vs secondary response).
- **Carbon cycle & deforestation** (sources/sinks).

11) Free, Reliable Resources

- **Biology Coursebook/Revision Guide** (your series).
- **BBC Bitesize Biology (GCSE)** — quick topic refreshers.
- **S-cool / Save My Exams / NotesNinja** — summaries + questions.
- **Khan Academy / Amoeba Sisters** — clear animations.
- **Past papers & mark schemes** — gold standard practice.

12) Self-Check Before a Test

- I can **define** key terms precisely (osmosis, active transport, transpiration) []
- I can **label** all core diagrams from memory []
- I can **calculate** magnification, % change, and simple rates []
- I can **explain** processes using correct mechanisms (e.g., enzyme denaturation) []
- I can **plan/evaluate** a simple investigation (variables, controls, improvements) []
- I have done **one timed past-paper section** this week []

Biology – Extended Supercurriculum

(Year 11, 0610 / 0970)

1. Aim

To go beyond content recall — learning to **analyse**, **evaluate**, and **synthesise** biological information.

Top-grade students *explain* why processes happen, interpret unfamiliar data, and connect molecular detail with real-world systems.

Advanced Understanding & Thinking

A. Cell Biology & Biochemistry

- Compare **plant**, **animal**, **fungus**, **bacterial** and **viral** structures.

- **Microscopy:** calculate magnification from image or scale bar; interpret micrographs.
- **Enzymes:** describe *lock-and-key* and *induced fit* models; explain how temperature and pH alter **hydrogen bonds** and **active site shape**.
- **Diffusion vs osmosis vs active transport:** design experiments to test each (potato chips, dyes, root hairs).

B. Biological Molecules

- Link structure → function:
 - Carbohydrates → energy (starch/glycogen).
 - Lipids → insulation, membranes.
 - Proteins → enzymes, structure, hormones.
- **Test results:**
 - Starch → iodine → blue-black
 - Glucose → Benedict's → orange-red
 - Protein → Biuret → lilac
 - Lipid → ethanol + water → cloudy white.

2. Human Physiology (Depth)

System	Key Details	Higher-Level Analysis
Digestive	Enzymes: amylase, protease, lipase	Conditions (pH, temperature); enzyme inhibition
Circulatory	Double system, heart structure	Effect of exercise, coronary disease causes/prevention
Respiratory	Alveoli adaptations	Role of cilia, smoking effects, emphysema
Excretory	Kidney: filtration, reabsorption	ADH role (negative feedback), dialysis vs transplant
Nervous	Neurones, synapse	Electrical → chemical → electrical transmission
Endocrine	Insulin, adrenaline	Compare nervous vs hormonal coordination

3. Plants & Transport Systems

- **Xylem:** lignified walls, cohesion–tension, transpiration pull.
- **Phloem:** translocation (source → sink), active loading.
- **Investigations:** effect of humidity/wind/temp on transpiration rate.
- **Adaptations:** xerophytes (thick cuticle, small leaves), hydrophytes (stomata on upper surface).

4. Gas Exchange & Respiration

- **Equation:**

Aerobic: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$

Anaerobic:

- Humans: $C_6H_{12}O_6 \rightarrow 2C_3H_6O_3 + \text{energy}$
- Yeast: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$
- **Gas exchange:** alveoli & stomata comparison; link structure–function.
- **Practical:** limewater CO_2 test, hydrogen carbonate indicator, effect of exercise.

5. Genetics, Variation & Evolution

- **DNA structure:** double helix, complementary base pairing (A–T, C–G).
- **Gene** → **protein:** transcription → translation (RNA & ribosomes).
- **Monohybrid inheritance:**
 - Punnett squares → ratios.
 - Genotype vs phenotype, dominant/recessive alleles.
- **Sex determination:** XX, XY; chance = 50%.
- **Mutation:** causes (radiation, chemicals), beneficial vs harmful, sickle-cell as example.
- **Variation:** genetic vs environmental; continuous vs discontinuous.
- **Natural selection:** adaptation → survival → reproduction → evolution.

6. Ecology & Human Impact (Applied Biology)

- **Energy flow:** 10% rule; energy losses at each trophic level.
- **Pyramids:** of numbers, biomass, energy (draw & interpret).
- **Nutrient cycles:** carbon & nitrogen (decomposers, nitrifying/denitrifying bacteria).
- **Population graphs:** lag, log, stationary, death phases.
- **Biodiversity:** importance + threats (deforestation, pollution, global warming).
- **Sustainable development:** renewable energy, recycling, crop rotation.

7. Experimental Design & Evaluation (Paper 5/6 Core)

A. Planning

- **Aim:** specific & measurable.
- **Variables:**
 - Independent (change)
 - Dependent (measure)
 - Control (keep constant).
- **Method:** numbered, includes safety & accuracy.

B. Data Handling

- **Tables:** clear headings, units, means.
- **Graphs:**
 - Line: continuous data
 - Bar: discrete data
 - Axes labelled + units

- Use >50% of grid
- Identify anomalies
- Draw best-fit line/curve.

C. Evaluation

Error Source	Improvement
Measuring volume inaccurately	Use pipette/burette
Temperature changes	Use water bath
Inconsistent specimens	Same age/size species
Human timing error	Use datalogger or same observer
Uncontrolled light	Use lamp + light meter

Conclusion = *Pattern* + *Scientific Reason*.

8. Skills for Top Marks

- Use **scientific terminology** precisely.
- Connect **process** → **structure** → **function** → **evidence**.
- State **units, labels, and directions** clearly.
- Evaluate **method reliability**, not just repeatability.
- Support all reasoning with **data trends**, not opinions.

9. Weekly Challenge Rotation (6 Weeks)

Week	Focus	Output
1	Cell & Enzymes	Plan + evaluate practical
2	Human Physiology	Data interpretation task
3	Genetics	3 inheritance crosses + reflection
4	Ecology	Draw pyramid + energy calculation
5	Transport & Respiration	Compare plant vs human transport
6	Review	Mixed paper timed + analysis of errors

Biology — Student Guide

(Biology 0610 / 0970)

Exam Map

Route	Paper	Focus	Time	Marks	Weight
Core	1	Multiple choice	45 min	40	30%
	3	Theory	1h15	80	50%
	5/6	Practical / Alternative to Practical	1h	40	20%
Extended	2	Multiple choice	45 min	40	30%
	4	Extended theory	1h15	80	50%
	5/6	Practical / Alternative to Practical	1h	40	20%

2. Command Words

Word	What to Do
State	Short fact, no explanation
Describe	Say what happens / pattern
Explain	Give reason or mechanism
Compare	Similarities & differences
Outline	Main steps only
Suggest	Apply knowledge to new data
Define	Exact wording of biological term
Calculate	Show method + unit
Evaluate	Judgement with data & reasoning

3. Timing Strategy

Paper	Marks	Time	Plan
MCQ (1/2)	40	45 min	1 min/question
Theory (3/4)	80	75 min	~1 min/mark
Practical (5/6)	40	1 h	Plan 10m, data 30m, evaluation 20m

Tip: Attempt all — no negative marking.

4. Formula & Data Handling (Recall)

Topic	Formula	Units
Magnification	Image / Actual	×
Rate	Change / Time	cm ³ /min
% change	(Final – Initial)/Initial × 100	%
Energy (food)	4.2 × mass water × ΔT	J
SA:V ratio	Surface area / Volume	—

5. Practical Skills & Mark-Winning Habits

- Identify **CIV** (Control / Independent / Dependent).
- Always include **units** in tables & graphs.
- Repeat & calculate **mean** for reliability.
- **Label** all diagrams & apparatus.
- **Explain** *why* each step improves accuracy.

6. Frequent Mistakes & Fixes

Mistake	Fix
Mixing diffusion & osmosis	Osmosis = water only
Forgetting units	Add to headings in table
Poor graph	Axis titles, units, full grid
Unclear conclusion	Always link to hypothesis
Describing not explaining	Add “because” & mechanism
Mislabelled heart flow	Memorise route & valve names

7. Revision Structure (4 Weeks)

Week	Focus	Output
1	Cells, Enzymes, Nutrition	2 short papers
2	Physiology & Transport	Diagram recall + data
3	Genetics & Ecology	Inheritance tasks
4	Mixed Paper	Full timed paper + reflection

8. Vocabulary & Precision Checklist

Define: osmosis, active transport, enzyme, photosynthesis, respiration, mitosis, meiosis, gene, allele, ecosystem.

Use correct terms (not “food pipe” → *oesophagus*).

Write units & directions (CO₂ out, O₂ in).

For practicals: “valid” = controlled, “reliable” = repeatable.

9. Resources

- **Biology Coursebook / Revision Guide**
- **BBC Bitesize / Save My Exams / Physics & Maths Tutor**
- **Khan Academy / Amoeba Sisters / FuseSchool**
- **PhET & BioMan Biology** (virtual labs)
- **Cambridge past papers & mark schemes** (core & extended)

📌 60-Second Exam Check

Label diagrams accurately

Write units in every answer

Include “because” in explanations

Add +C in calculations where needed

Circle data patterns before describing

Read *every* question stem twice

Leave 5 min for graph/unit check

Combined Science

Basic Supercurriculum

Why Supercurriculum?

To move from **memorising** to **thinking like a scientist** — using concepts from **Biology, Chemistry, and Physics** to explain real situations, handle data, and plan investigations.

1) Core Knowledge — Must Master

📌 Biology (basics to recall + explain)

- **Cells & transport:** cell structures; **diffusion, osmosis, active transport** (mechanism + examples).
- **Enzymes & molecules:** effects of **temperature/pH**; food tests (iodine, Benedict's, Biuret, ethanol).

- **Human systems:** gas exchange (alveoli adaptations), heart & blood (double circulation), digestion (villi).
- **Plants:** photosynthesis (word/symbol equations), limiting factors, transpiration vs translocation.
- **Genetics & ecology:** genes/alleles, monohybrid crosses, food chains/webs, energy loss, cycles (carbon/nitrogen basics).

2 Chemistry

- **Particles & states:** diffusion, heating/cooling curves.
- **Atomic structure & bonding:** protons/neutrons/electrons, **ionic vs covalent**, simple molecular vs giant ionic/covalent.
- **Equations & moles:** balance equations; $n = \frac{m}{M_r}$; $n = \frac{m}{M_r}$; solutions $c = \frac{n}{V}$; $c = \frac{n}{V}$ (dm³).
- **Acids, bases & salts:** indicators, neutralisation, simple titration idea, salt prep routes.
- **Rates & energetics:** collision theory; exo/endo energy profiles.
- **Electrolysis:** products at electrodes (molten vs aqueous).
- **Periodic trends & organic intro:** G1/G7 trends; alkanes/alkenes; polymer idea.

2 Physics

- **Motion & forces:** $v = \frac{s}{t}$; $v = \frac{s}{t}$; $a = \frac{v - u}{t}$; $a = \frac{v - u}{t}$; $a = \frac{v - u}{t}$; **Newton's laws.**
- **Energy & power:** $KE = \frac{1}{2}mv^2$; $KE = \frac{1}{2}mv^2$; $GPE = mgh$; $GPE = mgh$; $GPE = mgh$; $P = \frac{E}{t}$; $P = \frac{E}{t}$; $P = \frac{E}{t}$; **efficiency %.**
- **Density & pressure:** $\rho = \frac{m}{V}$; $\rho = \frac{m}{V}$; $p = \frac{F}{A}$; $p = \frac{F}{A}$; $p = \frac{F}{A}$; **liquids $p = h\rho g$; $p = h\rho g$; $p = h\rho g$.**
- **Waves & light:** $v = f\lambda$; $v = f\lambda$; $v = f\lambda$; **reflection/refraction basics, TIR idea.**
- **Electricity:** $V = IR$; $V = IR$; $V = IR$; **series vs parallel rules, $Q = It$; $Q = It$; $Q = It$; $E = VQ$; $E = VQ$; $E = VQ$.**
- **Magnetism:** magnets & electromagnets, motor effect idea.

2) Mini Practicals (15–20 min, home/school-friendly)

- **Biology:** potato **osmosis** (mass change vs sugar conc); **enzyme** (yeast + H₂O₂ bubbles vs temp).
- **Chemistry:** **rate** (marble chips + acid, chip size effect); **chromatography** of inks (R_f).
- **Physics:** **ramp** acceleration (timed run); **I–V** for a resistor; **refraction** with ray box or phone torch + ruler.

Safety first: goggles, tiny quantities, labelled containers, dry hands near electrics.

3) Calculation Mini-Drills (5–10 min)

- **Bio:** magnification

$$= \frac{\text{image}}{\text{actual}} = \frac{\text{image}}{\text{actual}} = \frac{\text{image}}{\text{actual}}$$

$$= \Delta \times 100\% = \frac{\Delta}{\text{initial}} \times 100\% = \frac{\Delta}{\text{initial}} \times 100\%$$

- **Chem:** $n = \frac{m}{M_r}$; $c = \frac{n}{V}$; gas $\approx 24 \text{ dm}^3 \text{ mol}^{-1}$ at r.t.p.
 - **Phys:** $v = \frac{s}{t}$; $a = \frac{v-u}{t}$; $F = ma$; $P = VI$; $E = Pt$
- Habit:** write the **formula first**, include **units every step**, round at the end.

4) Diagrams to Sketch from Memory (weekly)

- **Bio:** alveolus + capillary; heart with flow arrows; leaf cross-section; villus; plant cell vs animal cell.
- **Chem:** energy profile (exo/endo); electrolysis cell; titration setup (burette/pipette).
- **Phys:** distance–time and velocity–time graphs; ray refraction; series/parallel circuit with meter placement.

5) Exam Power Lines (use these phrases)

- **Bio diffusion/osmosis:** “net movement down a concentration/water potential gradient (osmosis via a partially permeable membrane).”
- **Chem bonding:** “Giant ionic has strong electrostatic attractions → high mp, conducts when molten/aq.”
- **Chem intermolecular:** “Simple molecular → weak intermolecular forces, low mp, poor conductor.”
- **Phys forces:** “Unbalanced force → acceleration because $F = ma$.”
- **Phys electricity:** “Series: current same; Parallel: potential difference same.”

6) Common Pitfalls & Fixes

Pitfall	Fix
Mixing diffusion & osmosis	Osmosis = water only through partially permeable membrane
Saying “weak bonds between molecules”	Say weak intermolecular forces , not “weak covalent bonds”
Wrong electrolysis product (aq)	Cathode: least reactive cation/ H_2 ; Anode: halogen else O_2
Speed/velocity confusion	Velocity has direction ; use signs on v–t graphs
Missing units/labels	Add units in tables/axes and in final answers

7) Weekly Study Rhythm (40–60 min/day)

Day	Focus	Task
Mon	Content recall	12 flashcards per science (36 total, spaced repetition)
Tue	Calculations	6 chem + 6 phys + 3 bio maths
Wed	Diagrams	Draw/label 3 diagrams from memory
Thu	Past-paper shorts	12–15 mixed marks (all three sciences)
Fri	Practical skills	Plan an investigation (aim, variables, method, risks)
Weekend	Extension	Watch one science demo; write 6 bullet takeaways

8) Graphs & Data Skills (mark-winners)

- Axes with **name + symbol + unit**; use >50% of grid.
- Choose **line** for continuous, **bars** for categories.
- **Gradient** and **area** meaning (e.g., v–t area = distance).
- Identify **anomalies**, suggest **improvements** (repeats, finer scale, temperature control).
- Conclusions must be **pattern + scientific reason**.

9) Resource Shortlist (free & reliable)

- **Combined Science Coursebook/Revision Guide**
- **BBC Bitesize** (Bio/Chem/Phys GCSE pages)
- **Khan Academy, FuseSchool, Cognito** (videos)
- **PhET Simulations** (circuits, waves, gases)
- **Cambridge past papers & mark schemes** (your series)

Self-Check Before Any Test

- ☐ I can **define** key terms (osmosis, activation energy, acceleration) precisely.
- ☐ I can **balance** equations and do **mole/units** problems.
- ☐ I can **rearrange** physics formulas and include **units**.
- ☐ My **diagrams** are accurate and labelled.
- ☐ I can **plan/evaluate** a simple investigation (CIV, improvements).
- ☐ I've completed a **timed mixed-topic set** this week.

Combined Science
Extended Supercurriculum – Year 11
(Syllabus 0653 / 0973)

1. Aim

To integrate **Biology, Chemistry, and Physics** knowledge and reasoning into complete scientific explanations.

Students learn to **analyse unfamiliar data, interpret experiments, and connect mechanisms** across disciplines.

BIOLOGY — Advanced Connections

A. Cells & Enzymes

- **Cell specialisation:** red blood cell (surface area & haemoglobin), root hair cell (absorption).

- **Microscopy:** use magnification formula; estimate actual sizes from scale bars.
- **Enzymes:**
 - Describe *induced-fit model* (active site flexibility).
 - Graph: rate vs temperature (optimum, denaturation).
 - Explain denaturation via **breakage of H-bonds**, permanent active site change.

B. Transport & Circulation

- **Diffusion gradients** → alveoli & capillaries.
- **Heart:** route = body → right atrium → right ventricle → lungs → left atrium → left ventricle → body.
- **Blood:** roles of plasma, RBCs, WBCs, platelets.
- **Xylem/Phloem:** compare directions, substances, processes (passive vs active).

C. Inheritance & Genetics

- **DNA structure:** complementary base pairing (A-T, C-G).
- **Monohybrid crosses:** genotype/phenotype ratios; dominant vs recessive.
- **Sex-linked traits** (simple interpretation).
- **Mutation:** causes (ionising radiation, mutagens), types (gene/chromosomal).
- **Variation:** continuous (height), discontinuous (blood group).

D. Ecology & Environment

- **Energy flow:** 10% transfer efficiency; biomass pyramids (scale correctly).
- **Population graphs:** lag, log, stationary, decline.
- **Carbon cycle:** link respiration, photosynthesis, combustion, decay.
- **Sustainability:** recycling, renewable energy, deforestation impacts.

2. CHEMISTRY — Advanced Mastery

A. Atomic Structure & Bonding

- Subatomic particles: relative mass & charge; isotopes.
- **Electron arrangement:** first 20 elements; valency links to group.
- **Bonding summary:**

Type	Particles	Forces	Example	Property
Ionic	ions	electrostatic	NaCl	high mp, conducts molten
Covalent	atoms	shared electrons	H ₂ O	low mp, insulator
Metallic	atoms	delocalised e ⁻	Cu	conductor, malleable

B. Equations, Moles, & Calculations

- $n = \frac{m}{M_r}$ $n = \frac{V(\text{dm}^3)}{V_m}$ $c = \frac{n}{V(\text{dm}^3)}$ $c = \frac{V(\text{dm}^3)}{V_m} n$ $\text{gas vol.} = 24 \times n$ $\text{gas vol.} = 24 \times n$
- % yield & atom economy basics.
- Empirical formula from % composition.
- **Ionic equations:** cancel spectator ions.

C. Energetics & Rates

- Exo/endo profiles; activation energy (E_a).
- Rate = $1/\text{time}$; factors = temperature, concentration, surface area, catalyst.
- Collision theory explanation (frequency + energy).

D. Electrolysis & Periodicity

- **Electrolysis of brine:** anode Cl_2 , cathode H_2 , solution NaOH .
- **Reactivity series:** $\text{K}-\text{Na}-\text{Ca}-\text{Mg}-\text{Al}-(\text{C})-\text{Zn}-\text{Fe}-(\text{H})-\text{Cu}$.
- **Displacement:** more reactive metal forms ion, less reactive leaves solution.
- **Group trends:**
 - $G1 \rightarrow$ down = more reactive (outer e^- further).
 - $G7 \rightarrow$ down = less reactive (weaker attraction).

E. Organic & Environmental Chemistry

- **Alkanes/Alkenes:** $\text{C}_n\text{H}_{2n+2}$ vs C_nH_{2n} ; test = bromine water decolourises.
- **Combustion:** complete $\rightarrow \text{CO}_2 + \text{H}_2\text{O}$; incomplete $\rightarrow \text{CO} + \text{soot}$.
- **Polymers:** monomer $\rightarrow$ repeating unit; identify addition polymer.
- **Air pollution:** CO , SO_2 , $\text{NO}_x \rightarrow$ acid rain; catalytic converters.

3. PHYSICS — Advanced Connections

A. Motion, Forces, Energy

- $v = stv = \frac{s}{t}$; $v = ts$; $a = \frac{v-u}{t}$; $a = \frac{v-u}{t}$; $F = ma$; $F = ma$; $F = ma$.
- **Graphs:**
 - distance-time $\rightarrow$ gradient = speed
 - velocity-time $\rightarrow$ gradient = acceleration, area = distance.
- **Energy:**
 - $\text{KE} = \frac{1}{2}mv^2$; $\text{GPE} = mgh$; $\text{GPE} = mgh$; $\text{GPE} = mgh$.
 - Conservation of energy in pendulum, roller coaster.
- **Momentum:** $p = mv$; $p = mv$; $F = \frac{\Delta p}{\Delta t}$; $F = \frac{\Delta p}{\Delta t}$; $F = \frac{\Delta p}{\Delta t}$.

B. Thermal Physics

- Kinetic model: heating $\rightarrow$ particles move faster, spread further.
- Conduction (solids), convection (fluids), radiation (infrared).
- **Change of state:** latent heat — no temperature change.
- **Insulation:** poor conductor; reduces heat loss by convection & radiation.

C. Waves, Light, Sound

- $v = f\lambda$; $v = f\lambda$; $v = f\lambda$.
- Reflection: angle i = angle r .
- Refraction: $n = \frac{\sin i}{\sin r}$; $n = \frac{\sin i}{\sin r}$; $n = \frac{\sin i}{\sin r}$.
- Total internal reflection (optical fibres).
- Sound: longitudinal; speed measured by echo/time method.

D. Electricity & Magnetism

- $V = IR$; $V = IR$; $V = IR$; $Q = It$; $Q = It$; $Q = It$; $P = VI$; $P = VI$; $P = VI$; $E = VIt$; $E = VIt$; $E = VIt$.
- Series: current same; Parallel: voltage same.
- **Magnetism:**
 - Field lines $\text{N} \rightarrow \text{S}$;
 - Right-hand rule for current in wire;
 - Motor effect: $F = BIL$; $F = BIL$; $F = BIL$.

- **Electromagnet:** increase turns/current → stronger field.

4. Integrated Science Thinking

Examples:

- Respiration + combustion → same chemical equation type, different energy use.
- Photosynthesis + electrolysis → both need energy input (light/electric).
- Circulation + electricity → current = charge flow, heart = biological pump.
- Ecology + energy transfer = same principle as energy conservation.

5. Practical & Analytical Skills (Paper 5/6 level)

Plan: aim, variables, method, control of factors, risks.

Measure: tables with headings + units; mean values.

Graph: correct scale, label axes, use $> \frac{1}{2}$ grid, draw best-fit line.

Conclude: pattern + scientific reason.

Evaluate: identify **major error sources** & **practical improvements**.

Error Type	Example	Improvement
Random	Timing reaction	Use same observer or timer
Systematic	Parallax in measuring liquid	Read at eye level
Environmental	Temperature fluctuation	Use water bath
Biological	Inconsistent leaf size	Use same species/area

6. Skills for A–A* Success

✓ State + explain (don't just describe).

✓ Use **data values** in every graph answer.

✓ In long answers, use **connectors**: because / therefore / resulting in.

✓ Define + give example + apply to context.

✓ Link between disciplines (e.g., energy flow in food chain ↔ conservation law).

7. Weekly Challenge Rotation (6 Weeks)

Week	Focus	Output
1	Cells & Particles	3 short planning questions
2	Equations & Calculations	15-min calculation quiz
3	Energy & Rates	Graph + interpretation
4	Ecology & Circuits	Practical plan & sketch
5	Inheritance & Motion	Mixed-paper mini test
6	Review	Full 1-hour timed paper

Combined Science — Student Guide

(0653 / 0973)

Exam Map

Route	Paper	Focus	Time	Marks	Weight
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Core	1	Multiple choice	45 min	40	30%
	3	Theory (short/structured)	1h15	80	50%
	5 / 6	Practical / Alternative	1h	40	20%
Extended	2	Multiple choice	45 min	40	30%
	4	Extended theory	1h15	80	50%
	5 / 6	Practical / Alternative	1h	40	20%

Choose **Core** (1+3+5) or **Extended** (2+4+5/6).

2. Command Words

Word	Meaning
State / Define	short fact or definition
Describe	what happens / trend
Explain	reason using science
Suggest	apply knowledge to new context
Calculate	show working + unit
Compare	similarities + differences
Outline	main points only
Evaluate	judge accuracy/reliability
Determine	find from graph/data

3. Timing & Strategy

- **MCQ (1/2):** ~1 min per question.
- **Theory (3/4):** 1 mark = ~1 min.
- **Practical (5/6):**
 - 10 min: read & plan
 - 35 min: data collection
 - 15 min: analysis & evaluation

Tip: never leave blanks; mark-scheme rewards partial process marks.

4. Equations to Memorise

Physics

$v = st$
 $a = \frac{v - u}{t}$
 $F = ma$
 $p = F/A$
 $P = E/t$
 $V = IR$
 $Q = It$
 $E = VQ$
 $\rho = \frac{m}{V}$

Chemistry

$n = \frac{m}{M_r}$
 $c = \frac{n}{V}$
 $\text{gas vol.} = 24n$
 $\text{Rate} = \frac{\text{change}}{\text{time}}$

Biology

$\text{Magnification} = \frac{\text{image}}{\text{actual}}$
 $\text{Rate} = \frac{\text{change}}{\text{time}}$

Percentage change = $\frac{\Delta \text{initial} \times 100\%}{\text{initial}}$
 Percentage change = $\frac{\Delta}{\text{initial}} \times 100\%$

5. Data & Graph Skills

- Use **ruler** for straight lines, **curves** for biological patterns.
- Always include **units** in headings.
- Label **axes & lines** clearly (e.g. enzyme rate vs temperature).
- **Gradient = rate; area under curve = quantity** (Physics).
- Identify **anomalies**; justify with science.

6. Frequent Mistakes & Fixes

Mistake	Quick Fix
Forgetting units	Always attach at every step
Wrong scale on graph	Use $> \frac{1}{2}$ grid, consistent intervals
Vague “it increases”	Add data + reason (“due to more collisions”)
Mixing diffusion/osmosis	Water only = osmosis
Wrong circuit reading	Check parallel/series rules

7. High-Scoring Answer Frames

Explain pattern:

As temperature increases, enzyme activity increases because particles move faster and collisions with active sites occur more often.

Evaluate experiment:

The results are reliable as values are close together; could be improved by repeating and averaging to reduce random error.

Describe data:

The graph shows a steady rise from 20–40°C followed by a rapid decrease above 50°C.

8. Revision Cycle (4 Weeks)

Week	Focus	Practice
1	Bio Core	20 short-answer Qs
2	Chem Core	15 calculations + equations
3	Phys Core	15 numerical/data Qs
4	Mixed Paper	timed past paper (Core/Extended)

9. Resources

- **Combined Science Coursebook / Workbook**
- **BBC Bitesize (Combined Science)**
- **Khan Academy, FuseSchool, Cognito**
- **Cambridge past papers & mark schemes**
- **PhET Simulations** for practical training

📌 Final 60-Second Exam Check

- ✓ Write **units** on all answers
- ✓ Label **axes** and **diagrams** clearly
- ✓ Use **data** in every explanation
- ✓ Quote **laws or formulas**
- ✓ Don't skip the **evaluation** question
- ✓ Read every question stem **twice**

COMPUTER SCIENCE

Basic Supercurriculum

Why Supercurriculum?

Computer Science teaches how to **think logically**, **solve problems systematically**, and **use code creatively**.

This subject blends **mathematics, logic, and real-world technology**, developing digital fluency and computational confidence — skills essential for all STEM fields.

Core Knowledge — Must Master

📌 1. Data Representation

- **Binary**: base-2 system; convert between binary ↔ decimal ↔ hexadecimal.

- **Hexadecimal:** compact way to represent binary (used in colours, memory addresses).
- **Binary addition:** carry over when ≥ 2 ; overflow errors.
- **Characters:** ASCII and Unicode → code tables linking binary to symbols.
- **Images:**
 - Pixels = smallest element.
 - Colour depth = bits per pixel.
 - File size = width $\times$ height $\times$ colour depth.
- **Sound:** sampled at regular intervals; quality depends on sample rate and bit depth.

2. Computer Hardware

- **Input devices:** keyboard, mouse, microphone, sensors.
- **Output devices:** monitor, printer, actuator, speaker.
- **Storage:** primary (RAM, ROM), secondary (HDD, SSD), offline (optical, USB).
- **CPU structure:**
 - ALU (calculations)
 - CU (control flow)
 - Registers (storage during execution)
- **Fetch–Decode–Execute cycle:**
 - *Fetch:* instruction from memory
 - *Decode:* control unit interprets
 - *Execute:* ALU performs operation
- **Clock speed, cores, cache** affect performance.

3. Networks

- **LAN vs WAN:** local vs wide area.
- **Hardware:** router, switch, hub, NIC, modem.
- **Topologies:** star (central switch), bus (shared cable).
- **Protocols:**
 - TCP/IP (transfer)
 - HTTP/HTTPS (web pages)
 - FTP (file transfer)
 - SMTP/POP/IMAP (email)
- **Security:** firewalls, encryption, strong passwords, access rights.

4. Security & Ethics

- **Threats:** malware, phishing, SQL injection, brute force.
- **Prevention:** antivirus, 2FA, backups, encryption.
- **Ethics:** privacy, data protection (GDPR ideas), intellectual property, digital divide.

5. Algorithms & Logic

- **Flowcharts** (ovals = start/end, parallelograms = input/output, diamonds = decision).
- **Pseudocode** (Cambridge style):
- IF mark $\geq$ 50 THEN
- OUTPUT "Pass"
- ELSE
- OUTPUT "Fail"
- ENDIF
- **Sequence / Selection / Iteration (looping)** – the three building blocks of programming logic.
- **Dry run table:** trace variables step by step.

7. Programming Concepts (Python, VB, or pseudocode)

- **Variables & constants:** store and name data.
- **Data types:** integer, real, Boolean, string, char.
- **Operators:** +, -, *, /, MOD, DIV, ^ (power), AND, OR, NOT.
- **Input/output** routines.
- **Conditional statements** (IF / ELSE / CASE).
- **Loops:** FOR (fixed), WHILE (conditional).
- **Arrays/lists:** store multiple items under one name.
- **Subroutines:** functions/procedures for modular design.

8. Databases

- **Field / record / file** definitions.
- **Primary key** = unique identifier.
- **Validation:** format check, range check, presence check.
- **Verification:** double entry, visual confirmation.
- **SQL basics:**
- SELECT Name, Age FROM Students WHERE Age > 15;

9. Binary Logic

Gate	Symbol	Function
NOT	$\neg A$	Inverts input
AND	$A \cdot B$	1 if both 1
OR	$A + B$	1 if any 1
NAND / NOR	Negated AND/OR	1 if condition false

- **Truth tables:** list all input combinations and outputs.

10. Software Development

- **Stages:** Analysis → Design → Development → Testing → Implementation → Evaluation → Maintenance.
- **Testing:** normal, boundary, erroneous data.
- **Validation vs Verification:** correctness vs accuracy.
- **Documentation:** flowcharts, pseudocode, variable lists.

10. The Big Picture

- **Operating system** manages hardware/software.
- **Utility programs** (defragmentation, compression, anti-virus).
- **Cloud computing**: remote storage, benefits (scalability), risks (security).
- **AI & emerging tech**: automation, robotics, ethics of data collection.

2. “Think Like a Computer Scientist”

Skill	Practice
Decompose	Break big tasks into smaller ones
Abstract	Ignore unnecessary detail
Algorithmic thinking	Create repeatable, step-by-step processes
Debug systematically	Predict → test → adjust

3. Mini Challenges (15–20 min each)

Topic	Task
Binary	Convert $156_{10} \rightarrow$ binary $\rightarrow$ hex
Logic	Draw truth table for (A AND B) OR NOT C
Programming	Write a pseudocode quiz with scoring
SQL	Retrieve students with mark > 80
Flowchart	Design algorithm to calculate average of 5 numbers
Network	Label LAN diagram & describe purpose of router
Security	Explain how encryption protects online banking

4. Common Pitfalls & Fixes

Pitfall	Fix
Mixing AND/OR order	Use brackets for clarity
Forgetting indentation	Use clear structure in pseudocode
Writing vague comments	Add purpose (“calculate total score”)
Confusing validation/verification	Validation = data rules; Verification = correctness check
Overcomplicating loops	Start simple: test on paper first

5. Weekly Study Rhythm (45–60 min/day)

Day	Focus	Task
Monday	Data & binary	Conversions + short quiz
Tuesday	Programming logic	5 pseudocode problems
Wednesday	Hardware & systems	Diagram + short paragraph
Thursday	Security & networks	Identify 3 vulnerabilities + fixes
Friday	Past paper section	1 long structured Q
Weekend	Creative challenge	Design a small game algorithm or database

6. Vocabulary Bank (core 20)

Binary, hexadecimal, variable, algorithm, array, Boolean, validation, verification, loop, pseudocode, database, field, record, CPU, cache, router, encryption, malware, iteration, subroutine.

7. Real-World Applications

- **Gaming:** algorithms & object logic.
- **Finance:** encryption & data validation.
- **Medicine:** sensors, data logging.
- **AI/Robotics:** feedback systems.
- **Web Design:** databases + client-server models.

8. Self-Check Before a Test

- ☐ I can convert binary ↔ hex ↔ decimal correctly.
- ☐ I can design and trace an algorithm using pseudocode.
- ☐ I can explain how the CPU executes instructions.
- ☐ I can describe LAN, WAN, router, and protocols.
- ☐ I can explain security measures and threats.
- ☐ I can write simple SQL queries.
- ☐ I have practiced one past-paper question per week.

9. Recommended Resources

- **Computer Science Coursebook (CUP)**
- **BBC Bitesize Computer Science (GCSE)**
- **W3Schools SQL / Python Basics**
- **CS Unplugged** (for logic thinking)
- **Cambridge Past Papers & Mark Schemes**
- **YouTube Channels:** Bro Code, Computer Science Tutor, Craig'n'Dave.

📅 Quick Revision Plan (2 Weeks)

Day	Topic	Output
1	Binary, logic, conversions	Worksheet
2	Hardware & CPU	Summary sheet
3	Networks & security	Diagram & quiz
4	Programming logic	Flowchart
5	Databases & SQL	3 practice queries
6	Past paper	1 full paper (1h)
7	Review & reflection	10 key definitions

COMPUTER SCIENCE

Extended Supercurriculum — Year 11

(Syllabus 0478 / 0984)

1. Aim

To move from “writing code that works” → to “designing systems that *think*”.

Students learn to reason algorithmically, analyse problems, and evaluate computational efficiency — combining **programming, logic, and data systems** for creative real-world use.

Data Representation — Beyond the Basics

A. Number Systems

- **Binary ↔ Decimal ↔ Hexadecimal conversions**
 - Binary to decimal: multiply by powers of 2.
 - Hex to binary: convert each hex digit to 4-bit binary.
- **Binary arithmetic:**

- Handle overflow detection (carry beyond 8 bits).
- Understand significance of MSB = sign bit in two's complement.
- **Negative numbers (8-bit two's complement):**
 - Invert bits + add 1 → represents negative value.
- **Floating point:**
 - Represents very large/small numbers with mantissa & exponent.
 - Understand trade-off between precision and range.

B. Data Types & Files

- **Bit, Byte, Kilobyte, ... Terabyte** ($\times 1024$).
- **Text encoding:** ASCII (7 bit) vs Unicode (UTF-8).
- **Image storage:**
File Size = Width $\times$ Height $\times$ Colour Depth (bits).
- **Sound files:** sample rate $\times$ bit depth $\times$ duration.
- **Compression:**
 - *Lossless* (e.g. Run-Length Encoding, Huffman).
 - *Lossy* (MP3, JPEG).
 - Evaluate trade-offs (quality vs size).

2. Algorithms & Problem-Solving

A. Algorithm Design Principles

1. **Input** → **Process** → **Output** structure.
2. Use **flowcharts** or **structured pseudocode** (Cambridge syntax).
3. Apply **iteration, selection, and recursion** (for challenge students).
4. Ensure each step is *finite, deterministic, and clearly defined*.

B. Tracing & Debugging

- **Dry runs:** build trace tables showing variable values per step.
- Identify **logical errors** (wrong algorithm) vs **syntax errors** (invalid code).

C. Algorithm Patterns

Pattern	Description	Example
Count / Sum	Totals items in loop	Add exam scores
Search	Linear / binary search	Find record
Sort	Bubble / insertion / merge	Ascending order
Validation	Ensures correct input	Range check

Nested iteration	Loop inside loop	Multiplication table
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3. Programming Concepts — Advanced Logic

A. Variables & Data Structures

- **1D/2D arrays:**
 - 1D → store single list; 2D → tables (rows/columns).
 - Example pseudocode:


```
FOR row = 0 TO 2
  FOR col = 0 TO 2
    OUTPUT Table[row][col]
  NEXT col
NEXT row
```
- **Records:** store mixed-type fields (Name, Age, Mark).
- **Files:** sequential read/write operations (OPENFILE, READFILE, WRITEFILE, CLOSEFILE).

B. Subroutines

- **Procedures** (no return) and **Functions** (return a value).
- Modular programming → easier testing & reuse.
- Example:


```
FUNCTION Average(x, y)
  RETURN (x + y) / 2
ENDFUNCTION
```

C. Validation & Testing

Validation Type	Example
Range check	$1 \leq \text{Mark} \leq 100$
Presence	Data not blank
Type check	Must be integer
Format check	Email contains @
Length check	≤ 8 characters

Testing types: normal, boundary, erroneous, extreme, invalid.

4. Computational Logic — Thinking in Circuits

Logic Gate	Symbol	Expression	Output = 1 when...
NOT	$\neg A$	—	$A = 0$
AND	$A \cdot B$	—	$A \text{ and } B = 1$
OR	$A + B$	—	$A \text{ or } B = 1$
NAND	$\neg(A \cdot B)$	—	not both 1
NOR	$\neg(A + B)$	—	neither 1
XOR	$A \oplus B$	—	one 1 only

- Build **truth tables** for up to 3 inputs.
- Simplify logic using **De Morgan's laws** and **Boolean algebra**.
- Apply to real systems: alarms, sensors, control circuits.

5. Networks & the Internet

A. Hardware

- Router (connects networks), switch (directs within LAN), NIC (network card).
- Wired (Ethernet) vs Wireless (Wi-Fi, Bluetooth).
- Transmission media: copper (cheap), fibre (optical, fast, long range).

B. Protocols & Security

Protocol	Function
TCP/IP	Packet transfer
HTTP/HTTPS	Web data (secure = SSL/TLS)
FTP	File transfer
SMTP/POP/IMAP	Email systems
DNS	Translates domain names to IP

- **Encryption:** symmetric vs asymmetric (public/private keys).
- **Firewall / proxy / anti-malware** protection.
- **MAC address filtering** to limit devices.

6. Databases — From Concept to SQL

- Table structure: *fields, records, primary key*.
- **Relationships:** one-to-many via foreign key.
- **Normalisation:** reduce duplication.
- **SQL commands:**
- SELECT Name, Mark FROM Students
- WHERE Mark >= 80
- ORDER BY Name ASC;
- INSERT INTO Students VALUES ('Ali', 17, 91);
- UPDATE Students SET Mark=100 WHERE Name='Ali';
- DELETE FROM Students WHERE Mark < 50;

7. Software Development Cycle — Full View

Stage	Description
Analysis	Identify problem, inputs, outputs, constraints
Design	Flowcharts, pseudocode, HIPO diagrams
Implementation	Code development
Testing	Normal, boundary, erroneous
Evaluation	Compare to success criteria
Maintenance	Updates, debugging, user requests

Documentation: variable tables, user guides, technical guides.

8. Ethical, Legal, & Environmental Impact

- **Ethical:** privacy, bias, plagiarism, misuse of AI.
- **Legal:** data protection, copyright, computer misuse laws.
- **Environmental:** e-waste, energy consumption, recycling.

- **Cultural:** digital divide, unequal access.

9. Weekly Challenge Cycle (6 Weeks)

Week	Focus	Output
1	Binary & Logic	Conversions + truth tables
2	Algorithms	Design + dry-run 3 problems
3	Programming	Write + test small program
4	Databases	Create table + SQL queries
5	Networks & Security	Diagram + encryption paragraph
6	Full Integration	Mini project: Student Grade System

2 Student Guide

(Computer Science 0478 / 0984)

Exam Overview

Paper	Focus	Time	Marks	Weight
Paper 1	Theory of Computer Science	1 h 45 min	75	50 %
Paper 2	Problem-Solving & Programming	1 h 45 min	75	50 %

2. Command Words

Term	What to Do
State / Define	One precise sentence
Describe	Say <i>what</i> happens
Explain	Give <i>how/why</i> reasoning
Trace	Step through logic or pseudocode
Identify	Name key term
Discuss	Give advantages + disadvantages
Evaluate	Make judgement with evidence
Construct	Draw or write code/diagram accurately

3. Time & Strategy

- 75 marks → ~1 min per mark.
- Always read the **context paragraph** twice.
- For pseudocode: use consistent indentation and variable names.
- For logic questions: build a truth table before drawing the circuit.
- Use **annotated steps** even for short answers (method marks awarded).

4. Programming Pseudocode Quick Reference

```

IF <condition> THEN
    <statements>
ELSE
    <statements>
ENDIF

```

```
FOR i ← 1 TO n
    <loop body>
NEXT i
```

```
WHILE <condition>
    <loop body>
ENDWHILE
```

```
REPEAT
    <loop body>
UNTIL <condition>
```

```
PROCEDURE Display(Name)
    OUTPUT "Hello " & Name
ENDPROCEDURE
```

5. Frequent Mistakes & Fixes

Mistake	Fix
Forgetting indentation	Align blocks for readability
Wrong logic condition	Trace with table before coding
Poor variable naming	Use meaningful names (Score, Total)
Missing ELSE path	Always handle both outcomes
SQL syntax errors	Check semicolons & quotes
Overflow/rounding errors	State limits clearly (8-bit, 2 d.p.)

6. Data Representation Cheatsheet

Concept	Formula / Fact
Binary → Decimal	Multiply bits by powers of 2
File size (image)	width × height × colour depth
Sound file size	sample rate × bit depth × seconds
ASCII	7-bit per character
Hex to Binary	Each hex digit = 4 bits
Checksum	Detects data transmission errors

7. Revision Plan (4 Weeks)

Week	Focus	Task
1	Data & Logic	Conversion & Boolean quizzes
2	Programming	5 algorithm problems
3	Databases & Security	3 SQL queries + discussion
4	Mixed Paper	1 timed past paper + review

8. Top-Mark Strategy

Quote relevant *keywords* (variable, iteration, array, etc.)
 Draw logic clearly with symbols, not prose.
 Annotate pseudocode; comment purpose lines.
 Always show data conversions.
 When unsure — outline reasoning (partial marks!).
 Leave 5 min to check brackets, indentation, and units.

9. Recommended Digital Tools

- **CS Coursebook** (CUP)
- **Python Tutor** (visual step tracing)
- **Repl.it / Trinket.io** (online practice)
- **W3Schools SQL / Python**
- **Craig'n'Dave / Computer Science Tutor (YouTube)**
- **CS Unplugged** (offline algorithm games)

🔍 60-Second Final Exam Check

Read every question twice
 Pseudocode = indented + clear
 Tables & graphs = labels & units
 Show all binary steps
 Include comments for logic
 Check 8-bit limits (overflow)
 No blank answers — *attempt everything!*

Year 11 – Edexcel GCSE / IGCSE Art & Design (Basic)

Why Supercurriculum?

Art is not only about creating images — it's about **thinking visually**, **experimenting confidently**, and **communicating ideas** with skill and imagination.

The supercurriculum helps students move beyond classroom tasks to **personal expression**, **cultural awareness**, and **technical growth**.

Core Knowledge – Must Master

1. Assessment Objectives (AOs)

AO	Focus	What You Must Show
AO1	Develop ideas	Use artists and research to inspire your work
AO2	Experiment	Explore materials, techniques, and processes
AO3	Record	Draw, photograph, and annotate ideas accurately
AO4	Present	Create a personal and meaningful final piece

Success = showing *progress and connection* between all 4 AOs.

2. Core Concepts

📌 Visual Elements

- **Line:** contour, cross-hatching, gesture.
- **Shape/Form:** geometric, organic, positive/negative space.
- **Tone:** light, shadow, contrast.
- **Texture:** visual vs tactile (actual materials).
- **Colour:** primary, complementary, warm/cool, saturation.
- **Composition:** balance, rhythm, focal point, rule of thirds.

📌 Media Techniques

- Drawing (graphite, ink, pastel)
- Painting (acrylic, watercolour, mixed media)
- Collage & printmaking
- Photography (composition, light, perspective)
- Sculpture / 3D (wire, clay, paper, found materials)
- Digital art (Photoshop basics, layering, filters)

3. Study Habits for Visual Success

Habit	Description
Sketch daily	10–15 mins observational sketches (objects, textures, faces).
Annotate	Explain <i>why</i> you used certain materials or techniques.
Research	Include artist references (1–2 per theme).
Experiment	Try new materials — at least 3 for every project.
Reflect	After each task, write: “What worked? What would I change?”

4. Theme Examples (Year 11 Level)

Theme	Ideas to Explore
Identity	Portraits, culture, memories, symbolism
Nature & Growth	Patterns, organic forms, macro textures
Urban Life	Architecture, graffiti, movement, contrast
Time & Change	Decay, metamorphosis, reflections, nostalgia
Emotion & Expression	Colour psychology, body language, abstraction
Objects & Meaning	Still life, consumerism, storytelling through objects

5. Key Artists & References

Style	Artists to Explore
Impressionism	Monet, Degas, Pissarro
Expressionism	Munch, Kirchner, Kandinsky
Cubism	Picasso, Braque
Surrealism	Dali, Magritte
Pop Art	Warhol, Lichtenstein, Hockney
Modern / Contemporary	Yayoi Kusama, Banksy, Kara Walker, Ai Weiwei

For each artist, analyse:

“What techniques? What message? How can I apply this in my work?”

6. Developing Your Portfolio

Each project should include:

1. **Title Page:** theme, personal statement, goals.
2. **Research Pages:** artist studies + visual analysis.
3. **Experimentation:** materials, colour swatches, small studies.
4. **Development Work:** combining ideas, refining composition.
5. **Final Outcome:** resolved piece showing AO1–4 connection.

7. Annotation Framework (use short sentences)

- *Intent:* “I chose to explore ... because ...”
- *Process:* “I tested acrylic vs watercolour to see which created more texture.”
- *Evaluation:* “The composition feels balanced, but I could deepen tone for contrast.”
- *Artist link:* “Inspired by Van Gogh’s brushwork to show emotion through texture.”

8. Mini Weekly Challenges

Week	Focus	Task
1	Line & texture	5 object sketches using different mark-making tools
2	Colour & emotion	Paint same scene with warm vs cool palette
3	Composition	Photograph 10 frames exploring angles
4	Artist inspiration	Copy 1 small section of a famous artwork in your style
5	Mixed media	Combine collage + paint in 1 A4 experiment
6	Reflection	Write 100 words about what your art says about you

9. Self-Check Before Assessment

Skill	Question to Ask
AO1	Have I connected my ideas to an artist or theme?
AO2	Did I test multiple materials or techniques?
AO3	Did I draw/photograph accurately and annotate clearly?
AO4	Does my final piece communicate my main idea?

☑ Materials Checklist

☑ Sketchbook (A3)

☑ Acrylics / watercolours

⇒ Graphite, charcoal, fine liners

☑ Access to camera or phone for photography

☑ Glue, scissors, craft knife

☑ Brushes, palette, tape

☑ Optional: Photoshop, Canva, Krita (free digital software)

1 Recommended Resources

- Edexcel GCSE / IGCSE Art & Design Specification
- BBC Bitesize – Art & Design
- Tate Kids / Google Arts & Culture / The Art Story
- Pinterest or Behance for inspiration boards
- Local galleries / exhibitions (document in sketchbook!)

12. Self-Reflection Prompts

“What am I trying to communicate through my work?”

“Which artist or idea influenced me the most?”

“What do my colours, textures, and choices say about me?”

“Does my final piece show growth and confidence?”

Edexcel GCSE / IGCSE Art & Design Extended Supercurriculum – Year 11

1. Aim

To move from “producing art” → to **thinking, researching, and communicating visually** as an artist.

Top-level students demonstrate **depth of idea, technical control, and a personal visual voice** through consistent reflection, experimentation, and presentation.

Understanding the Creative Journey A. The Artistic Process (AO1–AO4 in sequence)

Stage	Focus	Example
Inspiration (AO1)	Research artists, cultures, and social issues.	Study Hockney’s landscapes or Greek mosaics.
Exploration (AO2)	Experiment with materials, techniques, composition.	Layer ink + collage + acrylic to test textures.
Recording	Observational drawing,	Photograph street textures →

(AO3)	photography, annotation.	draw 3 studies.
Resolution (AO4)	Develop & refine final outcome with intention.	Combine collage & painting to express identity.

☞ Each project must tell a visual story from idea → outcome.

2. Advanced Composition & Design

The “Visual Grammar” of Art

Principle	How to Apply
Balance	Distribute visual weight (symmetry / asymmetry).
Contrast	Mix light/dark, texture/smooth, scale differences.
Emphasis	Lead the eye with focus points, colour, or line.
Rhythm	Repetition, movement, or pattern flow.
Unity	Repetition of elements for harmony.
Space	Foreground, middle, background, or abstract depth.

Task: Analyse a favourite artwork — identify *at least three* principles and recreate a sketch using the same composition structure.

3. Deep Dive into Materials & Techniques

Medium	Key Skills	Extension Tasks
Drawing	Expressive mark-making, tone, proportion	Combine graphite & ink on textured paper
Painting	Acrylic layering, blending, impasto	Recreate a small masterwork using palette knife
Printmaking	Mono, lino, collagraph	Create 3 colour variations of one design
Photography	Depth of field, composition	Series of 5 thematic images (e.g. reflections)
Sculpture	Wire, clay, papier-mâché	Create a small 3D study inspired by an artist
Digital Media	Layers, filters, montage	Merge digital collage + hand-drawn elements

4. Analytical Thinking – Art Language for Annotation

Use this **4-Step Critical Framework** (for sketchbook pages):

1. **Describe** – What do you see? (materials, composition, subject)

2. **Analyse** – How has it been made? (techniques, style, texture)
3. **Interpret** – What could it mean or express? (emotion, idea, symbol)
4. **Evaluate** – What do you think? How can it inspire your own work?

Example Annotation:

“I admire how Kara Walker uses silhouettes to contrast beauty and violence. Her high contrast and historical imagery influence my own exploration of identity.”

5. Artist Research — Building Context & Depth

Include at least **2 artists or movements per project**.

Analyse *style + message + influence*.

Theme	Artist Examples
Identity	Frida Kahlo, Kehinde Wiley, Zanele Muholi
Emotion & Colour	Mark Rothko, Edvard Munch, Georgia O’Keeffe
Social Commentary	Banksy, Barbara Kruger, Ai Weiwei
Nature & Landscape	Monet, Hockney, Ansel Adams
Abstraction & Form	Kandinsky, Mondrian, Bridget Riley

Enrichment Task:

Visit an online gallery (Tate, MoMA, Uffizi) and write a 150-word reflection:

“Which artwork spoke to me and why?”

6. Experimentation Logs (AO2 Evidence)

Each experiment page should show:

- **Aim:** what you tested (e.g. “light effects with watercolour wash”).
- **Process photos or samples.**
- **Outcome notes:** what succeeded, what failed.
- **Next step:** how you’ll adapt or refine the idea.

Use small **contact sheets**, swatches, and overlays — avoid perfect pages; show *active exploration*.

7. Portfolio Structure (Component 1)

Section	What to Include
Introduction Page	Theme, title, short statement of intent
Artist Studies	Visual copies, analysis, reflection
Experimentation Pages	4–6 materials/technique trials
Development Pages	Sketches combining different ideas
Refinement	Selection of strongest visual direction
Final Piece	Well-planned composition + clear AO links
Evaluation Page	Reflection on success + artist influence

8. The Externally Set Assignment (Component 2)

- Choose **1 theme** from the exam paper (given by Edexcel).
- Develop sketchbook over ~8–10 weeks.
- Produce final piece in **10-hour supervised exam**.
- Marks split evenly across AO1–AO4 (just like coursework).

Tip:

Select a theme that connects with your *existing strengths* — e.g. photography, surrealism, or mixed media.

9. Building a Personal Visual Voice

- Use recurring motifs or textures that represent *you* (e.g., hands, eyes, buildings, maps).
- Incorporate words, digital fragments, or real materials (sand, thread, newspaper).
- Link with *issues you care about* — identity, environment, technology, memory.
- Each page should reflect your **personality and curiosity**.

📅 Weekly Challenge Cycle (6 Weeks)

Week	Focus	Output
1	Artist research	Write 150-word visual analysis + mini copy
2	Technique expansion	2 A4 experiments with new materials
3	Theme connection	Annotated mood board + keywords
4	Composition development	3 thumbnail sketches exploring layout
5	Final idea planning	Colour tests, material notes, mock-up
6	Reflection	Write 250 words: “How did my work evolve?”

📖 Student Guide

(Edexcel GCSE / IGCSE Art & Design – Year 11)

Assessment Overview

Component	Focus	Marks	Weight
Component 1	Personal Portfolio (coursework)	72	60%
Component 2	Externally Set Assignment (exam)	72	40%

Total = 144 marks (AOs equally weighted).

2. Command Words & Phrases

Command	What It Means
Develop	Build ideas from research & imagination
Refine	Improve technique or concept
Record	Draw, photograph, annotate
Analyse	Break down style & meaning

Present	Curate final piece showing intention
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Use these in annotation sentences for clarity and to show AO coverage.

3. Visual & Written Presentation

- Use **A3 sketchbook** or **mounted sheets**.
- Title every page.
- Keep layout dynamic (overlaps, layers, textures).
- Use arrows, boxes, and small notes for process flow.
- Handwriting can be simple — clarity over decoration.
- Include **dates** to show timeline progression.

4. Time Management for Final Year

Term	Goal
Autumn	Complete first project (AO1–4)
Winter	Develop portfolio depth & finish annotation
Spring	Prepare for Externally Set Assignment
April–May	Produce final 10-hour piece
June	Final review, mounting, evaluation

5. Common Mistakes & Fixes

Mistake	Fix
Pages too neat but lack process	Show evidence of <i>experimentation and revision</i>
Artist copied without reflection	Add 3 lines of personal connection (“I used their technique to express...”)
Missing annotation	Minimum 2–3 short notes per page
Final piece disconnected	Show clear visual link from research → outcome
No photography evidence	Add process photos (stages, setup, textures)

6. Key Vocabulary Bank

Texture, contrast, abstraction, realism, tone, proportion, perspective, mood, composition, juxtaposition, context, symbolism, conceptual, technique, process, refinement, distortion, layering, narrative, aesthetic.

7.

Creative Habits of A* Artists

sketch daily (observational or imaginative)

Visit galleries (in person or online)

Reflect after each piece — not just at the end

Use your phone as a creative tool (photo diaries)

Include failed experiments (they show growth)

Link art to *current issues* (environment, identity, technology)

8. Recommended Resources

- Edexcel GCSE Art & Design Specification
- BBC Bitesize – Art & Design
- Tate.org.uk / Google Arts & Culture
- The Art Story (movements & analysis)
- YouTube: The Art Assignment / Sycra / Draw with Chris
- Apps: Procreate Pocket, Krita, Canva, ibisPaint (for digital exploration)

9. Self-Evaluation Prompts

“What do I want the viewer to feel or think?”

“Have I shown clear progression of ideas?”

“Which artist influenced my visual language?”

“How do my materials support my message?”

“Does my portfolio reflect *who I am* as a young artist?”

📋 Final 60-Second Check Before Submission

All AOs covered in each project

Artist studies annotated

Clear evidence of experimentation

Date & title on every page

Final piece mounted and labelled

200–300-word evaluation written

Sketchbook shows creative *journey*, not just result

YEAR 11 – EDEXCEL GCSE / IGCSE SPANISH

Basic Supercurriculum

Why Supercurriculum?

To move from “learning lists” to **communicating confidently**: clear messages, accurate grammar, and natural phrases across **listening, speaking, reading, writing**.

1) What to Master (Core Content)

A. High-Frequency Topics (build 10–15 key ideas per theme)

- **Identity & culture:** family, friends, technology, leisure, food, festivals.
- **Local area & travel:** town, environment, holidays, transport.
- **School & future:** subjects, rules, plans, work.
- **Global issues:** health, poverty, climate, volunteering.

B. Essential Grammar (Year-11 toolkit)

- **Regular verbs:** -AR, -ER, -IR (present / preterite / imperfect).
- **Key irregulars:** *ser, estar, tener, ir, hacer, poder, querer, decir, haber*.
- **Near future:** *ir a + infinitivo* → *Voy a estudiar*.

- **Simple future:** *estudiaré*; **Conditional:** *estudiaría*.
- **Perfect:** *he hecho, hemos ido*.
- **Opinions + reasons:** *me gusta / prefiero / odio + porque...*
- **Comparatives/superlatives:** *más/menos... que; el más...*
- **Time markers:** *ayer, la semana pasada, mañana, siempre, desde hace...*
- **Connectives:** *además, sin embargo, por eso, aunque, ya que.*
- **Por vs para, ser vs estar, negatives** (*nunca, nadie*).

Stretch phrase: **Ojalá (que) + subj.** (*Ojalá pueda viajar a España.*)

2) Speaking & Writing Frames (plug-and-play)

Opinión + razón + ejemplo

*En mi opinión, el deporte es **importante** porque me **mantiene en forma**. Por ejemplo, juego al baloncesto **tres veces a la semana**.*

Pasado → Presente → Futuro (sound advanced!)

*El año pasado **fui** a Sevilla; **ahora estudio** español todos los días; **el año que viene voy a hacer** prácticas en una empresa.*

Comparar

*Mi pueblo **es más tranquilo que** la ciudad, **pero** hay **menos** oportunidades.*

Dar soluciones

*Para mejorar el medio ambiente, **deberíamos** reciclar **más** y **usar menos** coches.*

Role-play polite set

¿Podría ayudarme? / Me gustaría... / ¿Cuánto cuesta? / ¿A qué hora abre?

3) Listening & Reading Tactics

- **Skim → Scan → Detail** (3 passes).
- Highlight **negatives, quantities, time phrases**.
- Expect **synonyms** (e.g., *trabajo* = empleo).
- If stuck, grab **gist** and **eliminate** distractors.

4) Pronunciation Cheats (big wins)

- **Vowels** always pure: a-e-i-o-u.
- **H** is silent; **LL** ≈ y; **J** like harsh English *h*.
- Stress normally on the **second-to-last** syllable (unless accent).

5) Weekly Study Rhythm (30-40 min/day)

Day	Focus	Quick Task
Mon	Vocab set (topic)	12 items → Quiz + 4 sentences
Tue	Grammar drill	10 verbs (3 tenses) in mini-table
Wed	Listening	10-12 min clip → 6 notes (ES-EN)
Thu	Speaking	90-sec monologue using PPF (past-present-future)
Fri	Writing	90-120 words using 5 connectives
Weekend	Culture	Article/video + 5 bullet facts (ES)

6) High-Value Vocab Packs (learn in chunks)

- **Opinion bank:** *me encanta, me fastidia, me preocupa, es imprescindible, vale la pena.*
- **Time & frequency:** *a menudo, casi nunca, de vez en cuando, desde hace dos años.*
- **Boosters:** *además, por otro lado, sin duda, por lo tanto, aunque.*
- **Environment/health:** *reciclar, ahorrar energía, dieta equilibrada, hacer ejercicio.*
- **School/work:** *asignatura, aprobar/suspender, bachillerato, prácticas, sueldo.*

7) Common Pitfalls & Fixes

Pitfall	Fix
Mixing ser/estar	<i>Ser</i> = identity/traits; <i>Estar</i> = states/locations
Forgetting agreement	Check gender/number (adjectives & past participles)
One-tense answers	Always add otra frase in a different tense
Literal translation	Use chunks (e.g., <i>tener ganas de, suele + inf.</i>)
Weak reasons	Add porque + detalle (specific benefit/example)

8) Mini Practice Tasks (ready to set)

- **90-word blog:** *Tu rutina saludable.* Include 3 tenses + opinion + connective.
- **Photocard speaking:** 3 facts (present) → 1 past → 1 future + opinion.
- **Role-play:** booking a hotel room; ask 2 questions + respond to a problem.
- **Translation (EN→ES):** “Last year I worked in a shop; it was tiring but I learned a lot.”

9) Assessment Snapshot (Edexcel)

(Centres vary; typically each skill ≈ 25% — confirm with your teacher.)

- **Listening** — extracts from everyday contexts.
- **Speaking** — role-play, photocard, conversation (prepare bullet ideas).
- **Reading** — multiple choice, short answers, translation into English.
- **Writing** — 90/150-word tasks, sometimes structured + translation into Spanish.

Exam habits: underline the **task bullets**, tick off **each bullet**, aim for **3+ time frames**.

📁 Resources (free & reliable)

- **Pearson/Edexcel past papers** (spec pack)
- **BBC Bitesize GCSE Spanish**
- **LyricsTraining / News in Slow Spanish** (listening)
- **WordReference** (verbs forum), **SpanishDict** (conjugator)
- **Duolingo/Memrise/Quizlet** (spaced vocab)
- **Coffee Break Spanish / Dreaming Spanish** (graded videos)

11) Self-Check Before You Hand In

- ☐ I used **two tenses** (better: three).
- ☐ I included **opinions + because**.
- ☐ I linked ideas with **connectors**.
- ☐ Adjective and verb **agreements** checked.
- ☐ **Spelling/accents** (j, ñ, á/é/í/ó/ú, ñ).
- ☐ I ended with a **future/conditional** sentence.

12) Handy One-Page Verb Box (learn cold)

Infinitive	Presente (yo)	Pretérito (yo)	Futuro (yo)	Cond. (yo)
ser	soy	fui	seré	sería
estar	estoy	estuve	estaré	estaría
tener	tengo	tuve	tendré	tendría
ir	voy	fui	iré	iría
hacer	hago	hice	haré	haría
poder	puedo	pude	podré	podría
querer	quiero	quise	querré	querría

Edexcel GCSE / IGCSE Spanish

Extended Supercurriculum – Year 11

1. Aim

To move from “exam survival” → **real fluency with control**.

You’ll learn to build **layered arguments**, **precise grammar**, **natural flow** and cultural awareness — to write, speak, and understand *like a young native speaker*.

Core Mastery Goals

Skill	Advanced Focus
Grammar range	All 5 main tenses + subjunctive + imperatives
Complex structures	<i>Aunque, cuando + subj., si tuviera..., acabo de + inf.</i>
Accuracy	Agreements, accent marks, pronouns, connectors
Style	Natural linking, idioms, register awareness
Culture	Authentic examples from Spain & LatAm life

2. Grammar & Structures to Master

Function	Example
Sequence of tenses	<i>Si tuviera más tiempo, estudiaría más.</i>
Perfect + imperfect	<i>He visitado Madrid muchas veces, pero cuando era niño</i>

contrast	<i>no me interesaba.</i>
Passive voice	<i>Se venden libros / fue construido en 2005.</i>
Relative clauses	<i>El chico que vive allí es mi primo.</i>
Subjunctive triggers	<i>Espero que / dudo que / es posible que / Ojalá que...</i>
Idiomatic phrases	<i>dar una vuelta, estar a tope, me da igual, vale la pena</i>

Aim to include 3+ *advanced structures* per 150-word writing or 2-minute speaking task.

3. Advanced Speaking Techniques

📌 A. Photo Card Strategy (Foundation → Grade 9)

1. **Describe (3 details)** – use adjectives + position (*A la izquierda hay...*)
2. **Infer** – opinion or action (*Creo que están celebrando un cumpleaños.*)
3. **Add time frame** – *el año pasado / ahora / mañana voy a...*
4. **Justify** – *porque / ya que / debido a que*
5. **Turn the table** – ask examiner 1 question (*¿Te gusta celebrar con tu familia?*)

B. Conversation Flow Model

Use **P.P.F.** (Past – Present – Future)

El verano pasado fui a Málaga / Ahora estudio español / En el futuro me gustaría trabajar en España.

📌 C. Pronunciation Polish

- Emphasise open vowels: *a e i o u*.
- Roll *r* / *rr*, soften *d* between vowels (*ciudad* → *thyoudad* in Spain).
- Avoid English intonation – keep sentences musical but flat.

4. Writing for Top Bands

📌 Structure for 150-Word Task

Section	Function	Example
Introduction	Context + opinion	<i>Normalmente paso mis vacaciones en la costa porque adoro el mar.</i>
2. Past tense	Event + details	<i>El año pasado fui a Valencia y probé la paella auténtica.</i>
3. Present	Routine + feelings	<i>Hoy en día estudio mucho porque tengo exámenes importantes.</i>
4. Future / Conditional	Ambition + dream	<i>En el futuro me gustaría viajar por Sudamérica si tengo dinero.</i>

5. Complex finisher	Justify + reflect	<i>Aunque fue caro, valió la pena porque aprendí mucho español.</i>
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Target: 5 tenses + 5 connectives + 3 opinions + 2 justifications.

5. Listening & Reading at Advanced Level

Strategies

- Predict key words before audio.
- Focus on **verbs & negatives**.
- Note *quantity, time, place*.
- Expect synonyms: *feliz* → *contento*, *trabajo* → *empleo*.
- Annotate options; don't rely only on first sentence.

Extension Practice

- Watch short **RTVE Noticias** / **DW Español** clips with subtitles.
- Summarise in English (gist) then in Spanish (2 sentences).
- Shadow-read aloud to mimic rhythm and accent.

6. Translation & Grammar Precision

Common Traps

English idea	Correct Spanish
"I'm cold / hot"	<i>Tengo frío / calor</i>
"I like to swim"	<i>Me gusta nadar</i> (no "yo")
"I've been for two years"	<i>Llevo dos años + gerundio</i>
"People are friendly"	<i>La gente es amable</i> (singular)
"I need to / I have to"	<i>Necesito / tengo que + inf.</i>

Translation Routine

1. Highlight verbs → choose tense first.
2. Mark connectors → ensure flow.
3. Check agreements.
4. Re-read for logic, then for grammar.

7. Cultural & Authentic Learning

Theme	Example Project
Festivals	<i>La Tomatina</i> / <i>Las Fallas</i> / <i>El Día de los Muertos</i> presentation
Music	Analyse lyrics by <i>Shakira</i> / <i>Rosalía</i> → vocab extraction
Cinema	Watch <i>Coco</i> , <i>Campeones</i> , or <i>Volver</i> → 10 sentence review
Society	Read news on <i>El País</i> / <i>BBC Mundo</i> → summarise one article

Always note 5 new expressions + 1 cultural fact.

8. Weekly Challenge Cycle (6 Weeks)

Week	Focus	Output
1	Grammar boost	Mini test (5 tenses × 2 verbs)
2	Listening	10-min news clip summary
3	Writing	150-word blog post
4	Speaking	2-min presentation + feedback
5	Culture	Research & present a festival
6	Mixed	Past paper (1 section per skill)

7 Student Guide

(Edexcel GCSE / IGCSE Spanish – Year 11)

Exam Overview

Paper	Skill	Time	Marks	Weight
1	Listening	~45 min	50	25 %
2	Speaking	10–12 min	70	25 %
3	Reading (+ EN→ES translation)	1 h	60	25 %
4	Writing (+ ES→EN translation)	1 h 15	60	25 %

2. Command Words

Word	Task
Describe / Say what...	Give facts or details
Give an opinion / Reason	Use <i>me gusta porque...</i>
Compare	Use <i>más/menos... que</i>
Translate	Keep tense + meaning exact
Write about	3 time frames + opinions
Answer in Spanish	Full sentences > single words

3. Top-Band Checklist for Grades 8–9

5 different tenses
 3 complex connectives (*aunque, cuando, ya que*)
 1 subjunctive or *si*-clause
 3 opinions + reasons
 Accurate spelling & accents
 Cultural reference (optional bonus)

4. Common Errors to Avoid

Error	Fix
Wrong gender	<i>la ciudad bonita</i> , not <i>el ciudad</i>

Forgetting accents	<i>más, cómo, habló</i>
Overusing “muy bueno”	Replace with <i>increíble, maravilloso, divertido</i>
Literal English	Use Spanish idioms (<i>tener éxito, pasarlo bien</i>)
No time frames	Add past + future even if brief

5. Independent Practice Resources

- Pearson Edexcel past papers
- BBC Bitesize – Spanish GCSE
- LyricsTraining / News in Slow Spanish
- Coffee Break Spanish Podcast
- WordReference / SpanishDict
- Quizlet “Edexcel GCSE Spanish Vocab 2024”
- Netflix audio switch: watch in Spanish + subtitles EN→ES

6. Self-Review Template (per topic)

Topic: _____

- 3 key vocab chunks → _____
- 2 grammar points → _____
- 1 cultural reference → _____
- Sentence showing 3 tenses → _____

Final Exam Minute Check

Accent marks added

Tense variety present

Opinions + connectives ✓

Past / Present / Future ✓

Cultural phrase (optional bonus)

Question answered fully ✓

Year 11 – Edexcel GCSE / IGCSE
Physical Education (Basic Supercurriculum)

Why Supercurriculum?

PE is about understanding **how the body works**, **how performance improves**, and **how lifestyle affects health**.

The supercurriculum moves you from *doing sport* to *thinking like an athlete, coach, and scientist*.

Core Areas (Theory Paper Overview)

Topic	Focus	Example Question
1. Anatomy & Physiology	Structure and function of body systems	<i>What is the role of the alveoli?</i>
2. Movement Analysis	Levers, planes, and axes	<i>What type of lever is a bicep curl?</i>
3. Physical Training	Principles of training, FITT, overload	<i>Explain how specificity improves performance.</i>
4. Health, Fitness & Wellbeing	Effects of lifestyle, diet, stress	<i>Give two benefits of regular exercise.</i>

5. Sport Psychology	Goal setting, arousal, motivation	<i>Define intrinsic motivation.</i>
6. Socio-cultural Influences	Ethics, media, participation	<i>How has technology impacted sport?</i>
7. Use of Data	Graphs, statistics, analysis	<i>Interpret data showing heart rate changes.</i>

2. Practical Component

- 3 sports: **team + individual + one extra** (best of three marks).
- Skills assessed in **performance and decision making**.
- Short **personal exercise programme (PEP)** based on training principles.

3. Body Systems — Must Know

System	Function	Example
Skeletal	Support, movement, protection, blood cell production	Cranium protects brain
Muscular	Movement, posture, temperature control	Biceps contract to flex elbow
Cardiovascular	Transport O ₂ & nutrients	Heart rate increases with exercise
Respiratory	Gas exchange (O ₂ in, CO ₂ out)	Alveoli diffusion
Energy systems	Aerobic & anaerobic	Sprint = anaerobic; long-distance = aerobic

4. Fitness Components

Component	Test Example
Strength	Handgrip dynamometer
Speed	30 m sprint test
Cardiovascular endurance	12-minute Cooper run
Flexibility	Sit and reach
Power	Vertical jump
Coordination	Wall toss
Reaction time	Ruler drop
Agility	Illinois agility test

Mnemonic: “Some People Can Feel Powerful, Coordinated, Rapidly Agile.”

5. Principles of Training (SPORT + FITT)

Principle	Meaning
S – Specificity	Match training to sport demands
P – Progression	Gradual increase in load
O – Overload	Work harder than normal
R – Reversibility	Stop training → loss of fitness
T – Tedium	Vary sessions to avoid boredom

FITT: Frequency, Intensity, Time, Type → used to design PEPs.

6. Training Methods (link to FITT)

Type	Description
Continuous	Steady aerobic work (running, cycling)
Fartlek	Speed play – mix of paces
Interval	Work + rest periods
Circuit	Series of exercise stations
Weight	Resistance training
Plyometric	Jump training (explosive power)
HIIT	High-intensity bursts + recovery

Include warm-up & cool-down in all plans.

7. Diet & Nutrition (Health & Wellbeing)

Nutrient	Function	Example
Carbohydrates	Energy	Pasta, bread
Protein	Growth, repair	Chicken, fish
Fat	Long-term energy	Nuts, oil
Vitamins & Minerals	Immune & muscle function	Fruit, veg
Water	Hydration	2–3L/day

Balance = 55% carbs, 30% fat, 15% protein (approx.)

8. Sports Psychology Basics

Concept	Key Idea	Example
Motivation	Intrinsic (enjoyment) vs extrinsic (rewards)	Winning medal vs self-satisfaction
Arousal	Inverted U theory	Too low = boredom; too high = anxiety
Goal Setting	SMART goals	“Improve 5k time by 2 min in 6 weeks.”
Feedback	Positive / Negative / Knowledge of performance	Coach praise or timing data
Personality	Introvert = calm sports; extrovert = dynamic team play	Golf vs football

9. Socio-Cultural Influences

- **Gender, age, ethnicity, disability, income** affect participation.
- **Media** can promote or stereotype.
- **Technology** improves analysis (VAR, Hawk-Eye) but raises fairness debates.
- **Ethics:** sportsmanship, gamesmanship, deviance (e.g., doping).

📌 Common Data Interpretation Tasks

- Graph: *Heart rate vs time* → identify trends.
- Table: *Performance stats* → calculate mean, range.
- Scenario: Apply principle (e.g., “Which principle explains the improvement?”).
- Command words: *Analyse, Evaluate, Justify*.

1 Weekly Study Rhythm

Day	Focus	Task
Mon	Body systems	Label diagram (lungs or heart)
Tue	Fitness testing	Research one test + purpose
Wed	Training principles	Design 10-min mini-circuit
Thu	Psychology	SMART goal + arousal example
Fri	Lifestyle	3 factors improving participation
Weekend	Sport log	Record heart rate before/after 20 min activity

12. Key Formulae to Remember

- **HR max** = $220 - \text{age}$
- **Training zone** = 60–80% of HR max
- **Speed** = $\text{distance} \div \text{time}$
- **Work** = $\text{force} \times \text{distance}$
- **Power** = $\text{work} \div \text{time}$
- **BMI** = $\text{weight} \div (\text{height}^2)$ (kg/m^2)

13. Practical Example (Link theory to sport)

Concept	Example
Lever	3rd class – bicep curl
Energy	Anaerobic – 100m sprint
Fitness test	Sit & reach for gymnastics
Overload	Increasing weight from 30→35kg
Feedback	Coach video analysis

14. Common Pitfalls & Fixes

Pitfall	Fix
Confusing aerobic/anaerobic	Aerobic = oxygen; anaerobic = short + intense
Forgetting data units	Use correct SI (kg, m, s)
Listing, not explaining	Add “because...” in every long answer
Ignoring applied examples	Always link theory to a real sport
Repeating answers	Use varied vocabulary (improves / increases / enhances)

15. Resources

- Edexcel GCSE/IGCSE PE Specification

- **BBC Bitesize – PE (Edexcel)**
- **EverLearner / PE Scholar / Seneca Learning**
- **Cambridge Go / Pearson Active Learn**
- **YouTube:** Mr Bruff PE / Craig'n'Dave PE Science
- **Apps:** MyFitnessPal, Nike Training, Strava (log training data)

16. Self-Check Before Exam

- ☐ I can describe the **five body systems**.
- ☐ I can explain **FITT & SPORT**.
- ☐ I can apply **training methods** to a chosen sport.
- ☐ I can interpret **graphs/data** accurately.
- ☐ I can define and apply **SMART goals**.
- ☐ I have practical evidence for **three activities**.

Edexcel GCSE / IGCSE Physical Education Extended Supercurriculum – Year 11

1. Aim

To progress from *understanding sport* → to *analysing performance scientifically*.
Students learn to link **body systems, training, psychology, and society** to real-world performance and develop a professional-level **Personal Exercise Programme (PEP)**.

Anatomy & Physiology – Advanced Concepts

A. The Musculoskeletal System

Focus	Key Detail	Example
Joint types	Hinge (knee), Ball & socket (shoulder), Pivot (neck), Condylod (wrist)	Shoulder = high mobility, low stability
Movement terms	Flexion/extension, adduction/abduction, rotation, plantar flexion	Knee flexion in sprint start
Antagonistic pairs	Agonist contracts / antagonist relaxes	Biceps-triceps during push-up
Planes of movement	Sagittal, Frontal, Transverse	Cartwheel = frontal plane
Axes of rotation	Longitudinal, Transverse, Frontal	Somersault = transverse axis

☞ *Top Tip:* In data Qs, describe the **joint, muscle, movement, and plane/axis** together for full marks.

B. The Cardiovascular & Respiratory Link

- **Double circulation** → pulmonary + systemic.
- **Oxygen pathway:** mouth → trachea → bronchi → bronchioles → alveoli → capillaries → blood → muscles.
- **Gaseous exchange:** O₂ diffuses into blood; CO₂ diffuses out.
- **During exercise:** HR ↑, SV ↑, CO ↑ (Cardiac output = HR × SV).
- **Recovery:** Oxygen debt → lactic acid removal → heart rate returns to resting.

C. Short- and Long-Term Effects of Exercise

System	Immediate Effect	Long-Term Adaptation
Muscular	Lactic acid buildup	Greater strength & endurance
Cardiovascular	Increased heart rate	Lower resting HR, increased SV
Respiratory	Increased rate/depth	Increased vital capacity
Skeletal	Synovial fluid production	Stronger ligaments/bones
Energy	Depletion of glycogen stores	Better energy efficiency

2. Biomechanics & Movement Analysis

Levers

Class	Structure	Sporting Example	Mechanical Advantage
1st	Effort–Fulcrum–Load	Heading a football	Variable
2nd	Fulcrum–Load–Effort	Calf raise	>1 (power)
3rd	Fulcrum–Effort–Load	Bicep curl	<1 (speed)

Planes and Axes Quick Recall

Plane	Movement	Axis
Sagittal	Flexion/Extension	Transverse
Frontal	Abduction/Adduction	Frontal
Transverse	Rotation	Longitudinal

3. The Energy Systems (Advanced)

System	Duration	Energy Source	Sporting Example	Waste Product
ATP-PC	0–10s	Phosphocreatine	100m sprint	None
Anaerobic glycolysis	10s–2min	Glucose (no O ₂)	400m sprint	Lactic acid
Aerobic	2min+	Carbs → fats (O ₂)	Marathon	CO ₂ + H ₂ O

“The body always uses all three — but one dominates depending on intensity and duration.”

4. Principles & Methods of Training (Advanced Application)

Principle	Advanced Application Example
Specificity	Tailor drills to sport positions (e.g., winger sprints)
Progression	Gradual overload to avoid injury
Overload	Increase intensity 5–10% every 2 weeks
Reversibility	2 weeks rest = VO ₂ max drop
Tedium	Add HIIT variety or cross-training

FITT Precision Example:

To improve aerobic endurance: 4x/week, 70% HRmax, 40 min continuous run.

Methods Summary

Method	Target	Example
Continuous	Aerobic	30-min run
Fartlek	Aerobic/anaerobic mix	Hill sprints
Interval	Power/speed	6×200m with rest
Circuit	All-round fitness	8 stations x 1 min
Weight	Strength	3 sets × 8 reps @ 75% 1RM
Plyometric	Power	Box jumps, bounding
HIIT	Cardiovascular endurance	20s work / 10s rest × 8

5. Designing a Personal Exercise Programme (PEP)

Step-by-Step Framework

1. **Choose aim** – e.g., improve cardiovascular endurance.
2. **Initial testing** – record baseline (e.g., Cooper test).
3. **Plan 6-week programme** (FITT + methods).
4. **Record progress weekly** (HR, time, perceived effort).
5. **Final retest** – compare results & evaluate.

Include: warm-up/cool-down, SMART targets, safety notes, data graphs.

6. Sports Psychology – Extended

Concept	Detail	Application
Arousal	Inverted U – too low or high = poor performance	Yoga vs crowd hype
Drive theory	Linear ↑ arousal = ↑ performance (simple skills)	Sprinting
Intrinsic motivation	Internal satisfaction	Enjoying training itself
Extrinsic motivation	External reward	Medal, applause
Goal setting	SMART (Specific, Measurable, Achievable, Realistic, Time-bound)	“Improve VO ₂ max by 10% in 6 weeks”
Mental	Visualising performance	Gymnast before routine

rehearsal		
Selective attention	Filtering distractions	Footballer ignoring crowd noise

7. Socio-Cultural & Ethical Influences

Area	Examples / Key Points
Participation barriers	Gender, age, ethnicity, cost, access
Commercialisation	Sponsorship, media, player image rights
Technology	VAR, heart rate monitors, biomechanical sensors
Deviance	Doping, match-fixing, aggression
Sportsmanship	Fair play, respect for officials
Gamesmanship	Bending rules (time-wasting)

Case Study Task:

Analyse one real incident (e.g., VAR controversy or doping scandal) → impact on fairness, spectators, ethics.

8. Data Analysis & Application

- Interpret heart rate / recovery graphs.
- Calculate averages and ranges.
- Recognise **correlations** between variables (training → performance).
- Evaluate validity and reliability of fitness tests.

Key Formulae Recap:

- $HR_{max} = 220 - \text{age}$
- Training zone = 60–80% HR_{max}
- $\text{Speed} = \text{distance} \div \text{time}$
- $\text{Power} = \text{work} \div \text{time}$
- $\text{Work} = \text{force} \times \text{distance}$
- $\text{BMI} = \text{weight} \div (\text{height}^2)$

9. Exam Strategy & Command Words

Word	What to Do
Define / State	1-sentence fact
Describe	Step-by-step detail
Explain	Use “because” or “this means that...”
Analyse	Break down + interpret
Evaluate	Judge effectiveness (advantages vs disadvantages)
Apply	Link to a named sport

Example:

Explain how the respiratory system responds to exercise (3 marks)

✓ Breathing rate increases → more oxygen to alveoli → more O_2 diffuses into blood → muscles get more oxygen for energy.”

📅 Weekly Challenge Cycle (6 Weeks)

Week	Focus	Output
1	Anatomy	Label full skeleton + 10 muscles
2	Training	Create 1 FITT plan + PEP aim
3	Energy systems	Table + 3 sport examples
4	Psychology	Mind-map arousal/motivation
5	Data handling	Interpret training graph
6	Mixed review	20-mark mock exam

📖 Student Guide

(Edexcel GCSE / IGCSE Physical Education – Year 11)

Exam Overview

Paper	Content	Marks	Weight
Paper 1	Fitness & Body Systems	90	36%
Paper 2	Health & Performance	70	24%
Practical	3 Sports	70	30%
PEP Coursework	Training programme	20	10%

2. Exam Command-Word Framework (8–9 Strategy)

- 1-mark = definition
- 2–3 marks = describe or state + example
- 4–6 marks = explain using theory + applied example
- 9 marks = evaluate with conclusion and balance

Always:

State → Explain → Apply → Conclude

3. Top 10 Examiner Tips

- ✓ Always name **muscle + movement + joint** together.
- ✓ Link every physiology point to a **sport action**.
- ✓ Include **data or examples** in extended answers.
- ✓ Use **connectives**: “therefore”, “as a result”, “this leads to”.
- ✓ For 9-mark Qs, use a **mini paragraph per AO**.
- ✓ Use sport you play — it adds realism.
- ✓ Diagrams score marks — label neatly.
- ✓ Time your paper: 1 mark ≈ 1 minute.
- ✓ Check all command words.
- ✓ Leave 5 min for graph/data review.

4. Vocabulary for Top-Band Writing

Adaptation, alveoli, aerobic, anaerobic, antagonist, arousal, cardiovascular, diffusion, endorphins, hypertrophy, metabolism, oxygen debt, overload, proprioception, recovery, synovial, tendons, vascular shunt, vasodilation, VO₂ max.

5. Quick Revision Tasks (2–5 mins)

- Sketch the heart and label flow.
- Match 8 fitness components to sports.
- Name 3 principles of training + definitions.
- State 2 long-term effects of exercise.
- Describe how motivation affects performance.
- Evaluate a doping case in 3 sentences.

6. Resources for Independent Study

- **Edexcel GCSE PE Specification (Pearson)**
- **BBC Bitesize – PE (Edexcel)**
- **Seneca Learning – GCSE PE Full Course**
- **TheEverLearner / PE Scholar**
- **YouTube:** EverLearner PE, The PE Classroom, Matt Spalding PE
- **Apps:** MyFitnessPal, Strava, Human Anatomy Atlas

📌 Final 60-Second Exam Check

- ✓ Correct units (m/s, kg, bpm)
- ✓ Key words (muscle, joint, principle)
- ✓ Applied example for every 3+ mark Q

Year 11 – Edexcel GCSE / IGCSE

Chinese (Mandarin) – Basic Supercurriculum

Why Supercurriculum?

Mandarin Chinese is one of the most widely spoken languages in the world — learning it builds **communication, memory, and intercultural understanding**. This supercurriculum moves students from memorising phrases to **thinking and expressing ideas naturally** using key grammar, tones, and cultural insight.

Core Content Areas (Topics)

Focus on expressing **who you are, what you do, where you go, and why** — across three themes:

Theme	Focus Topics
1. Identity & Culture	Family, friends, daily life, technology, hobbies, food, festivals
2. Local Area, Travel & Tourism	Home, neighbourhood, environment, transport, holidays
3. Education & Future Plans	School, study habits, jobs, ambitions, world of work

📌 Use the **same ideas across all four skills** — listening, speaking, reading, writing.

2. Key Grammar & Sentence Patterns

Function	Structure	Example
Word order	Subject + Time + Place + Verb + Object	我 今天 在学校 学中文。 (I study Chinese at school today.)
Measure words	个, 本, 张, 只, 条, etc.	一个朋友, 一本书
Negation	不 (for present/future) / 没 (for past)	我不喜欢。 / 我没去。
Past tense	Verb + 了	我吃了午饭。
Future tense	要 / 会 / 打算 + verb	我会去中国。
Adverbs	也, 都, 常常, 一起	我们都喜欢打篮球。
Comparisons	A 比 B + adjective	他比我高。
Time duration	Verb + 了 + time	我学中文了三年。
'Because... therefore...'	因为... 所以...	因为我累, 所以我不想去。
'Although... but...'	虽然... 但是...	虽然下雨, 但是很热。

☞ Use at least one connective (因为/ 虽然/ 如果) in every extended answer.

3. Vocabulary Core (High-Value Words)

Category	Essential Vocab
Family	爸爸, 妈妈, 哥哥, 妹妹, 家人
School	学校, 科目, 作业, 老师, 同学
Food	米饭, 面条, 水果, 茶, 水, 早餐
Daily life	起床, 上学, 看电视, 玩手机
Weather	晴天, 下雨, 冷, 热
Transport	公共汽车, 地铁, 骑自行车
Places	学校, 商店, 公园, 医院, 饭馆
Verbs	是, 有, 去, 来, 看, 吃, 喝, 想, 喜欢
Time words	今天, 明天, 昨天, 上午, 下午, 晚上
Connectives	和, 但是, 因为, 所以, 虽然, 如果
Adjectives	大, 小, 漂亮, 忙, 好, 高兴, 累

4. Pronunciation & Tones

- Four tones:

- ˉ (high) → mā 妈 (mother)
- ˊ (rising) → má (hemp)
- ˇ (fall-rise) → mǎ (horse)
- ˋ (falling) → mà (scold)

- Tone marks change meaning — practice by *listening & repeating daily*.

- Use **Pinyin + character** together in notes until confident.
- Apps:** ChineseSkill, Duolingo, Pleco (dictionary + stroke order).

5. Character Writing Basics

- **Stroke order matters.**
- **Left to right, top to bottom, horizontal before vertical.**
- Learn **radicals** (部首) → meaning hints (e.g., 水 = water, 氵 = water radical).
- Target **5–10 new characters/day** → revise weekly.

✍ Practice short sentences daily, not single words.

6. Listening & Reading Strategy

- **Listen for tone, key words, numbers, and time expressions.**
- Common question words: 什么, 谁, 哪儿, 几, 多少, 为什么.
- Focus on **topic vocabulary** and **context clues** (tense/time).
- In reading: look for familiar radicals, guess meaning by structure.

7. Speaking & Writing Frames

A. Describe People / Places

我的妈妈很漂亮，她喜欢运动。

我的学校不大，但是很干净。

B. Talk about Activities

周末我常常和朋友一起打篮球。

我打算去中国旅行。

C. Express Opinions

我喜欢学中文，因为很有用。

虽然很难，但是很有意思。

D. Past–Present–Future Frame (use all 3 for top marks)

去年我去了北京，现在我在英国学习，明年我要去中国。

8. Cultural Understanding

Area	Example
Festivals	春节 (Spring Festival), 中秋节 (Mid-Autumn Festival), 端午节 (Dragon Boat Festival)
Symbols	红色 = good luck; 龙 = power
Customs	礼貌 (politeness), 家庭很重要 (family importance)
Food culture	饺子, 面条, 茶
Daily	双筷子 (chopsticks), early meals, WeChat use

habits	
---------------	--

⑦ Compare with your culture — mention this in speaking/writing for high marks.

9. Common Mistakes & Fixes

Mistake	Fix
Wrong tone or order	Always say aloud before writing
Forgetting measure words	Add: 一个朋友 / 一张票
Mixing 不 & 没	不 = habit/future, 没 = past
Missing 了	Use 了 for completed action
Literal translation	Learn chunks, not words: 想要 = would like to

⑦ Weekly Study Rhythm (30–40 min/day)

Day	Focus	Task
Mon	Characters	Write 10 + review old ones
Tue	Grammar	3 pattern drills + 2 sentences
Wed	Listening	Watch 10-min video + note 5 words
Thu	Speaking	2-min talk using picture/topic
Fri	Reading	Read short text + 3 questions
Weekend	Culture	Learn 1 festival + 5 facts

1 Assessment Snapshot (Edexcel IGCSE Chinese)

Skill	Focus	Marks	Weight
Listening	Short dialogues, announcements, longer passages	50	25%
Reading & Translation	Articles, adverts, Chinese→English translation	50	25%
Speaking	Picture + role play + discussion	70	25%
Writing	90 / 150 characters + English→Chinese translation	60	25%

12. Quick Revision Tasks

- Write 5 sentences about your weekend.
- Record 1-minute voice note describing your school.
- Create a mini travel dialogue (你去哪儿？我去商店。).
- Quizlet: family / food / time words.
- Label your room items with Chinese characters.

13. Resources

- **Edexcel GCSE / IGCSE Chinese Specification**
- **BBC Bitesize Mandarin**
- **ChinesePod / Yoyo Chinese / Ninchanese**
- **Pleco** (dictionary + handwriting)
- **Du Chinese / Chairman's Bao** (graded readers)

- **YouTube:** Chinese Zero to Hero / Miss Lin Mandarin

14. Self-Check Before Exam

I can write **90–150 characters** independently.

I can use **past + present + future** tenses.

I can describe **people, places, and daily life** clearly.

I can understand the **gist** of a short passage.

I know **at least 5 connectives** and **5 measure words**.

I can give **opinions + reasons**.

Edexcel GCSE / IGCSE Chinese

Extended Supercurriculum – Year 11

1. Aim

To move from “memorising words” → to *thinking and expressing ideas in Chinese naturally*.

This programme strengthens **complex grammar, authentic communication, cultural literacy**, and **exam mastery** across all four skills.

Core Competence Targets

Skill	Advanced Focus
Listening & Reading	Understand main ideas <i>and</i> implied meaning.
Speaking & Writing	Communicate complex ideas in paragraphs using connectives.
Grammar & Vocabulary	Use all time frames, connectors , and measure words accurately.
Culture & Context	Integrate cultural examples naturally in responses.

2. Advanced Grammar Patterns

Structure	Meaning	Example
是...的	Emphasises detail (how, when, where something happened)	我是昨天坐地铁去的。(It was yesterday that I went by metro.)

除了...以外, 还...	"In addition to..."	除了中文以外, 我还学法文。
一边...一边...	"While doing..." (simultaneous actions)	我一边听音乐, 一边做作业。
把 + Object + Verb	Emphasises result/action	请把书给我。
被 + Verb	Passive voice	我的手机被偷了。
如果...就...	"If... then..."	如果明天下雨, 我们就不去。
虽然...但是...	"Although... but..."	虽然很难, 但是很有意思。
又...又...	"Both...and..."	她又聪明又漂亮。
越来越...	"More and more..."	天气越来越冷。
不仅...而且...	"Not only...but also..."	他不仅会说中文, 而且会写汉字。

☑ Use at least three of these in every long writing or speaking task for top-band marks.

3. High-Impact Connectives (for fluency)

因为 (because), 所以 (therefore), 虽然 (although), 但是 (but), 如果 (if), 还是 (or), 而且 (and also), 然后 (then), 最后 (finally), 比如 (for example).

4. Building Longer Sentences

- Join **two simple ideas** with 因为/所以 or 虽然/但是.
- Add **time phrases** to show sequence: 以前, 现在, 将来.
- Use **opinion + reason + example** pattern:

我觉得学习汉语很有用, 因为我想在中国工作。比如, 我喜欢跟中国人聊天。

5. Writing Skills for Top Grades

A. Structure for 150-Character Task

Section	Purpose	Example
Opening	Context + opinion	我最喜欢的节日是春节。
2. Past	Describe event + feeling	去年我跟家人一起吃年夜饭, 看烟花。
3. Present	What happens now	现在我们每年都给红包。
4. Future	Plans or dreams	明年我打算在中国过春节。
5. Reflection	Opinion + cultural comment	虽然很忙, 但是我觉得春节让我们更亲近。

✓ 4-5 sentences per paragraph.

✓ Use 3+ connectives + at least one complex grammar pattern.

6. Listening & Reading – Advanced Strategies

Strategy	Technique
Prediction	Use question words (谁 / 哪儿 / 什么时候) to anticipate answers.
Key details	Focus on numbers, time, and negation words (不 / 没).
Synonyms	老师 = 教师, 学生 = 同学.

Context clues	If one word unknown, read before & after it.
Multiple-choice	Eliminate clearly wrong answers; trust logic.

📅 Practice: 5 mins/day with **short dialogues** → summarise orally in English + Chinese.

7. Speaking – Natural Flow & Pronunciation

A. Conversation Strategy

Use the “PPP Framework”:

Point – Past – Plan

我喜欢打篮球。去年我跟朋友打比赛。明年我想加入篮球队。

B. Pronunciation Tips

- Practice **tone pairs** (e.g., 你好, 学校, 中国).
- Avoid **flat intonation** — each tone has a direction.
- Record and self-correct with **Pleco** or **Speechling**.

C. Photo Card Technique

1. Describe 3 things in the picture (谁, 在哪儿, 做什么).
2. Give personal link (“我也喜欢...”).
3. Add feeling/opinion + reason.

8. Translation Skills

English → Chinese	Example
“I went to China last year.”	我去年去了中国。
“I have studied Chinese for three years.”	我学中文学了三年。
“I would like to visit the Great Wall.”	我想去长城。
“Although it was expensive, it was worth it.”	虽然很贵，但是很值得。
“If I have time, I will watch TV.”	如果有时间，我就看电视。

📅 Always identify **time** + **subject** + **verb** + **object** + **complement** order.

9. Cultural Competence & Intercultural Thinking

Theme	Chinese Example	How to Discuss
Festivals	春节, 中秋节, 端午节	Compare with local holidays (e.g., “像我们的圣诞节一样...”)
School Life	早上上课, 午休, 晚自习	Compare school schedules
Food Culture	饺子, 火锅, 米饭	Describe family meal customs
Technology	微信, 支付宝	Link to global digital habits
Values	家庭, 尊敬长辈, 集体活动	Reflect on similarities/differences

📅 Add one cultural note in writing/speaking for A★ fluency range.

📅 Idiomatic & Advanced Expressions

Expression	Meaning	Use
马马虎虎	So-so	Describe feelings or weather
一心一意	Wholeheartedly	我一心一意地学习汉语。
对...感兴趣	To be interested in	我对音乐很感兴趣。
不但...而且...	Not only...but also	他不但聪明，而且努力。
越来越...	More and more	我的中文越来越好。
一边...一边...	Doing two actions	我一边听音乐，一边写作业。

1 Exam Overview (Edexcel)

Paper	Focus	Time	Weight
1	Listening	~35 min	25 %
2	Speaking	10–12 min	25 %
3	Reading & Translation	~50 min	25 %
4	Writing & Translation	~70 min	25 %

12. Top-Band Criteria (Grades 8–9 / A–A★)

Communicates detailed ideas clearly.

Wide range of vocab & structures (10+ per task).

3+ connectives, 3+ tenses, 1+ complex grammar pattern.

Few character or tone errors.

Confident cultural reference.

13. Common Pitfalls & Fixes

Pitfall	Fix
English word order	Always: <i>Time</i> → <i>Place</i> → <i>Verb</i> → <i>Object</i>
Forgetting tones	Say aloud when revising
Too few characters	Aim: 90–150+ characters per writing task
Literal translation	Think in ideas, not words
No conclusion	Always end with reflection or feeling

14. Weekly Challenge Cycle (6 Weeks)

Week	Focus	Task
1	Grammar	Write 10 sentences using 是...的 and 虽然...但是
2	Listening	Watch short video, note 8 new words
3	Writing	120 characters about your school life
4	Speaking	1-min self-recording using PPF
5	Culture	Research 1 festival & compare to yours
6	Translation	5 EN→CN + 5 CN→EN mini tasks

15. Independent Resources

- **Pearson Edexcel IGCSE Chinese Specification**
- **BBC Bitesize Mandarin / ChineseClass101 / Yoyo Chinese**
- **Pleco** (dictionary + flashcards)
- **HelloChinese / Du Chinese / The Chairman's Bao**
- **YouTube:** “Learn Chinese with Miss Lin” / “Chinese Zero to Hero”
- **Apps:** Skritter (characters), Speechling (speaking), HSK Online

Student Guide

(Edexcel GCSE / IGCSE Chinese – Year 11)

Command Words (Exam)

Command	Task
写 / 写作	Write — use 3+ time frames
说 / 谈谈	Speak — include opinion + reason
翻译	Translate — keep meaning & structure
阅读	Read — extract information
听 / 回答	Listen — note numbers, times, negations

2. Self-Evaluation Template

Topic: _____

3 time phrases

2 connectives

1 advanced grammar

1 idiom

1 cultural comparison

Checked for tone/character accuracy

3. Quick Exam Reminders

Practise tones daily.

Include **past** + **present** + **future** in every long task.

Use 2 connectives per paragraph.

Add cultural context.

End with opinion/reflection.

Always check measure words.

4. Cultural Bonus for A★

Mention one of:

- “我最喜欢的中国菜是...”
- “我觉得春节很热闹，因为...”
- “我希望将来在中国工作。”
- “在中国，家人常常一起吃饭。”

5. Example Closing Sentence Patterns

English	Chinese
In my opinion...	我觉得...
I hope that...	我希望...
It makes me happy because...	我很高兴，因为...
I learned a lot.	我学到了很多。
Although it was difficult, it was worth it.	虽然很难，但是很值得。

YEAR 11 – EDEXCEL GCSE / IGCSE GREEK (MODERN GREEK) – BASIC SUPERCURRICULUM

Why Supercurriculum?

Greek is one of Europe's richest languages — studying it strengthens **logic**, **culture**, and **expression**.

This supercurriculum helps students move from **word memorisation** to **confident communication**, using grammar, idioms, and examples from everyday Greek life.

Core Themes (as in the Edexcel syllabus)

Theme	Topics
1. Identity & Culture	Family, friends, leisure, technology, food, festivals, traditions
2. Local Area, Holiday & Travel	Town, transport, weather, environment, tourism
3. School & Future Aspirations	Subjects, teachers, rules, work, future plans
4. The World Around Us	Media, social issues, volunteering, global challenges

1. *Aim*: express your opinions naturally with reasons, time phrases, and connectives.

2. Grammar Core – Must Know

Function	Greek	English Example
To be / To have	είμαι / έχω	Είμαι μαθητής. Έχω δύο αδέρφια.
Present tense	-ω, -εις, -ει, -ουμε, -ετε, -ουν(ε)	Παίζω ποδόσφαιρο.
Past tense (aorist)	έπαιξα, έφαγα, πήγα	Πήγα στο σχολείο χτες.
Future tense	θα + verb	Αύριο θα διαβάσω.
Continuous / imperfect	διάβαζα, έτρωγα	Όταν ήμουν μικρός, έπαιζα κάθε μέρα.
Negatives	δεν / ποτέ / ούτε	Δεν μου αρέσει. Ποτέ δεν πάω.
Comparatives	πιο ... από / ο πιο ...	Είναι πιο μεγάλος από μένα.
Modal verbs	μπορώ / πρέπει / θέλω	Θέλω να πάω διακοπές.
Pronouns	μου, σου, του / αυτός, αυτή, αυτό	Ο φίλος μου είναι καλός.
Connectives	και, αλλά, γιατί, επειδή, αν, όταν, όμως	...αλλά δεν έχω χρόνο.

3. Vocabulary by Theme (high-frequency words)

Theme	Key Words
Family & Friends	οικογένεια, φίλος, γονείς, αδερφός, παιδί
Daily Life	πρωινό, σχολείο, διάβασμα, τηλεόραση, κινητό
Food	ψωμί, τυρί, σαλάτα, φρούτα, μακαρόνια, κρέας
Town & Travel	πόλη, χωριό, λεωφορείο, αυτοκίνητο, παραλία, ξενοδοχείο
School	μάθημα, καθηγητής, μαθητής, διάλειμμα, τεστ
Hobbies	μουσική, αθλητισμός, χορός, ταινίες, ίντερνετ
Weather	ήλιος, βροχή, χιόνι, ζέστη, κρύο
Work	δουλειά, γραφείο, λεφτά, αφεντικό, υπάλληλος

4. Essential Phrases for Speaking & Writing

Opinions & Reasons

- Μου αρέσει / Δεν μου αρέσει γιατί...
→ *Μου αρέσει η μουσική γιατί με χαλαρώνει.*
- Προτιμώ ... γιατί είναι πιο ενδιαφέρον.
- Νομίζω ότι... / Πιστεύω πως...

Time Expressions

- κάθε μέρα, χτες, αύριο, συνήθως, πέρσι, το καλοκαίρι, στο μέλλον

Comparisons

- πιο... από / τόσο... όσο / ο πιο...

Connectives

- και, αλλά, επειδή, ωστόσο, μετά, πριν, αν

Past–Present–Future Frame

Πέρσι πήγα στην Κρήτη.

Τώρα μένω στην Αθήνα.

Στο μέλλον θέλω να σπουδάσω στο εξωτερικό.

5. Listening & Reading Strategy

- Focus on **verbs, time phrases, and negation words** (δεν, ποτέ).
- Learn to identify **key connectors** (γιατί, επειδή, όταν).
- In **multiple-choice questions**, eliminate obvious distractors.
- Practise with **Greek news for children** or **Easy Greek YouTube**.

6. Speaking Essentials

Photo-card Strategy

Describe (3–4 details) → Στη φωτογραφία βλέπω ένα σχολείο.

2. Add opinions → Μου αρέσει γιατί φαίνεται όμορφο.

3. Add a reason or feeling → Οι άνθρωποι είναι χαρούμενοι.

4. Ask a question back → Εσύ πηγαίνεις στο σχολείο με τα πόδια;

Role-play tip: learn polite requests → Μπορώ να έχω...; / Θα ήθελα...

7. Writing Tips

- Use **3 tenses** (past–present–future).
- Include **opinions + reasons + examples**.
- Add **connectors** for flow.
- Use **paragraphs** (intro + main idea + conclusion).
- End with a **reflective comment**:

Ήταν υπέροχα και ελπίζω να ξαναπάω!

Mini model (90 words):

Το καλοκαίρι πήγαμε οικογενειακώς στη Ρόδο. Κάναμε μπάνιο, φάγαμε φρέσκο ψάρι και είδαμε πολλά μέρη. Μου άρεσε πολύ γιατί είχε ζέστη και ωραίους ανθρώπους. Στο μέλλον θέλω να μείνω εκεί μόνιμα.

8. Common Mistakes & Fixes

Mistake	Fix
Forgetting accents (τόνοι)	Always check last syllable!
Mixing tenses	Write in time order (χτες → σήμερα → αύριο)
Translating word-by-word	Think in Greek chunks
Forgetting articles	Always include: ο, η, το
Using English syntax	Follow Greek order (subject–verb–object)

9. Weekly Study Rhythm (30–40 min/day)

Day	Focus	Task
Mon	Vocabulary	10 new words + 3 sentences

Tue	Grammar	5 verbs in 3 tenses
Wed	Listening	10-min Easy Greek clip
Thu	Speaking	Record 1-min talk
Fri	Reading	Short article + 3 questions
Weekend	Culture	Learn one Greek celebration (π.χ. Πάσχα)

2 Greek Culture & Society

Theme	Example
Festivals	Πάσχα, Απόκριες, Πρωτοχρονιά
Food	Μουσακάς, γύρος, ντολμαδάκια
Music	Ρεμπέτικα, έντεχνα, λαϊκά
Places	Αθήνα, Θεσσαλονίκη, Σαντορίνη
Traditions	Χοροί, οικογενειακά τραπέζια, φιλοξενία

2 *Tip*: mention one cultural element in writing/speaking for top marks.

1 Assessment Snapshot (Edexcel GCSE / IGCSE Greek)

Paper	Skill	Time	Weight
Paper 1	Listening	~35-40 min	25%
Paper 2	Speaking	10-12 min	25%
Paper 3	Reading + Greek→English	1 h	25%
Paper 4	Writing + English→Greek	1 h 15	25%

12. Self-Check Before Any Task

Used **3 tenses** (past, present, future)

At least **2 connectives**

One **opinion** + **reason**

One **time phrase**

Spelling and accent marks correct

At least **90-150 words** for writing

13. Recommended Resources

- Edexcel GCSE/IGCSE Greek Specification
- BBC Languages Greek / LingQ / Duolingo Greek
- Easy Greek (YouTube) – street interviews & subtitles
- GreekPod101 / Glossika Greek
- WordReference Greek–English Dictionary
- EPT Παιδικά / ΣΚΑΪ Νέα (easy news for listening)

14. Quick Practice Ideas

- Write 5 sentences about your family.
- Record 1-minute talk about your hobbies.
- Translate: “Last year I went to Greece with my friends.”
- Describe your school in 5 lines.
- Learn 5 verbs + 5 nouns per week.

EDEXCEL GCSE / IGCSE GREEK

Extended Supercurriculum – Year 11

1. Aim

To move beyond basic communication and towards **authentic Greek expression** — accurate grammar, rich vocabulary, idiomatic phrasing, and cultural awareness across all four skills.

Core Mastery Goals

Skill	Advanced Focus
Grammar & Structure	Use all key tenses + complex clauses
Vocabulary	Develop thematic and idiomatic range
Fluency	Build natural flow with connectives
Culture	Integrate cultural, historical, and social awareness
Accuracy	Consistent spelling, accents, and gender agreement

2. Advanced Grammar Toolkit

Concept	Function	Example
Perfective vs Imperfective	Completed vs ongoing past actions	Διάβαζα (was reading) vs διάβασα (read)
Subjunctive (να + verb)	Wishes, purpose, obligation	Θέλω να πάω διακοπές.

Passive voice (-θηκ-forms)	Emphasises the receiver	Το σπίτι χτίστηκε το 1990.
Relative clauses (που)	Link sentences naturally	Ο φίλος που γνώρισα χτες είναι Ιταλός.
Conditional (θα + imperfect)	Hypothetical / polite	Θα ήθελα να ταξιδέψω στην Κύπρο.
Gerund-like (μιλώντας)	Describe simultaneous action	Μιλώντας με τη γιαγιά, έμαθα πολλά.
Conjunctions	επειδή, ώστε, παρόλο που, ενώ	Παρόλο που κουράστηκα, τελείωσα τη δουλειά.

☑ Include at least **three complex structures** (να-clauses, conditionals, or relative clauses) per extended answer.

3. Extended Connectives & Discourse Markers

Type	Examples
Adding	επίσης, ακόμη, επιπλέον
Contrasting	όμως, αντίθετα, παρόλο που
Sequencing	πρώτα, μετά, στη συνέχεια, τέλος
Explaining	δηλαδή, με άλλα λόγια
Concluding	γενικά, επομένως, τελικά

4. High-Level Vocabulary Themes

Theme	Examples
Environment	ανακύκλωση, μόλυνση, καθαρή ενέργεια, υπερθέρμανση
Technology	διαδίκτυο, μέσα κοινωνικής δικτύωσης, εφαρμογές
Culture & Arts	θέατρο, ποίηση, μουσική, παράδοση, πολιτισμός
Society	μετανάστευση, εκπαίδευση, υγεία, οικονομία
Emotions	χαρά, λύπη, ενθουσιασμός, ανησυχία, υπερηφάνεια

☑ Use specific vocabulary in context:

“Η ανακύκλωση είναι σημαντική γιατί βοηθάει το περιβάλλον και μειώνει τα σκουπίδια.”

5. Writing for Top Grades

A. Essay Structure (150–180 words)

Section	Focus	Example
Introduction	Set theme & opinion	Το αγαπημένο μου μέρος στην Ελλάδα είναι η Κρήτη.
2. Past	Give event & experience	Πέρσι πήγα με την οικογένειά μου και επισκεφθήκαμε το Ηράκλειο.
3. Present	Add description + emotions	Το νησί είναι όμορφο και οι άνθρωποι πολύ φιλικοί.
4. Future	Ambition / plan	Στο μέλλον θέλω να σπουδάσω εκεί.
5.	Reflection	Παρόλο που είναι μακριά, αξίζει να το

Conclusion	επισκεφτεί κανείς.
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B. Style Tips

- ✓ Use **paragraphs** clearly.
- ✓ Add **idioms** or **colloquial phrases** (μου ήρθε στο μυαλό, ταξίδι ζωής).
- ✓ Avoid repetition — use synonyms (όμορφος → υπέροχος → καταπληκτικός).
- ✓ End with opinion/reflection.

6. Speaking Techniques

A. Conversation Flow Model

Use **PPP + O** = Point → Past → Plan → Opinion

Μου αρέσει ο αθλητισμός. Πέρσι έπαιξα βόλεϊ στο σχολείο και του χρόνου θα γράφτώ σε ομάδα, γιατί με βοηθά να χαλαρώνω.

B. Photo Card Strategy

Describe the image (3 details).

2. Link with personal experience.
3. Give opinion + reason.
4. Add question back to examiner.

Example:

Στη φωτογραφία βλέπω μια οικογένεια που τρώει μαζί. Μου αρέσουν τα οικογενειακά τραπέζια γιατί περνάμε όμορφα. Εσύ πόσο συχνά τρως με την οικογένειά σου;

7. Listening & Reading – Advanced Skills

- **Anticipate content:** read question before track.
- Recognise **synonyms:** δουλειά = εργασία; παιδί = νέο άτομο.
- Identify **tenses & negations:** δεν πήγε / δεν θα πάει.
- **Summarise gist** before choosing answer.
- Keep a “list of traps” (double negatives, dates, opposite options).

8. Cultural Understanding

Area	Example	How to Use in Exams
Festivals	Πάσχα, Απόκριες, Δεκαπενταύγουστος	Describe celebrations & compare with your own culture
Music & Film	Θεοδωράκης, Χαρούλης, “Πολίτικη Κουζίνα”	Mention as cultural examples in writing
Cuisine	μεζές, φέτα, μπουγάτσα	Use for cultural comparison tasks
Values	οικογένεια, φιλοξενία, παράδοση	Add to opinion writing (“Η φιλοξενία είναι βασικό στοιχείο του ελληνικού πολιτισμού.”)
Tourism & Modern Greece	Αθήνα, Θεσσαλονίκη, νησιά, μετανάστευση	Integrate factual references for authenticity

✦ One cultural detail = instant examiner bonus.

9. Idiomatic & Expressive Phrases

Greek	Meaning	Use
κάνω βόλτα	go for a walk	Κάθε μέρα κάνω βόλτα στην παραλία.
περνάω τέλεια	have a great time	Πέρσι στις διακοπές πέρασα τέλεια.
μου ήρθε στο μυαλό	it came to mind	Μου ήρθε στο μυαλό η παιδική μου ηλικία.
δεν βλέπω την ώρα	can't wait	Δεν βλέπω την ώρα να ξαναπάω στην Ελλάδα!
αξίζει τον κόπο	it's worth it	Αν και ήταν δύσκολο, άξιζε τον κόπο.

10 Common Pitfalls & Fixes

Pitfall	Fix
Missing accent marks	Always check final draft aloud
Gender agreement	Match adjectives with nouns (όμορφος – όμορφη – όμορφο)
Literal translation from English	Use Greek idioms or natural phrasing
Forgetting cases	Learn nominative/genitive/accusative endings
No tense variety	Include at least 3 time frames

1 Weekly Challenge Cycle (6 Weeks)

Week	Focus	Output
1	Grammar	10 advanced verbs in 3 tenses
2	Writing	150-word essay “Το αγαπημένο μου μέρος”
3	Listening	Watch Easy Greek episode + 10 vocab notes
4	Speaking	Record 1-min talk with 2 idioms
5	Culture	Compare Greek and local celebrations
6	Mixed	Practice past paper (Paper 3 Reading)

12. Exam Overview (Edexcel)

Paper	Focus	Time	Weight
1	Listening	~35 min	25%
2	Speaking	10–12 min	25%
3	Reading + Greek→English translation	~1h	25%
4	Writing + English→Greek translation	~1h 15	25%

11 Student Guide

(Edexcel GCSE / IGCSE Greek – Year 11)

Command Words (Exam Vocabulary)

Command	Meaning	Tip
Γράψε / Περιέγραψε	Write / Describe	Include details + opinions

Πες / Μίλησε	Say / Speak	Add examples + emotions
Σύγκρινε	Compare	Use “όπως / αντίθετα / παρόμοιο”
Εξήγησε / Ανάλυσε	Explain / Analyse	Give reason + result
Μετάφρασε	Translate	Keep sense, not word order

2. Top-Band Criteria (Grades 8–9 / A–A★)

Wide range of structures (subjunctive, conditionals, relative clauses)

At least 3 time frames

5+ connectives and idioms

Few spelling/grammar errors

Clear organisation & logical flow

Natural, expressive tone

3. Revision Plan (4 Weeks)

Week	Focus	Activity
1	Grammar review	Past vs perfective practice
2	Writing	2 sample essays + feedback
3	Speaking	3 picture prompts + idioms
4	Mock exam	Full paper timed practice

4. Cultural Reflection Prompts

«Τι σημαίνει για σένα η ελληνική φιλοξενία;»

«Πώς μοιάζει η ελληνική οικογένεια με τη δική σου;»

«Γιατί αξίζει να μάθει κανείς ελληνικά;»

«Ποια είναι η αγαπημένη σου ελληνική γιορτή;»

5. Quick Exam Checklist

Used **να**-clauses and **complex connectives**

Included **cultural example**

3 tenses (past, present, future)

Accent marks checked

Paragraphs structured logically

Proofread aloud (τονισμός και ρυθμός)

6. Recommended Resources

- Edexcel GCSE/IGCSE Greek Specification
- BBC Languages – Greek
- Easy Greek (YouTube)
- GreekPod101 / LingQ / Duolingo Greek
- WordReference Greek–English
- ΣΚΑΪ News / ERT Παιδικά / TaNea.gr (simple news)
- Apps: Clozemaster, Quizlet Greek GCSE

